# 13 Cardiovascular system

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## Introduction

Recent decades have seen major changes in patterns of cardiovascular disease. In the developed world, syphilitic and tuberculous involvement of the cardiovascular system has become rare, and the incidence of rheumatic disease has declined considerably. Myocardial and conducting tissue diseases are diagnosed with increasing frequency and the importance of arterial hypertension has become recognized. Coronary artery disease has emerged as the major cardiovascular disorder of the era, becoming the most common cause of premature death throughout Europe, North America and Australasia. In the last 30 years, there has been a steady fall in age-specific death rates from coronary artery disease in Western societies, but elsewhere its prevalence is increasing; in the underdeveloped world it now threatens to overtake malnutrition and infectious disease as the major cause of death.

As patterns of cardiovascular disease have changed, so have the cardiologist's diagnostic tools, although a good history and thorough clinical examination remain cornerstones of the assessment of patients with cardiovascular disease. A century that started with the stethoscope, the sphygmomanometer, the chest X-ray and a very rudimentary electrocardiogram saw the development of a variety of new imaging modalities, using ultrasound, radioisotopes, X-rays and magnetic resonance. This non-invasive capability was complemented by the introduction of the catheterization laboratory, permitting angiographic imaging, electrophysiological recording and tissue biopsy of the heart. Add to this the resources of the chemical pathology, bacteriology and molecular biology laboratories, and the array of diagnostic technology available to the modern cardiologist becomes almost overwhelming.

## The cardiac history

The history should record details of presenting symptoms, of which the most common are chest pain, fatigue and dyspnoea, palpitations, and presyncope or syncope (see below and Box 13.1). Previous illness should also be recorded, as it may provide important clues about the cardiac diagnosis; thyroid, connective tissue and neoplastic disorders, for example, can all affect the heart. Rheumatic fever in childhood is important because of its association with valvular heart disease, and diabetes and dyslipidaemias because of their association with coronary artery disease. Smoking is a major risk factor for coronary artery disease. Alcohol abuse predisposes to cardiac arrhythmias and cardiomyopathy. The cardiac history should quantify both habits in terms of pack-years smoked and units of alcohol consumed. The use of other recreational drugs (in particular cocaine) can be associated with acute presentations of chest pain and intravenous drug use is an increasingly important cause of infective endocarditis.

The family history should always be documented because coronary artery disease and hypertension often run in families, as do some of the less common cardiovascular disorders, such as hypertrophic cardiomyopathy. Indeed, in patients with hypertrophic cardiomyopathy, a family history of sudden death is probably the single most important indicator of risk. Finally, the drug history should be recorded, because many commonly prescribed drugs are potentially cardiotoxic. β-Blockers and some calcium channel blockers (diltiazem, verapamil), for example, can cause symptomatic bradycardias, and tricyclic antidepressants and β agonists can cause tachyarrhythmias. Vasodilators cause variable reductions in blood pressure, which can lead to syncopal attacks, particularly in patients with aortic stenosis. The myocardial toxicity of certain cytotoxic drugs (notably doxorubicin and related compounds) is an important cause of cardiomyopathy.

## Chest pain

Myocardial ischaemia, pericarditis, aortic dissection and pulmonary embolism are the most common causes of acute, severe chest pain. Chronic, recurrent chest pain is usually caused by angina, oesophageal reflux or musculoskeletal pain.

### Myocardial ischaemia

Ischaemia of the heart results from an imbalance between myocardial oxygen supply and demand,

**Box 13.1 Structure for the cardiac history**

**Presenting complaint (PC)**

- The symptom that prompts the patient to seek medical attention—commonly chest pain, breathlessness (dyspnoea), palpitation, dizziness or blackouts (syncope)

**History of presenting complaint (HPC)**

- This should define the nature of the symptoms, initially through open questioning. Closed questions are used to elicit the presence or absence of features which help to differentiate between diagnoses:
  - Chest pain: site, radiation, character, duration, provoking and relieving factors, associated symptoms?
  - Breathlessness: orthopnoea, paroxysmal nocturnal dyspnoea, ankle swelling, cough, wheeze, haemoptysis?
  - Palpitation: sudden onset and offset, 'thumps' or 'pauses', presyncope or syncope?
  - Dizziness/syncope: provoking factors, warning, duration, recovery?

**Risk factors for cardiovascular disease**

- Smoking, hypertension, hypercholesterolaemia, diabetes, family history of premature vascular disease

**Past medical history (PMH)**

- Stroke or transient ischaemic attack (TIA), renal impairment, rheumatic fever, peripheral vascular disease, other
- Operations, hospital clinic attendances

**Family history**

- Cardiac disease, sudden death

**Drug history**

- Include quantification of alcohol intake
- If a patient with known cardiovascular disease is not taking the recognized standard treatment, the reason for this should be established. For example, why no statin treatment in a patient with previous myocardial infarction?—'Because it caused muscle pains'.

producing pain called angina (Boxes 13.2 and 13.3). Angina is usually a symptom of atherosclerotic coronary artery disease, which impedes myocardial oxygen supply. Other causes of coronary artery disease (Box 13.4) are rare. However, it is important to be vigilant for causes of angina owing to increased myocardial oxygen demand, such as aortic stenosis. The history is diagnostic for angina if the location of the pain, its character, its relation to exertion and its duration are typical. The patient describes retrosternal pain, which may radiate into the arms, the throat or the jaw. It has a constricting character, is provoked by exertion and relieved within minutes by rest. The patient's threshold for angina is typically reduced after eating or in cold weather owing to the diversion of blood to the gut and the increased myocardial work consequent upon peripheral vasoconstriction, respectively. Occasionally angina is

**Box 13.2 Angina**

**Typical patient**

- Middle-aged or elderly man or woman often with a family history of coronary heart disease and one or more of the major reversible risk factors (smoking, hypertension, hypercholesterolaemia, diabetes)

**Major symptoms**

- Exertional chest pain and shortness of breath. Pain often described as 'heaviness' or 'tightness', and may radiate into arms, neck or jaw. Pain comes predictably after exertion

**Major signs**

- None, although hypertension and signs of hyperlipidaemia (xanthelasmata, xanthomas) may be present
- Peripheral vascular disease, evidenced by absent pulses or arterial bruits, is commonly associated with coronary heart disease

**Diagnosis**

- Typical history is most important diagnostic tool
- Electrocardiogram (ECG): often normal; may show Q waves in patients with previous myocardial infarction
- Exercise ECG test: exertional ST depression
- Isotope or magnetic resonance perfusion scan: stress-induced perfusion defects
- Coronary angiogram: confirms coronary artery disease

**Additional investigations**

- Blood sugar and lipids to rule out diabetes and dyslipidaemia

**Comments**

- A careful history is the single most important means of diagnosing angina

**Box 13.3 Causes of angina**

**Impaired myocardial oxygen supply**

- Coronary artery disease:
  - atherosclerosis
  - arteritis in connective tissue disorders
  - diabetes mellitus
- Coronary artery spasm
- Congenital coronary artery disease:
  - arteriovenous fistula
  - anomalous origin from pulmonary artery
- Severe anaemia or hypoxia

**Increased myocardial oxygen demand**

- Left ventricular hypertrophy:
  - hypertension
  - aortic valve disease
  - hypertrophic cardiomyopathy
- Tachyarrhythmias

**Box 13.4 Causes of coronary artery disease**

- Atherosclerosis
- Arteritis:
  - systemic lupus erythematosus
  - polyarteritis nodosa
  - rheumatoid arthritis
  - ankylosing spondylitis
  - syphilis
  - Takayasu's disease
- Coronary dissection:
  - spontaneous
  - catheter or angioplasty induced
- Embolism:
  - infective endocarditis
  - left atrial/ventricular thrombus
  - left atrial/ventricular tumour
  - prosthetic valve thrombus
  - paradoxical embolism
  - complication of cardiac catheterization
- Coronary mural thickening:
  - amyloidosis
  - radiation therapy
  - Hurler's disease
  - pseudoxanthoma elasticum
- Other causes of coronary luminal narrowing:
  - aortic dissection
  - coronary spasm
- Congenital coronary artery disease:
  - anomalous origin from pulmonary artery
  - arteriovenous fistula

provoked only by the first significant activity of the day, a phenomenon known as the 'warm-up effect', which is caused by myocardial preconditioning. Less commonly, myocardial ischaemia may manifest as breathlessness, fatigue or symptoms that the patient finds difficult to describe—'I just have to stop'—in which case the clues to the diagnosis are the relation of the symptoms to exertion, the presence of risk factors for coronary artery disease and the absence of an alternative explanation for the symptoms, such as heart failure.

## Acute coronary syndromes

In acute coronary syndromes, which are life-threatening cardiac emergencies, the pain is similar in location and character to angina, but is usually more severe, more prolonged and unrelieved by rest (Box 13.5).

## Pericarditis

Pericarditis causes central chest pain, which is sharp in character and aggravated by deep inspiration, cough or postural changes. Characteristically, the pain is exacerbated by lying recumbent and reduced by sitting forward. Pericarditis is usually idiopathic or caused by Coxsackie B infection. It may also occur as a complication of myocardial infarction, but other causes are seen less commonly (Box 13.6). Myocardial involvement—myocarditis—is increasingly recognized. It is associated with the release of cardiac enzymes and abnormalities on the electrocardiogram (ECG) that can be indistinguishable from myocardial infarction.

**Box 13.5 Acute coronary syndromes**

**Typical patient**

- Middle-aged (male) or elderly (either sex) patient, often with a family history of coronary artery disease and one or more of the major reversible risk factors (smoking, hypertension, hypercholesterolaemia, diabetes). Care must be taken in female patients: coronary artery disease occurs frequently and is often under-diagnosed.
- In many patients, there is no preceding history of angina.

**Major symptoms**

- Chest pain and shortness of breath. Pain usually prolonged and often described as 'heaviness' or 'tightness', with radiation into arms, neck or jaw. Alternative descriptions include 'congestion' or 'burning', which may be confused with indigestion.

**Major signs**

- Frequently none
- Autonomic disturbance, sweating, vomiting – implies myocardial infarction
- Tachycardia (anterior myocardial infarction), bradycardia (inferior myocardial infarction)
- Fourth heart sound, dyskinetic precordial impulse, pulmonary oedema with large infarcts

**Diagnosis**

(STEMI = ST elevation myocardial infarction)

| | STEMI | Non-STEMI | Unstable angina |
|---|---|---|---|
| ECG | ST elevation | Normal, ST depression, T-wave inversion | Normal, ST depression, T-wave inversion |
| Cardiac biomarkers (e.g. troponin I or T) | Raised | Raised | Normal |

**Additional investigations**

- Biochemistry: blood sugar and lipids to rule out diabetes and dyslipidaemia
- Risk stratification: echocardiogram (left ventricular function), coronary angiogram in high-risk patients, perfusion imaging in low-risk patients

**Comments**

- History and troponin testing most useful diagnostic tools in non-ST elevation acute coronary syndromes

## Aortic dissection

Aortic dissection produces severe tearing pain in either the front or the back of the chest. The onset

**Box 13.6 Causes of acute pericarditis**

- Idiopathic
- Infective:
  - viral (Coxsackie B, influenza, herpes simplex)
  - bacterial (*Staphylococcus aureus*, *Mycobacterium tuberculosis*)
- Connective tissue disease:
  - systemic lupus erythematosus
  - rheumatoid arthritis
  - polyarteritis nodosa
- Uraemia
- Malignancy (e.g. breast, lung, lymphoma, leukaemia)
- Radiation therapy
- Acute myocardial infarction
- Post-myocardial infarction/cardiotomy (Dressler's syndrome)

**Box 13.7 Aortic dissection**

**Typical patient**

- Middle-aged or elderly patient with a history of hypertension or arteriosclerotic disease
- Occasionally younger patient with aortic root disease (e.g. Marfan's syndrome)

**Major symptoms**

- Chest pain, typically interscapular

**Major signs**

- Often none
- Sometimes regional arterial insufficiency (e.g. occlusions of coronary artery causing myocardial infarction, carotid or vertebral artery causing stroke, spinal artery causing hemi- or quadriplegia, renal artery causing renal failure); subclavian artery occlusion may cause differential blood pressure in either arm; aortic regurgitation; cardiac tamponade; sudden death.

**Diagnosis**

- Chest X-ray: widened mediastinum, occasionally with left pleural effusion
- Transoesophageal echocardiogram, computed tomography (CT) scan or magnetic resonance imaging (MRI) scan: confirms dissection

**Comments**

- Type A dissections involve the ascending aorta and are usually treated surgically. Type B dissections involve the arch and/or descending aorta and are usually managed medically or with an endovascular stent.

is abrupt, unlike the crescendo quality of ischaemic cardiac pain (Box 13.7).

## Pulmonary embolism

Peripheral pulmonary embolism causes sudden-onset sharp, pleuritic chest pain, breathlessness and haemoptysis. Major, central pulmonary embolism presents with breathlessness, chest pain that can be indistinguishable from ischaemic chest pains and syncope. Risk factors for pulmonary embolism should be sought in the history (Box 13.8).

**Box 13.8 Pulmonary embolism**

**Typical patient**

- Recent surgery, lower limb fracture or long-distance air travel; obese; sedentary; heart failure; malignancy

**Major symptoms**

- Chest pain, dyspnoea, haemoptysis, syncope

**Major signs**

- Peripheral emboli: pleural rub
- Large, central emboli: tachycardia, hypotension, cyanosis, raised jugular venous pressure (JVP)

**Diagnosis**

- D-Dimer: a negative D-dimer in a low-risk patient makes pulmonary embolism very unlikely.
- ECG: sinus tachycardia; right bundle branch block (RBBB); classic 'S1, Q3, T3' pattern uncommon
- Chest X-ray: normal; wedge-shaped peripheral opacification; absent pulmonary vascular markings
- Echocardiogram: dilated right heart in some cases of large central pulmonary embolism
- CT pulmonary angiogram: has superseded V/Q scanning as the diagnostic test of choice.

**Comments**

- Suspect pulmonary embolism in patients with unexplained hypoxia. Thrombolytic therapy should be considered for patients with pulmonary embolism associated with shock and/or a dilated right heart on echo. Patients with no risk factors for pulmonary embolism should be investigated for prothrombotic states.

Rare cardiovascular causes of chest pain include mitral valve disease associated with massive left atrial dilatation. This causes discomfort in the back, sometimes associated with dysphagia owing to oesophageal compression. Aortic aneurysms can also cause pain in the chest owing to local compression.

## Dyspnoea

Dyspnoea is an abnormal awareness of breathing occurring either at rest or at an unexpectedly low level of exertion. It is a major symptom of many cardiac disorders, particularly left heart failure (Table 13.1), but its mechanisms are complex. In acute pulmonary oedema and orthopnoea, dyspnoea is caused mainly by elevated left atrial pressure that characterizes left heart failure (Box 13.9). This produces a corresponding elevation of the pulmonary capillary pressure and increases transudation into the lungs, which become oedematous and stiff. Oxygenation of blood in the pulmonary arterioles is reduced, causing hypoxaemia, and this, together

**Table 13.1** Causes of heart failure

| Ventricular pathophysiology | Clinical examples |
|---|---|
| Restricted filling | Mitral stenosis<br>Constrictive pericarditis<br>Restrictive cardiomyopathy<br>Hypertrophic cardiomyopathy |
| Pressure loading | Hypertension<br>Aortic stenosis<br>Coarctation of the aorta |
| Volume loading | Mitral regurgitation<br>Aortic regurgitation |
| Contractile impairment | Coronary artery disease<br>Dilated cardiomyopathy<br>Myocarditis |
| Arrhythmia | Severe bradycardia<br>Severe tachycardia |

**Box 13.9** Acute left ventricular failure

**Typical patient**

- Patient with acute myocardial infarction or known left ventricular disease

**Major symptoms**

- Severe dyspnoea, orthopnoea, frothy sputum

**Major signs**

- Low-output state (hypotension, oliguria, cold periphery); tachycardia; S3; sweating; crackles at lung bases

**Diagnosis**

- Chest X-ray: bilateral air space consolidation with typical perihilar distribution
- Echocardiogram: usually confirms left ventricular disease

**Additional investigations**

- ECG: may show evidence of acute or previous myocardial infarction
- Blood gas analysis: shows variable hypoxaemia

**Comments**

- Although most cases are caused by acute myocardial infarction or advanced left ventricular disease, it is vital to exclude valvular disease, which is potentially correctable by surgery.

**Box 13.10** Congestive heart failure

**Typical patient**

- Middle-aged (male) or elderly (either sex) patient with a history of myocardial infarction or longstanding hypertension
- In cases where there is no clear cause, always enquire about alcohol consumption.

**Major symptoms**

- Exertional fatigue and shortness of breath, with orthopnoea and paroxysmal nocturnal dyspnoea in advanced cases

**Major signs**

- Fluid retention: basal crackles, raised JVP, peripheral oedema
- Reduced cardiac output: cool skin, peripheral cyanosis
- Other findings: third heart sound

**Diagnosis**

- ECG: usually abnormal; often shows Q waves (previous myocardial infarction), left ventricular hypertrophy (hypertension), or left bundle branch block (LBBB)
- Chest X-ray: cardiac enlargement, congested lung fields
- Echocardiogram: left ventricular dilatation with regional (coronary artery disease) or global (cardiomyopathy) contractile impairment

**Additional investigations**

- Raised B-type natriuretic peptide useful in cases of diagnostic uncertainty
- Renal function as prelude to diuretic and angiotensin converting enzyme (ACE) inhibitor therapy
- Blood count to rule out anaemia

**Comments**

- The echocardiogram is the single most important diagnostic test in the patient with heart failure.

**Table 13.2** NYHA classification

| NYHA class | Description |
|---|---|
| I | Asymptomatic |
| II | Symptoms on normal exertion (e.g. walking up a flight of stairs) |
| III | Symptoms on minimal exertion, e.g. getting dressed |
| IV | Symptoms at rest |

NYHA, New York Heart Association.

with the extra effort required to ventilate the stiff lungs, causes dyspnoea.

## Exertional dyspnoea

Exertional dyspnoea is the most troublesome symptom in heart failure (Box 13.10). Exercise causes a sharp increase in left atrial pressure and this contributes to the pathogenesis of dyspnoea by causing pulmonary congestion (see above). However, the severity of dyspnoea does not correlate closely with exertional left atrial pressure, and other factors must therefore be important. These include respiratory muscle fatigue and the effects of exertional acidosis on peripheral chemoreceptors. As left heart failure worsens, exercise tolerance deteriorates. In advanced disease, the patient is dyspnoeic at rest.

Breathlessness in heart failure can be simply classified by use of the New York Heart Association Classification (Table 13.2). It is simple to acquire

and despite poor reproducibility and its subjective nature provides powerful prognostic information. For example, patients with Class IV heart failure have a very poor outlook.

### Orthopnoea

In patients with heart failure, lying flat causes a steep rise in left atrial and pulmonary capillary pressure, resulting in pulmonary congestion and severe dyspnoea. To obtain uninterrupted sleep, extra pillows are required, and in advanced disease, the patient may choose to sleep sitting in a chair.

### Paroxysmal nocturnal dyspnoea

Frank pulmonary oedema on lying flat wakes the patient from sleep with distressing dyspnoea and fear of imminent death. The symptoms are corrected by standing upright, which allows gravitational pooling of blood to lower the left atrial and pulmonary capillary pressure, the patient often feeling the need to obtain air at an open window.

## Fatigue

Exertional fatigue, an important symptom of heart failure, is particularly troublesome towards the end of the day. Its aetiology is complex but is caused partly by deconditioning and muscular atrophy.

## Palpitation

Awareness of the heartbeat is common during exertion or heightened emotion. Under other circumstances it may be indicative of an abnormal cardiac rhythm. A description of the rate and rhythm of the palpitation is essential as are exacerbating behaviours—such as exercise, caffeine intake. Extrasystoles are common but rarely signify important heart disease. They usually are experienced as 'missed' or 'dropped' beats; the forceful beats that follow may also be noticed. Rapid irregular palpitation is typical of atrial fibrillation. Rapid regular palpitation of abrupt onset occurs in atrial, junctional and ventricular tachyarrhythmias.

## Dizziness and syncope

Cardiovascular disorders produce dizziness and syncope by transient hypotension, resulting in abrupt cerebral hypoperfusion. For this reason, patients who experience cardiac syncope usually describe either brief lightheadedness or no warning symptoms at all prior to their syncopal attacks. Recovery is usually rapid, unlike with other common causes of syncope (e.g. stroke, epilepsy, overdose).

### Postural hypotension

Syncope on standing upright reflects inadequate baroreceptor-mediated vasoconstriction. It is common in the elderly in whom it is frequently compounded by medication. Abrupt reductions in blood pressure and cerebral perfusion cause the patient to fall to the ground, whereupon the condition corrects itself.

### Vasovagal syncope

Vasovagal syncope is caused by autonomic overactivity, usually provoked by emotional or painful stimuli, less commonly by coughing or micturition ('cough syncope' or 'micturition syncope'). Only rarely are syncopal attacks so frequent as to be significantly disabling ('malignant' vasovagal syndrome). Vasodilatation and inappropriate slowing of the pulse combine to reduce blood pressure and cerebral perfusion. Recovery is rapid if the patient lies down.

### Carotid sinus hypersensitivity

Exaggerated vagal discharge following external stimulation of the carotid sinus (e.g. from shaving or a tight shirt collar) causes reflex vasodilatation and slowing of the pulse. These may combine to reduce blood pressure and cerebral perfusion in some elderly patients, causing loss of consciousness.

### Valvular obstruction

Fixed valvular obstruction in aortic stenosis may prevent a normal rise in cardiac output during exertion, such that the physiological vasodilatation that occurs in exercising muscle produces an abrupt reduction in blood pressure and cerebral perfusion, resulting in syncope. Vasodilator therapy may cause syncope by a similar mechanism. Intermittent obstruction of the mitral valve by left atrial tumours (usually myxoma or thrombus) may also cause syncopal episodes (Fig. 13.1).

### Stokes–Adams attacks

Stokes–Adams attacks are caused by self-limiting episodes of asystole (Fig. 13.2) or rapid tachyarrhythmias (including ventricular fibrillation). The loss of cardiac output causes syncope and striking pallor. Following restoration of normal rhythm, recovery is rapid and associated with flushing of the skin as flow through the dilated cutaneous bed is re-established.

# The cardiac examination

A methodical approach is recommended, starting with inspection of the patient and proceeding to examination of the radial pulse, measurement of heart rate and blood pressure, examination of the neck (carotid pulse, jugular venous pulse), palpation of the anterior chest wall, auscultation of the heart, percussion and auscultation of the lung bases and, finally, examination of the peripheral pulses and auscultation for carotid and femoral arterial bruits (Box 13.11).

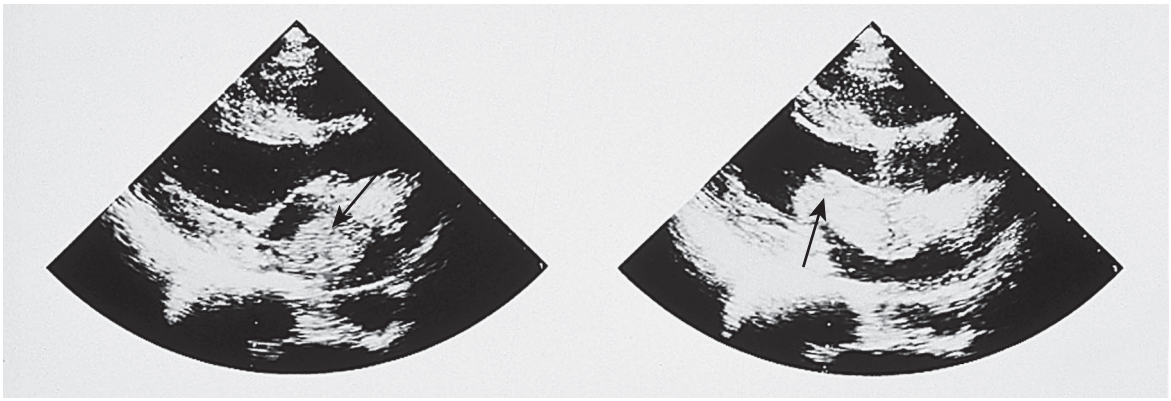


**Figure 13.1** Left atrial myxoma: 2D echocardiogram (long-axis view). During diastole, the tumour (*arrowed*) prolapses through the mitral valve and obstructs left ventricular filling.

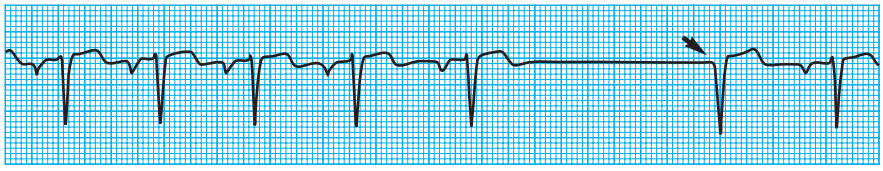


**Figure 13.2** Prolonged sinus arrest. After the fifth sinus beat there is a pause of about 1.8 seconds terminated by a nodal escape beat (*arrowed*) before sinus rhythm resumes.

**Box 13.11** Routine for the cardiovascular system examination

- Wash hands
- Introduce yourself to patient
- Recline patient at 45°
- Observe general appearance: comfortable, breathless, pale?
- Inspect the hands for clubbing, splinter haemorrhages, nicotine staining
- Examine the radial pulse(s) for symmetry, rate, rhythm, character (collapsing?)
- Measure the blood pressure
- Assess the height and waveform of the JVP
- Examine the carotid pulse character (slow rising?) and volume (Corrigan's sign?)
- Inspect the face, eyes and mucous membranes for xanthelasma, corneal arcus and anaemia, and cyanosis, respectively
- Inspect the chest for scars and pulsations
- Assess the position and character of the apex beat
- Palpate the praecordium for heaves and thrills
- Auscultate the heart
- Auscultate the lungs
- Examine the ankles and sacrum for oedema
- Examine the peripheral pulses

## Inspection of the patient

Chest wall deformities, such as pectus excavatum, should be noted, because these may compress the heart and displace the apex, giving a spurious impression of cardiac enlargement. The presence of a median sternotomy scar usually indicates previous coronary artery bypass graft (CABG) and/or cardiac valve surgery. The long saphenous vein is the standard conduit for vein grafts so patients with prior CABG often also have a scar along the medial aspect of one or both legs. A lateral thoracotomy scar may indicate previous mitral valvotomy. Large ventricular or aortic aneurysms may cause visible pulsations. Superior vena caval obstruction is associated with prominent venous collaterals on the chest wall. Prominent venous collaterals around the shoulder occur in axillary or subclavian vein obstruction.

Hypercholesterolaemia may be suggested by the presence of tendon and ocular xanthelasma.

Evidence of device implantation (permanent pacemaker or implantable defibrillator) is seen with a scar usually under the left clavicle. In thin patients a bulge from the generator may also be seen.

### Anaemia

This may exacerbate angina and heart failure. Pallor of the mucous membranes is a useful but sometimes misleading physical sign, and diagnosis requires laboratory measurement of the haemoglobin concentration.

### Cyanosis

Cyanosis is a blue discolouration of the skin and mucous membranes caused by increased concentration of reduced haemoglobin in the superficial blood vessels. Peripheral cyanosis may result when cutaneous vasoconstriction slows the

blood flow and increases oxygen extraction in the skin and the lips. It is physiological during cold exposure. It also occurs in heart failure, when reduced cardiac output produces reflex cutaneous vasoconstriction. In mitral stenosis, cyanosis over the malar area produces the characteristic mitral facies or malar flush.

Central cyanosis may result from the reduced arterial oxygen saturation caused by cardiac or pulmonary disease. It affects not only the skin and the lips but also the mucous membranes of the mouth. Cardiac causes include pulmonary oedema (which prevents adequate oxygenation of the blood) and congenital heart disease. Congenital defects associated with central cyanosis include those in which desaturated venous blood bypasses the lungs by ('reversed') shunting through septal defects or a patent ductus arteriosus (e.g. Eisenmenger's syndrome, Fallot's tetralogy).

The mouth should also be inspected for signs of poor dental hygiene.

### Clubbing of the fingers and toes

In congenital cyanotic heart disease, clubbing is not present at birth but develops during infancy and may become very marked. Infective endocarditis is the other cardiac cause of clubbing.

### Other cutaneous and ocular signs of infective endocarditis

Other cutaneous and ocular signs of infective endocarditis are caused by immune complex deposition in the capillary circulation. A vasculitic rash is common, as are splinter haemorrhages in the nail bed, although these are a very non-specific finding. Other 'classic' manifestations of endocarditis, including Osler's nodes (tender erythematous nodules in the pulps of the fingers), Janeway lesions (painless erythematous lesions on the palms) and Roth's spots (erythematous lesions in the optic fundi), are rarely seen now.

### Coldness of the extremities

In patients hospitalized with severe heart failure, coldness of the extremities is an important sign of reduced cardiac output. It is caused by reflex vasoconstriction of the cutaneous bed.

### Pyrexia

Infective endocarditis is invariably associated with pyrexia, which may be low grade or 'swinging' in nature if paravalvular abscess develops. Pyrexia also occurs for the first 3 days after myocardial infarction.

### Oedema

Subcutaneous oedema that pits on digital pressure is a cardinal feature of congestive heart failure. Pressure should be applied over a bony prominence (tibia, lateral malleoli, sacrum) to provide effective compression. Oedema is caused by salt and water retention by the kidney. Two mechanisms are responsible:

1. Reduced sodium delivery to the nephron. This is caused by reduced glomerular filtration owing to constriction of the preglomerular arterioles in response to sympathetic activation and angiotensin II production.
2. Increased sodium reabsorption from the nephron. This is the more important mechanism. It occurs particularly in the proximal tubule early in heart failure but, as failure worsens, renin-angiotensin activation stimulates aldosterone release, which increases sodium reabsorption in the distal nephron.

Salt and water retention expands plasma volume and increases the capillary hydrostatic pressure. Hydrostatic forces driving fluid out of the capillary exceed the osmotic forces reabsorbing it, so that oedema fluid accumulates in the interstitial space. The effect of gravity on capillary hydrostatic pressure ensures that oedema is most prominent around the ankles in the ambulant patient and over the sacrum in the bedridden patient. In advanced heart failure, oedema may involve the legs, genitalia and trunk. Transudation into the peritoneal cavity (ascites), the pleural and pericardial spaces may also occur.

## Arterial pulse

The arterial pulses should be palpated to evaluate rate, rhythm, character and symmetry.

### Rate and rhythm

By convention, both rate and rhythm are assessed by palpating the right radial pulse. Rate, expressed in beats per minute (bpm), is measured by counting the number of beats in a timed period of 15 seconds and multiplying by four. Normal sinus rhythm is regular, but in young patients may show phasic variation in rate during respiration (sinus arrhythmia). An irregular rhythm usually indicates atrial fibrillation, but it may also be caused by frequent ectopic beats or self-limiting paroxysmal arrhythmias. In patients with atrial fibrillation, the rate should be measured by auscultation at the cardiac apex, because beats that follow very short diastolic intervals may create a 'pulse deficit' by not generating sufficient pressure to be palpable at the radial artery.

### Character

Character is defined by the volume and waveform of the pulse and should be evaluated at the right carotid artery (i.e. the pulse closest to the heart and least subject to damping and distortion in the arterial tree). Pulse volume provides a crude indication of stroke volume, being small in heart failure and large in aortic regurgitation. The waveform of the pulse is of greater diagnostic importance (Fig. 13.3). Severe aortic stenosis produces a slow-rising carotid pulse; the fixed obstruction restricts the rate at which blood can be ejected from the left ventricle. In aortic regurgitation, in diastole, the left ventricle receives not only its normal pulmonary venous return but also a proportion of the blood ejected into the aorta during the previous systole as it flows back

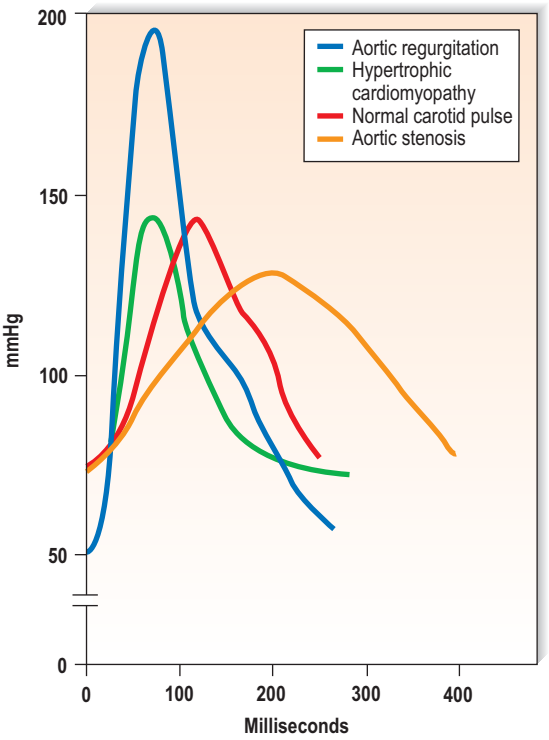



**Figure 13.3** The waveform of the pulse is characterized by the rate of rise of the carotid upstroke. Note that in aortic regurgitation, the upstroke is rapid and followed by abrupt diastolic 'collapse'. In hypertrophic cardiomyopathy, the upstroke is also rapid and the pulse has a jerky character. In aortic stenosis, the upstroke is slow with a plateau.

through an incompetent valve. The resultant large stroke volume, vigorously ejected, produces a rapidly rising carotid pulse, which collapses in early diastole owing to backflow through the aortic valve. This collapsing pulse can be exaggerated at the radial artery by lifting the arm. In mixed aortic valve disease, a biphasic pulse with two systolic peaks is occasionally found. Alternating pulse—alternating high and low systolic peaks—occurs in severe left ventricular failure, but the mechanism for this is unknown.

Paradoxical pulse refers to an inspiratory decline in systolic pressure greater than 10 mmHg (Fig. 13.4). In normal circumstances, inspiration results in an increase in venous return as blood is 'sucked into' the thorax by the decline in intrathoracic pressure. This increases right ventricular stroke volume, but left ventricular stroke volume falls slightly (ventricular interdependence). When the heart is constrained in a 'fixed box' by a pericardial effusion (cardiac tamponade) or by thickened pericardium (pericardial constriction), the increased inspiratory right ventricular blood volume reduces left ventricular compliance, resulting in a more pronounced reduction in left ventricular filling, stroke volume and systolic blood pressure during inspiration. 'Pulsus paradoxus' therefore represents an exaggeration of the normal inspiratory decline in systolic pressure and is not truly paradoxical. Pulsus paradoxus in acute severe asthma is thought to be caused by negative pleural

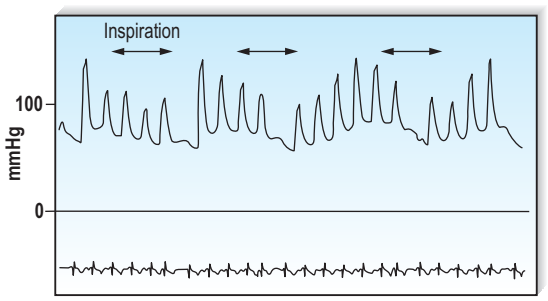



**Figure 13.4** Paradoxical pulse (radial artery pressure signal). The patient had severe tamponade. Note the exaggerated (>10 mmHg) decline in arterial pressure during inspiration.

pressure increasing afterload and thereby impedance to left ventricular emptying.

Blood pressure is measured by inflating a cuff until no sounds are heard. The pressure is then slowly decreased until systolic sounds are first heard during expiration but not during inspiration; note this reading. The pressure is slowly decreased further until sounds are heard throughout the respiratory cycle (inspiration and expiration); note this second reading. If the pressure difference between the two readings is above 10 mmHg, it can be classified as pulsus paradoxus.

### Symmetry

Symmetry of the radial, brachial, carotid, femoral, popliteal and pedal pulses should be confirmed. A reduced or absent pulse indicates an obstruction more proximally in the arterial tree, caused usually by atherosclerosis or thromboembolism, less commonly by aortic dissection. Coarctation of the aorta causes symmetrical reduction and delay of the femoral pulses compared with the radial pulses ('radio-femoral delay'), a sign that should be looked for in younger patients with hypertension. Bruits from collateral vessels may also be heard over the back of such patients.

## Measurement of blood pressure

Blood pressure is measured indirectly, traditionally by sphygmomanometry, but automated blood pressure monitors are being used increasingly in clinical practice. The principle of manual blood pressure measurement is that turbulent flow through a partially compressed artery (typically the brachial) creates noises that can be auscultated with a stethoscope and the points at which these noises (called Korotkoff sounds) change in intensity correlate with systemic arterial pressures. Accurate blood-pressure measurement requires careful technique; patients should be sitting or lying at ease because significant changes in arterial pressure occur with exertion, anxiety and changes in posture. The manometer should be at the same level of the cuff on the patient's arm and the observer's eye. For most adult patients, a standard cuff (12 cm width) is appropriate, but obese subjects require use of a wider (thigh) cuff of 15 cm or the blood pressure

will be overestimated. For children, various sized cuffs are available; select the one that covers most of the upper arm leaving a gap of 1 cm or so below the axilla and above the antecubital fossa.

Palpate the radial pulse as the cuff is inflated to a pressure of 20 mmHg above the level at which radial pulsation can no longer be felt. Place the stethoscope lightly over the brachial artery and reduce the pressure in the cuff at a rate of 2–3 mmHg/second until the first sounds are heard. This, the first Korotkoff sound, correlates with systolic blood pressure as flow is just possible through the pressure applied by the compressive cuff. As the pressure is lowered further, subtle changes in pitch and volume occur; these are the second and third Korotkoff sounds and are not important clinically. With further lowering of the pressure in the cuff, the artery becomes less compressed, flow becomes less turbulent and the sounds over the brachial artery become muffled. This is the fourth Korotkoff sound. Shortly after this (usually 1–10 mmHg lower), the sounds die away completely as flow is unimpeded by the cuff; this, the fifth Korotkoff sound, correlates most accurately with diastolic blood pressure. Its identification is also less subjective than the fourth, but in some conditions (aortic regurgitation, arteriovenous fistula, pregnancy), the Korotkoff sounds remain audible despite complete deflation of the cuff. In such situations, phase 4 must be used for the diastolic measurement. Both systolic and diastolic values are recorded; the difference between these two values is called the pulse pressure. Certain conditions of the aortic valve may cause important abnormalities of pulse pressure. Supine and erect blood pressure measurements provide an assessment of baroreceptor function, a postural drop being defined by a fall in systolic blood pressure on standing. It is essential to work swiftly as well as accurately, as compression of a limb, by itself, will cause a rise in blood pressure. If several successive measurements are made, the air pressure in the cuff should be allowed to fall to zero between readings.

## Jugular venous pulse

Fluctuations in right atrial pressure during the cardiac cycle generate a pulse that is transmitted backwards into the jugular veins. It is best examined in good light while the patient reclines at 45°. If the right atrial pressure is very low, however, visualization of the jugular venous pulse may require a smaller reclining angle. Alternatively, manual pressure over the upper right side of the abdomen may be used to produce a transient increase in venous return to the heart, which elevates the jugular venous pulse (hepatojugular reflux).

## Jugular venous pressure

The jugular venous pressure (JVP) should be assessed from the waveform of the internal jugular vein, which lies adjacent to the medial border of the sternocleidomastoid muscle. Distention of the external jugular vein is a useful clue to an elevated JVP but, strictly speaking, it should not be used because it can be compressed as it passes under the clavicle. The JVP is measured in centimetres vertically from the sternal angle to the top of the venous waveform. The normal upper limit is 4 cm. This is about 9 cm above the right atrium and corresponds to a pressure of 6 mmHg. Elevation of the JVP indicates a raised right atrial pressure unless the superior vena cava is obstructed, producing engorgement of the neck veins (Box 13.12). During inspiration, the pressure within the chest decreases and there is a fall in the JVP. In constrictive pericarditis, and less commonly in tamponade, inspiration produces a paradoxical rise in the JVP (Kussmaul's sign) because the increased venous return that occurs during inspiration cannot be accommodated within the constrained right side of the heart (Fig. 13.5).

## Waveform of jugular venous pulses

In sinus rhythm, the jugular venous pulse has a double waveform attributable to the 'a' and 'v'

**Box 13.12** Causes of elevated jugular venous pressure

- Congestive heart failure
- Cor pulmonale
- Pulmonary embolism
- Right ventricular infarction
- Tricuspid valve disease
- Tamponade
- Constrictive pericarditis
- Hypertrophic/restrictive cardiomyopathy
- Superior vena cava obstruction
- Iatrogenic fluid overload, particularly in surgical and renal patients

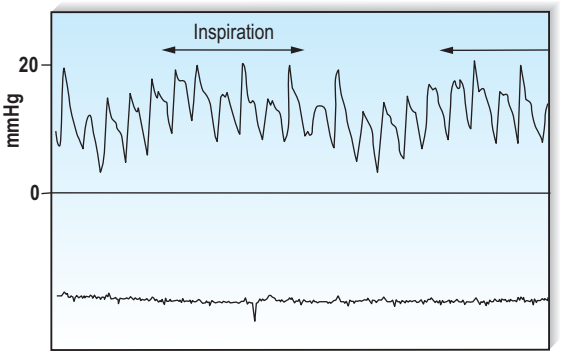



**Figure 13.5** Kussmaul's sign. Jugular venous pressure recording in a patient with tamponade. The venous pressure is raised and there is a particularly prominent systolic 'x' descent, giving the waveform of the JVP an unusually dynamic appearance. Note the inspiratory rise in atrial pressure (Kussmaul's sign) reflecting the inability of the tamponaded right heart to accommodate the inspiratory increase in venous return.

waves separated by the 'x' and 'y' descents. The 'a' wave is produced by atrial systole. It is followed by the 'x' descent (marking descent of the tricuspid valve ring), which is interrupted by the diminutive 'c' wave caused by tricuspid valve closure. Atrial pressure then rises again, producing the 'v' wave as the atrium fills passively during ventricular systole. The decline in atrial pressure as the tricuspid valve opens to allow ventricular filling produces the 'y' descent. Important abnormalities of the pattern of deflections are shown in Figure 13.6.

In atrial fibrillation, there is no atrial contraction. Consequently, there is no 'a' wave and the jugular venous pulse loses its double waveform. It is not always easy to differentiate venous from arterial pulsations in the neck, but several features help to distinguish the jugular venous pulse from the carotid arterial pulse (Box 13.13).

## Palpation of the chest wall

The apex beat is defined as the lowest and most lateral point at which the cardiac impulse can be palpated. Inferior or lateral displacement from its normal location in the fifth intercostal space in the mid-clavicular line usually indicates cardiac enlargement. Chronic volume loading of the left ventricle (mitral regurgitation, aortic regurgitation) causes left ventricular dilatation which can be appreciated clinically. In contrast, isolated pressure loading of the left ventricle (hypertension, aortic stenosis) causes left ventricular hypertrophy which does not

**Box 13.13** Characteristics of the jugular venous pulse

- Double waveform (in sinus rhythm)
- Varies with respiration
- Varies with posture
- Impalpable
- Obliterated by pressure at base of waveform
- Transient increase in volume and height with hepatojugular reflux

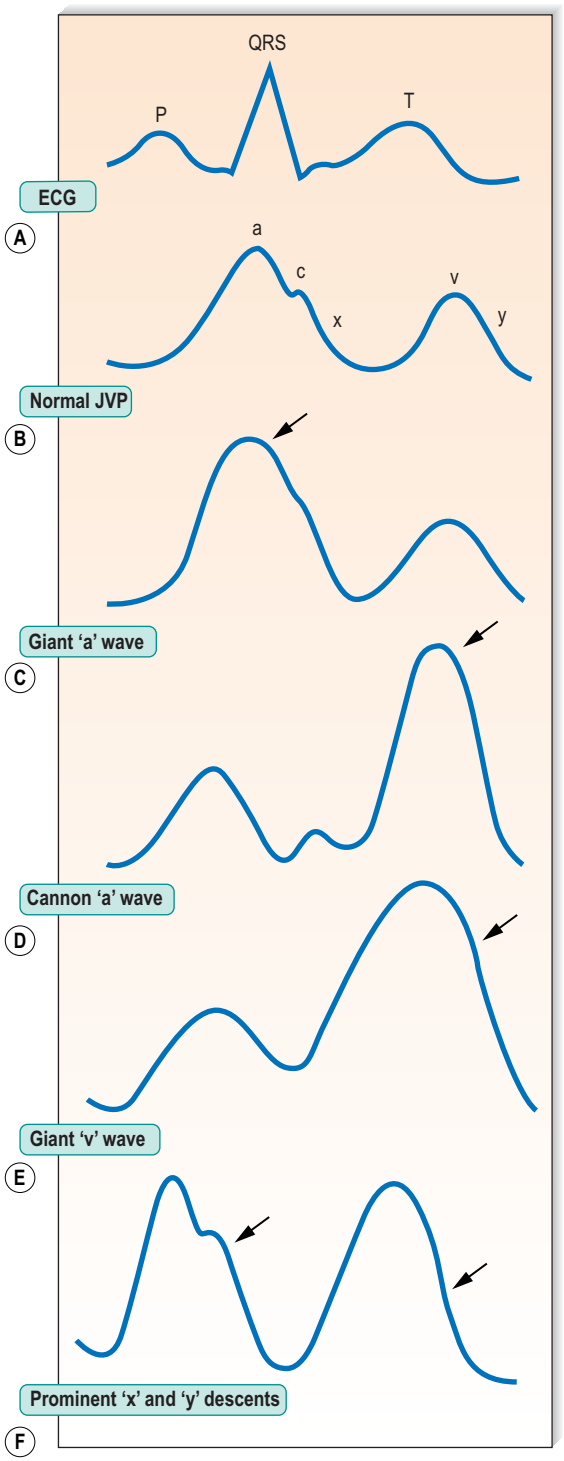



**Figure 13.6** Waveform of the jugular venous pulse. **(A)** The ECG is portrayed at the top of the illustration. Note how electrical events precede mechanical events in the cardiac cycle. Thus, the P wave (atrial depolarization) and the QRS complex (ventricular depolarization) precede the 'a' and 'v' waves, respectively, of the JVP. **(B)** Normal JVP. The 'a' wave produced by atrial systole is the most prominent deflection. It is followed by the 'x' descent, interrupted by the small 'c' wave marking tricuspid valve closure. Atrial pressure then rises again ('v' wave) as the atrium fills passively during ventricular systole. The decline in atrial pressure as the tricuspid valve opens produces the 'y' descent. **(C)** Giant 'a' wave. Forceful atrial contraction against a stenosed tricuspid valve or a non-compliant hypertrophied right ventricle produces an unusually prominent 'a' wave. **(D)** Cannon 'a' wave. This is caused by atrial systole against a closed tricuspid valve. It occurs when atrial and ventricular rhythms are dissociated (complete heart block, ventricular tachycardia) and marks coincident atrial and ventricular systole. **(E)** Giant 'v' wave. This is an important sign of tricuspid regurgitation. The regurgitant jet produces pulsatile systolic waves in the JVP. **(F)** Prominent 'x' and 'y' descents. These occur in constrictive pericarditis and give the JVP an unusually dynamic appearance. In tamponade, only the 'x' descent is usually exaggerated.

cause displacement of the apex beat. Palpable third and fourth heart sounds give the apical impulse a double thrust. In the past, considerable importance was attached to the character of the apical impulse ('thrusting' in aortic valve disease, 'tapping' in mitral stenosis), but this is of very limited practical value in the modern era.

Left ventricular aneurysms can sometimes be palpated medial to the cardiac apex. Right ventricular enlargement produces a systolic thrust (heave) in the left parasternal area. The turbulent flow responsible for murmurs may produce palpable vibrations (thrills) on the chest wall, particularly in aortic stenosis, ventricular septal defect and patent ductus arteriosus.

## Auscultation of the heart

The diaphragm and bell of the stethoscope permit appreciation of high- and low-pitched auscultatory events, respectively. The apex, lower left sternal edge, upper left sternal edge and upper right sternal edge should be auscultated in turn. These locations correspond respectively to the mitral, tricuspid, pulmonary and aortic areas, and loosely identify sites at which sounds and murmurs arising from the four valves are best heard (Box 13.14).

### First sound (S1)

The first sound corresponds to mitral and tricuspid valve closure at the onset of systole. It is accentuated in mitral stenosis because prolonged diastolic filling through the narrowed valve ensures that the thickened leaflets are widely separated at the onset of systole. Thus, valve closure generates unusually vigorous vibrations. In advanced mitral stenosis, the valve is rigid and immobile and S1 becomes soft again.

### Second sound (S2)

The second sound corresponds to aortic and pulmonary valve closure following ventricular ejection. S2 is single during expiration. Inspiration, however, causes physiological splitting into aortic followed by pulmonary components because increased venous return to the right side of the heart delays pulmonary valve closure. Important abnormalities of S2 are illustrated in Figure 13.7.

### Third and fourth sounds (S3, S4)

The third and fourth sounds are low-frequency sounds that occur early and late in diastole, respectively. When present, they give a characteristic 'gallop' to the cardiac rhythm. Both sounds are best heard with the bell of the stethoscope at the cardiac apex. They are caused by abrupt tensing of the ventricular walls following rapid diastolic filling. Rapid filling occurs early in diastole (S3) following atrioventricular valve opening, and again late in diastole (S4) owing to atrial contraction. S3 is physiological in children and young adults, but usually disappears after the age of 40. It also occurs in high-output states caused by anaemia, fever, pregnancy and thyrotoxicosis. After the age of 40, S3 is nearly always pathological, usually

**Box 13.14 Routine for auscultation of the heart**

- Auscultate at apex with diaphragm
- Reposition patient on left side: 'Please turn onto your left side'
- Listen with diaphragm (mitral regurgitation) and then bell (mitral stenosis)
- Return patient to original position, reclining at 45°
- Auscultate with diaphragm at lower left sternal edge (tricuspid regurgitation, tricuspid stenosis, ventricular septal defect)
- Auscultate with diaphragm at upper left sternal edge (pulmonary stenosis, pulmonary regurgitation, patent ductus arteriosus)
- Auscultate with diaphragm at upper right sternal edge (aortic stenosis, hypertrophic cardiomyopathy)
- Sit patient forward. Auscultate with diaphragm at lower left sternal edge in held expiration (aortic regurgitation): 'breathe in … breathe out … stop'
- Auscultate over the carotid arteries (radiation of murmur of aortic stenosis, carotid artery bruits)

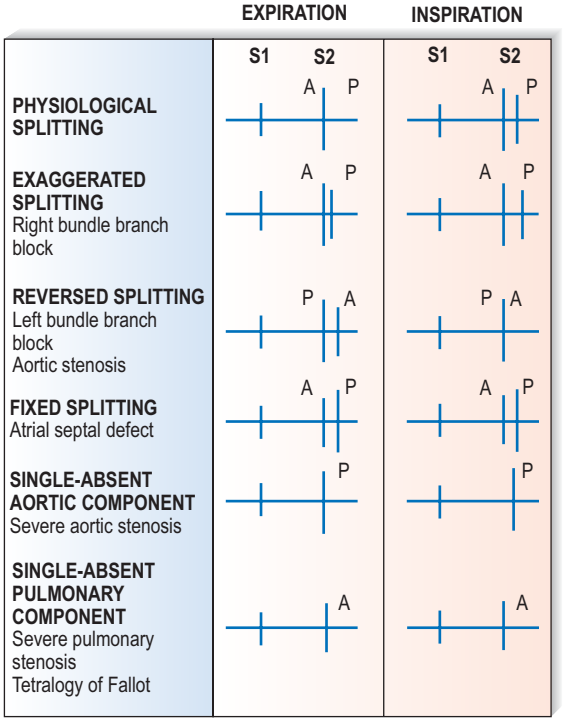



**Figure 13.7** Splitting of the second heart sound. The first sound, representing mitral and tricuspid closure, is usually single, but the aortic and pulmonary components of the second sound normally split during inspiration as increased venous return delays right ventricular emptying. Abnormal splitting of the second heart sound is an important sign of heart disease.

indicating left ventricular failure or, less commonly, mitral regurgitation or constrictive pericarditis. In the elderly, S4 is sometimes physiological. More commonly, however, it is pathological, and occurs when vigorous atrial contraction late in diastole is required to augment filling of a hypertrophied, non-compliant ventricle (e.g. hypertension, aortic stenosis, hypertrophic cardiomyopathy).

## Systolic clicks and opening snaps

Valve opening, unlike valve closure, is normally silent. In aortic stenosis, however, valve opening produces a click in early systole that precedes the ejection murmur. The click is only audible if the valve cusps are pliant and non-calcified, and is particularly prominent in the congenitally bicuspid valve. A click later in systole suggests mitral valve prolapse, particularly when followed by a murmur. In mitral stenosis, elevated left atrial pressure causes forceful opening of the thickened valve leaflets. This generates a snap early in diastole that precedes the mid-diastolic murmur.

## Heart murmurs

Heart murmurs are caused by turbulent flow within the heart and great vessels (Fig. 13.8). Occasionally the turbulence is caused by increased flow through a normal valve, usually aortic or pulmonary, producing an 'innocent' murmur. However, murmurs may also indicate valve disease or abnormal communications between the left and right sides of the heart (e.g. septal defects).

Rheumatic heart disease has become much less common in developed countries, although it remains common elsewhere, and is the cause of many of the classic heart murmurs (Box 13.15). Degenerative valve disease (calcific aortic stenosis, mitral regurgitation owing to chordal rupture) is increasingly common. Heart murmurs are defined by four characteristics: loudness, quality, location and timing.

The loudness of a murmur reflects the degree of turbulence. This relates to the volume and velocity of flow and not the severity of the cardiac lesion. Loudness is graded on a scale of 1 (barely audible) to 6 (audible even without application of the stethoscope to the chest wall). The quality of a murmur relates to its frequency and is best described as low-, medium- or high-pitched. The location of a murmur on the chest wall depends on its site of origin and has led to the description of four valve areas (see above). Some murmurs radiate, depending on the velocity and direction of the blood flow. The sound of the high-velocity systolic flow in aortic stenosis and mitral regurgitation, for example, is directed towards the neck and the axilla, respectively; that of the high-velocity diastolic flow in aortic regurgitation is directed towards the left sternal edge. Murmurs are timed according to the phase of systole or diastole during which they are audible. It is inadequate to describe the timing of a murmur as systolic or diastolic without more specific reference to the length of the murmur and the phase of systole or diastole during which it is heard: systolic

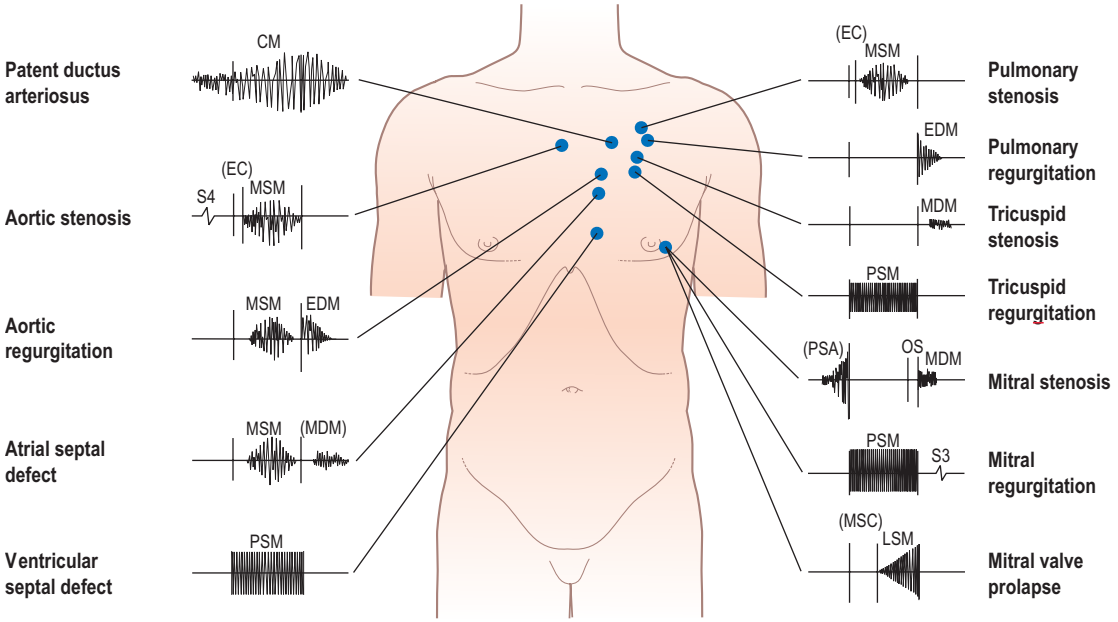



**Figure 13.8** Heart murmurs. These are caused by turbulent flow within the heart and great vessels, and may indicate valve disease. Heart murmurs may be depicted graphically as shown in this illustration. CM, continuous murmur; EC, ejection click; EDM, early diastolic murmur; LSM, late systolic murmur; MDM, mid-diastolic murmur; MSC, mid-systolic click; MSM, mid-systolic murmur; OS, opening snap; PSA, presystolic accentuation of murmur; PSM, pansystolic murmur; S3, third heart sound; S4, fourth heart sound. Parentheses indicate those auscultatory findings that are not constant.

**Box 13.15 Rheumatic heart disease**

**Typical patient**

- Middle-aged woman, less commonly man, with a history of childhood rheumatic fever (often not recognized or dismissed as trivial feverish illness). The mitral valve is almost invariably affected, commonly with associated aortic valve involvement. Right-sided valves less commonly affected. Presentation is usually with exertional dyspnoea, less commonly with unexplained atrial fibrillation or unheralded stroke.
- Pregnant woman presenting abruptly with atrial fibrillation and pulmonary oedema

**Major symptoms**

- Mitral valve disease: dyspnoea or symptoms of frank congestive failure
- Aortic valve disease: dyspnoea, sometimes angina or syncope

**Major signs**

- Mitral stenosis: atrial fibrillation, signs of fluid retention (raised JVP ± peripheral oedema and basal crackles in lung fields), loud S1, opening snap in early diastole followed by low-pitched mid-diastolic murmur best heard at cardiac apex. With increasing calcification, the valve gets more rigid and the loud S1 and opening snap disappear.
- Mitral regurgitation: atrial fibrillation, signs of fluid retention, pansystolic murmur best heard at cardiac apex, often with third heart sound
- Aortic stenosis: slow-rising carotid pulse, ejection systolic murmur best heard at base of heart
- Aortic regurgitation: fast-rising carotid pulse, ejection systolic murmur with early diastolic murmur best heard at left sternal edge

**Diagnosis**

- Echocardiogram: diagnostic of rheumatic heart disease, Doppler studies providing additional information about the severity of valvular stenosis or regurgitation

**Treatment**

- Diuretics for fluid retention and vasodilators to increase forward flow through regurgitant left-sided valves. Digoxin or β-blockers for rate control in atrial fibrillation plus warfarin to protect against embolism. Symptomatic mitral valve disease that fails to respond to treatment requires valve surgery (repair or replacement) or, in selected cases of mitral stenosis, percutaneous valvuloplasty. Symptomatic aortic valve disease always requires consideration for valve replacement.

murmurs are either mid-systolic, pansystolic or late systolic; diastolic murmurs are either early diastolic, mid-diastolic or presystolic in timing. Continuous murmurs are audible in both phases of the cardiac cycle.

A mid-systolic ('ejection') murmur is caused by turbulence in the left or right ventricular outflow tract during ejection. It starts following opening of the aortic or pulmonary valve, reaches a crescendo in mid-systole and disappears before the second heart sound. The murmur is loudest in the aortic area (with radiation to the neck) when it arises from the left ventricular outflow tract, and in the pulmonary area when it arises from the right ventricular outflow tract. It is best heard with the diaphragm of the stethoscope while the patient sits forward. Important causes of aortic ejection murmurs are aortic stenosis and hypertrophic cardiomyopathy.

Aortic regurgitation also produces an ejection murmur owing to increased stroke volume and velocity of ejection. Pulmonary ejection murmurs may be caused by pulmonary stenosis or infundibular stenosis (as in Fallot's tetralogy).

In atrial septal defect, the pulmonary ejection murmur results from right ventricular volume loading and consequent increased blood flow through the pulmonary valve and does not indicate organic valvular disease. 'Innocent' murmurs unrelated to heart disease are always mid-systolic in timing and are caused by turbulent flow in the left (sometimes right) ventricular outflow tract. In most cases, there is no clear cause, but they may reflect a hyperkinetic circulation in conditions such as anaemia, pregnancy, thyrotoxicosis or fever. They are rarely louder than grade 3, often vary with posture, may disappear on exertion and are not associated with other signs of heart disease.

Pansystolic murmurs are audible throughout systole from the first to the second heart sounds. They are caused by regurgitation through incompetent atrioventricular valves and by ventricular septal defects. The pansystolic murmur of mitral regurgitation, which is loudest at the cardiac apex, radiates into the left axilla. It is best heard using the diaphragm of the stethoscope with the patient lying on the left side. The murmurs of tricuspid regurgitation and ventricular septal defect are loudest at the lower left sternal edge. Inspiration accentuates the murmur of tricuspid regurgitation because the increased venous return to the right side of the heart increases the regurgitant volume. Mitral valve prolapse may also produce a pansystolic murmur but, more commonly, prolapse occurs in mid-systole, producing a click followed by a late-systolic murmur.

Early diastolic murmurs, which are high-pitched, start immediately after the second heart sound, fading away in mid-diastole. They are caused by regurgitation through incompetent aortic and pulmonary valves and are best heard using the diaphragm of the stethoscope while the patient leans forward. The early diastolic murmur of aortic regurgitation radiates from the aortic area to the left sternal edge, where it is usually easier to hear, in maintained expiration with the patient leaning forward. Pulmonary regurgitation is loudest at the pulmonary area. Mid-diastolic murmurs are caused by turbulent flow through the atrioventricular valves.

They start after valve opening, relatively late after the second sound, and continue for a variable period during mid-diastole. Mitral stenosis is the principal cause of a mid-diastolic murmur which is best heard at the cardiac apex using the bell of the stethoscope while the patient lies on the left side. Increased flow across a non-stenotic mitral valve occurs in ventricular septal defect and mitral regurgitation and may produce a mid-diastolic murmur. In severe aortic regurgitation, pre-closure of the anterior leaflet of the mitral valve by the regurgitant jet may produce mitral turbulence associated with a mid-diastolic murmur (Austin Flint murmur). A mid-diastolic murmur at the lower left sternal edge, accentuated by inspiration, is caused by tricuspid stenosis and also by conditions that increase tricuspid flow (e.g. atrial septal defect, tricuspid regurgitation).

In mitral or tricuspid stenosis, atrial systole produces a presystolic murmur immediately before the first heart sound. The murmur is perceived as an accentuation of the mid-diastolic murmur associated with these conditions. Because presystolic murmurs are generated by atrial systole, they do not occur in patients with atrial fibrillation.

Continuous murmurs are heard during systole and diastole; they are uninterrupted by valve closure. The most common cardiac cause is patent ductus arteriosus, in which flow from the high-pressure aorta to the low-pressure pulmonary artery continues throughout the cardiac cycle, producing a murmur over the base of the heart which, though continuously audible, is loudest at end systole and diminishes during diastole. Ruptured sinus of Valsalva aneurysm also produces a continuous murmur.

### Friction rubs and venous hums

A friction rub occurs in pericarditis. It is best heard in maintained expiration with the patient leaning forward as a high-pitched scratching noise audible during any part of the cardiac cycle and over any part of the left precordium. A continuous venous hum at the base of the heart reflects hyperkinetic jugular venous flow. It is particularly common in infants and usually disappears on lying flat.

### Finishing the cardiovascular examination

The assessment of the cardiovascular system should be concluded with the examination of the abdomen for organomegaly (hepatomegaly in heart failure, splenomegaly in infective endocarditis and renal abnormalities in hypertension) and abdominal aortic aneurysm, auscultation of the chest bases for crackles related to impaired left ventricle (LV) function and urinalysis for proteinuria and haematuria.

## The electrocardiogram

The electrocardiogram records the electrical activity of the heart at the skin surface. A good-quality 12-lead ECG is essential for the evaluation of almost all cardiac patients.

## Electrophysiology

### Generation of electrical activity

The stimulus for every normal ventricular contraction (sinus beat) begins with depolarization of an area of specialized conducting tissue high in the right atrium called the sinoatrial (SA) node. The depolarization spreads through the walls of the atria causing contraction of the atrial muscle before reaching another area of specialized conducting tissue in the lower part of the right atrium called the atrioventricular (AV) node. Conduction through the AV node is relatively slow, which allows atrial contraction to be completed and the ventricles to fill before depolarization travels down the bundle of His and then into the left and right bundle branches. The left bundle branch divides further into the left anterior fascicle and the left posterior fascicle. From here, the depolarization spreads through the Purkinje fibres in the ventricular muscle, which stimulates ventricular contraction. Once ventricular contraction has occurred, the muscle cells repolarize and the ventricles relax to allow ventricular filling to occur.

The wave of depolarization that spreads through the heart during each cardiac cycle has vector properties defined by its direction and magnitude. The net direction of the wave changes continuously during each cardiac cycle and the ECG deflections change accordingly, being positive as the wave approaches the recording electrode and negative as it moves away. Electrodes orientated along the axis of the wave record larger deflections than those orientated at right angles. Nevertheless, the size of the deflections is determined principally by the magnitude of the wave, which is a function of muscle mass. Thus, the ECG deflection produced by depolarization of the atria (P wave) is smaller than that produced by the depolarization of the more muscular ventricles (QRS complex). Ventricular repolarization produces the T wave.

### Inscription of the QRS complex

The ventricular depolarization vector can be resolved into two components:

1. Septal depolarization: spreads from left to right across the septum.
2. Ventricular free wall depolarization: spreads from endocardium to epicardium.

Left ventricular depolarization dominates the second vector component, the resultant direction of which is from right to left. Thus, electrodes orientated to the left ventricle record a small negative deflection (Q wave) as the septal depolarization vector moves away, followed by a large positive deflection (R wave) as the ventricular depolarization vector approaches. The sequence of deflections for

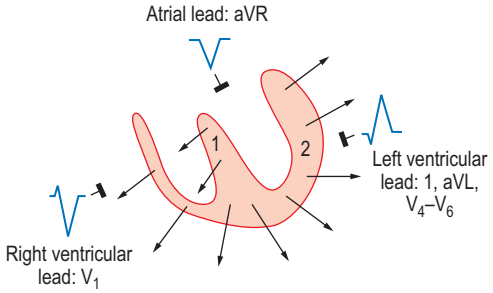



**Figure 13.9** Inscription of the QRS complex. The septal depolarization vector **(1)** produces the initial deflection of the QRS complex. The ventricular free-wall depolarization vector **(2)** produces the second deflection, which is usually more pronounced. Lead aVR is orientated towards the cavity of the left ventricle and records an entirely negative deflection.

electrodes orientated towards the right ventricle is in the opposite direction (Fig. 13.9).

Any positive deflection is termed an R wave. A negative deflection before the R wave is termed a Q wave (this must be the first deflection of the complex), whereas a negative deflection following the R wave is termed an S wave.

### Electrical axis

Because the mean direction of the ventricular depolarization vector (the electrical axis) shows a wide range of normality, there is a corresponding variation in QRS patterns consistent with a normal ECG. Thus, correct interpretation of the ECG must take account of the electrical axis. The frontal plane axis is determined by identifying the limb lead in which the net QRS deflection (positive and negative) is least pronounced. This lead must be at right angles to the frontal plane electrical axis, which is defined using an arbitrary hexaxial reference system (Fig. 13.10).

## Normal 12-lead ECG

A normal 12-lead ECG is illustrated in Figure 13.11. Leads I–III are the standard bipolar leads, which each measures the potential difference between two limbs:

- Lead I: left arm to right arm
- Lead II: left leg to right arm
- Lead III: left leg to left arm

The remaining leads are unipolar, connected to a limb (aVR to aVF) or to the chest wall ($V_1$–$V_6$). Because the orientation of each lead to the wave of depolarization is different, the direction and magnitude of ECG deflections is also different in each lead. Nevertheless, the sequence of deflections (P wave, QRS complex, T wave) is identical. In some patients, a small U wave can be seen following the T wave. Its orientation (positive or negative) is the same as the T wave, but its cause is unknown.

## Analysis of the ECG

### Heart rate

The ECG is usually recorded at a paper speed of 25 mm/s. Thus, each large square (5 mm) represents 0.20 s. The heart rate (bpm) is conveniently calculated by counting the number of large squares between consecutive R waves and dividing this into 300.

### Rhythm

In normal sinus rhythm, P waves precede each QRS complex and the rhythm is regular. Absence of P waves and an irregular rhythm indicate atrial fibrillation.

### Electrical axis

Evaluation of the frontal place QRS axis is described above.

### P-wave morphology

The duration should not exceed 0.10 s, prolongation indicating left atrial enlargement, often the result of mitral valve disease or left ventricular failure. Tall-peaked 'pulmonary' P waves indicate right atrial enlargement, caused usually by pulmonary hypertension and right ventricular failure.

### PR interval

The normal duration is 0.12–0.20 s measured from the onset of the P wave to the first deflection of the QRS complex. Prolongation indicates delayed atrioventricular conduction (first-degree heart block). Shortening indicates rapid conduction through an accessory pathway bypassing the atrioventricular node (Wolff-Parkinson-White (WPW) syndrome).

### QRS morphology

The QRS duration should not exceed 0.12 s. Prolongation indicates slow ventricular depolarization caused by bundle branch block (Fig. 13.12), pre-excitation (WPW syndrome), ventricular tachycardia or hypokalaemia.

Exaggerated QRS deflections indicate ventricular hypertrophy (Fig. 13.13). The voltage criteria for left ventricular hypertrophy are fulfilled when the sum of the S and R wave deflections in leads $V_1$ and $V_6$ exceeds 35 mm (3.5 mV). Right ventricular hypertrophy causes tall R waves in the right ventricular leads ($V_1$ and $V_2$). A dominant R wave in lead $V_1$ can also be caused by right bundle branch block, posterior myocardial infarction, WPW syndrome and dextrocardia. Diminished QRS deflections occur in myxoedema and also when pericardial effusion or obesity electrically insulates the heart. The presence of pathological Q waves (duration >0.04 s) usually indicates previous myocardial infarction.

### QT interval

The QT interval is measured from the onset of the QRS complex to the end of the T wave and represents

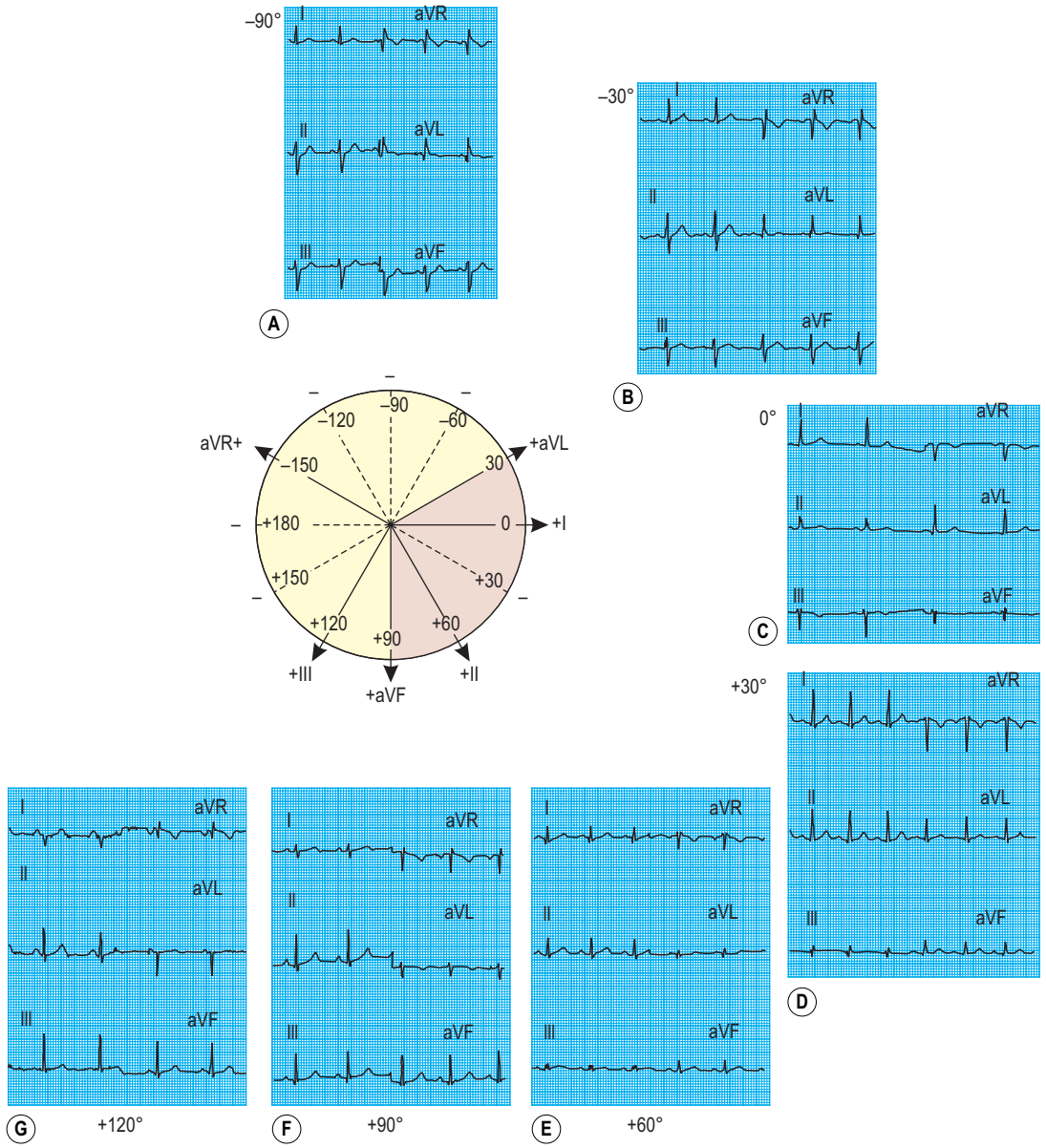



**Figure 13.10** Mean frontal QRS axis. This is the mean direction of the left ventricular depolarization vector in those leads (I–aVF) that lie in the frontal plane of the heart. It lies at right angles to the lead in which the net QRS deflexion is least pronounced. It is quantified using a hexaxial reference system. The QRS axis shows a wide range of normality from –30° to 90°. Thus, despite the different ECG patterns in this illustration, only recordings **(A)** and **(G)** are abnormal, owing to left and right axis deviation, respectively.

the duration of electrical systole (mechanical systole starts between the QRS complex and the T wave). The QT interval (0.35–0.45 s) is very rate sensitive, shortening as heart rate increases. A commonly used way to correct the QT interval (the so-called QTc) involves dividing the measured QT interval by the square root of the RR interval in seconds. As an example, at 60 bpm, the RR interval would be 1 second: the QTc would be equal to QT $\sqrt{1}$. Abnormal prolongation of the QT interval predisposes to ventricular arrhythmias and may be congenital or occur in response to hypokalaemia, rheumatic fever or drugs (e.g. quinidine, amiodarone, tricyclic antidepressants). Shortening of the QT interval is caused by hyperkalaemia and digoxin therapy.

## ST segment morphology

Minor ST elevation reflecting early repolarization may occur as a normal variant (Fig. 13.14), particularly in subjects of African or West Indian origin. Pathological elevation (>2.0 mm above the isoelectric line) occurs in acute myocardial infarction, variant angina and pericarditis. Horizontal ST depression indicates myocardial ischaemia. Other important causes of ST depression are digoxin therapy and hypokalaemia.

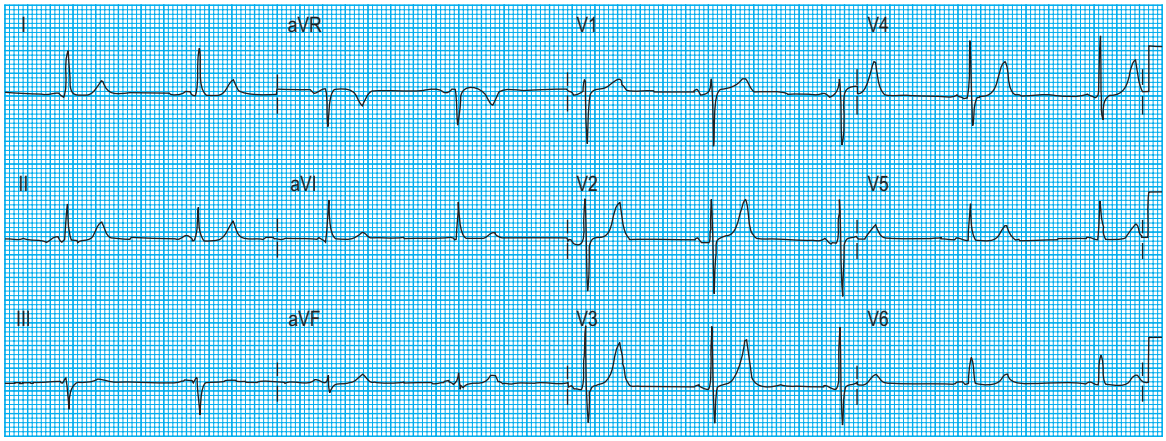



**Figure 13.11** Standard 12-lead ECG. This is a normal recording. The QRS deflections are equiphasic in lead aVF. This is at right angles to lead I (see Fig. 13.10), which is dominantly positive. The frontal plane QRS axis is, therefore, 0°. The square wave calibration signal is 1 mV.

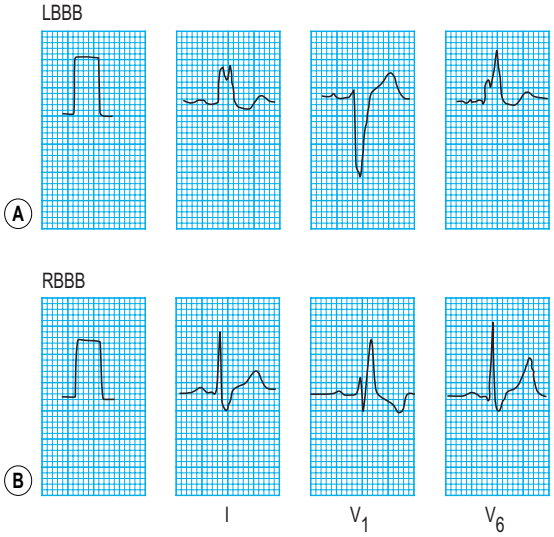



**Figure 13.12** Bundle branch block. **(A)** Left bundle branch block (LBBB); the entire sequence of ventricular depolarization is abnormal, resulting in a broad QRS complex with large slurred or notched R waves in I and $V_6$. **(B)** Right bundle branch block (RBBB); right ventricular depolarization is delayed, resulting in a broad QRS complex with an 'rSR' pattern in $V_1$ and prominent S waves in I and $V_6$.

### T-wave morphology

The orientation of the T wave should be directionally similar to the QRS complex. Thus T-wave inversion is normal in leads with dominantly negative QRS complexes (aVR, $V_1$ and sometimes lead III). Pathological T-wave inversion occurs as a non-specific response to various stimuli (e.g. viral infection, hypothermia). More important causes of T-wave inversion are ventricular hypertrophy, myocardial ischaemia and myocardial infarction. Exaggerated peaking of the T wave is the earliest ECG change in ST elevation myocardial infarction. It also occurs in hyperkalaemia.

## Clinical applications of ECG

### Diagnosis of coronary artery disease

The territories supplied by the three major coronary arteries, although variable, are highly circumscribed, the left anterior descending artery supplying the anterior wall, the circumflex artery the lateral wall and the right coronary artery the inferior wall of the left ventricle. The regional distribution of coronary flow has important implications for electrocardiography (and diagnostic imaging), patients with coronary artery disease showing regional electrocardiographic (or wall motion) abnormalities and patients with diffuse myocardial disease (e.g. cardiomyopathy) showing more widespread changes.

#### Stable angina

The ECG is often normal in patients with stable angina unless there is a history of myocardial infarction, when pathological Q waves or T-wave inversion may be present.

#### Exercise stress testing

The patient is usually exercised on a treadmill, the speed and slope of which can be adjusted to increase the workload gradually. In patients with coronary artery disease, exercise-induced increases in myocardial oxygen demand may outstrip oxygen delivery through the atheromatous arteries, resulting in regional ischaemia. This causes planar or downsloping ST segment depression, with reversal during recovery (Fig. 13.15). The ready availability of the exercise test means that it is one of the most widely used tests for evaluating the patient with chest pain, but its diagnostic accuracy is limited to about 70%, false-positive or false-negative results being common when the pre-test probability of coronary disease is very low (as in young women) or very high (as in elderly patients with typical symptoms), respectively. Non-invasive imaging modalities (see

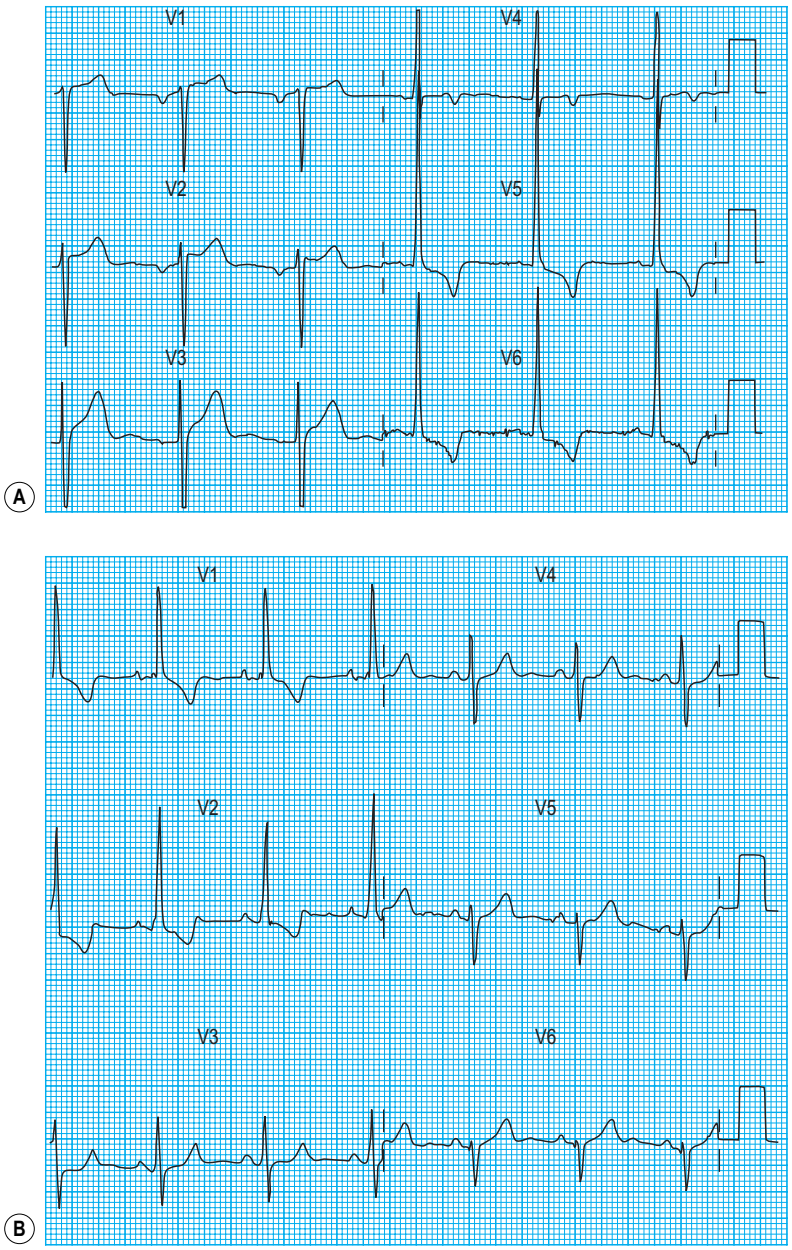



**Figure 13.13** Ventricular hypertrophy. **(A)** Left ventricular hypertrophy. The QRS voltage deflections are exaggerated such that the sum of S and R waves in $V_1$ and $V_6$, respectively, exceeds 35 mm. T-wave inversion in $V_5$ and $V_6$ indicates left ventricular 'strain'. **(B)** Right ventricular hypertrophy. Prominent R waves in $V_1$ and $V_2$ associated with T-wave inversion are shown.

later: magnetic resonance perfusion imaging, stress echocardiography, computed tomography (CT) coronary angiography) have higher rates of diagnostic accuracy and are now preferred in the assessment of patients with suspected coronary artery disease, where available. The exercise ECG also provides prognostic information in patients who are known to have coronary artery disease: an increased risk of myocardial infarction or sudden death is indicated by ST depression very early during exercise, by an exertional fall in blood pressure or by exercise-induced ventricular arrhythmias. In these cases, urgent coronary angiography should be considered.

## Acute coronary syndromes

Acute myocardial infarction and unstable angina present similarly with unprovoked—often severe—ischaemic cardiac pain. Reliable differentiation between the acute coronary syndromes on clinical grounds cannot be made and requires measurement of cardiac biomarkers, in particular troponin I or T (see below), and observation of the 12-lead ECG. The combination of typical symptoms plus raised troponins is diagnostic of myocardial infarction, which is categorized as ST elevation myocardial infarction (STEMI) or non-ST elevation myocardial infarction (non-STEMI) by the ECG findings.

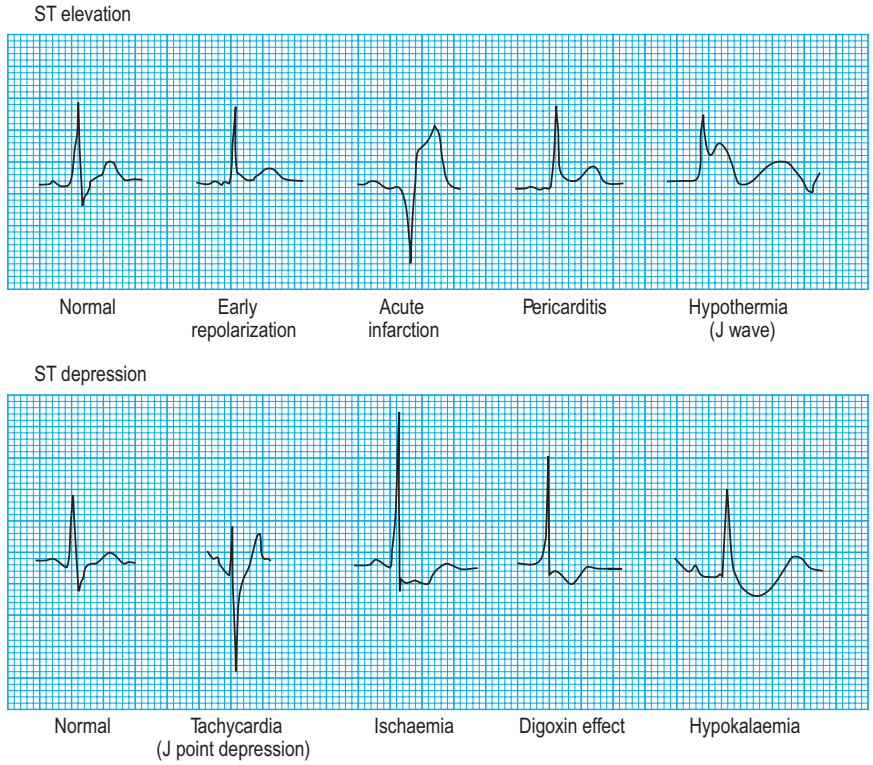



**Figure 13.14** ST segment morphology: common causes of ST segment elevation and depression. Note that depression of the J point (junction between the QRS complex and ST segment) is physiological during exertion and does not signify myocardial ischaemia. Planar depression of the ST segment, on the other hand, is strongly suggestive of myocardial ischaemia.

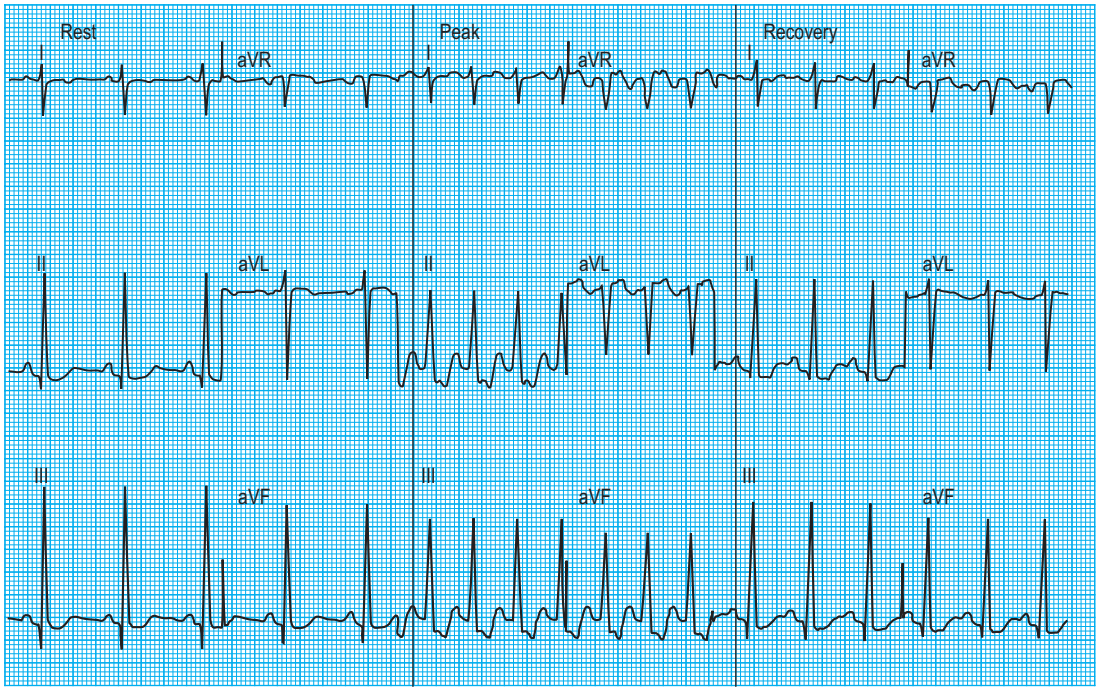



**Figure 13.15** Exercise ECG: ischaemic changes in inferior standard leads (II, III and aVF). At rest, the ST segments are isoelectric. Exercise causes tachycardia and provokes 3 mm of downsloping ST depression in leads II, III and aVF. The changes reverse during recovery. The findings suggest exertional ischaemia affecting the inferior wall of the heart. The probability of coronary artery disease is high.

Typical symptoms not associated with either troponin release or ST elevation are diagnosed as troponin-negative acute coronary syndrome or unstable angina. It therefore follows that acute myocardial infarction and unstable angina may be associated with a completely normal ECG or with ST depression (Fig. 13.16) or T-wave changes, the diagnosis depending on the presence or absence of raised troponins.

In STEMI, the evolution of ECG changes is characteristic, although it may be aborted by timely reperfusion therapy (thrombolysis or primary stenting). Peaking of the T wave followed by ST segment elevation occurs during the first hour of pain (Fig. 13.17). The changes are regional, and reciprocal ST depression may be seen in the opposite ECG leads. Usually a pathological Q wave develops during the following 24 hours and persists indefinitely. The ST segment returns to the isoelectric line within 2–3 days, and T-wave inversion may occur. The ECG is a useful indicator of infarct location. Changes in leads II, III and aVF indicate inferior infarction (Fig. 13.18), whereas changes in leads $V_1$–$V_6$ indicate anteroseptal ($V_1$–$V_3$) or anterolateral ($V_1$–$V_6$) infarction (Fig. 13.19). When the infarct is located posteriorly, ECG changes may be difficult to detect, but dominant R waves in leads $V_1$ and $V_2$ often develop (see Fig. 13.18).

## Detection of cardiac arrhythmias

Electrocardiographic documentation of the arrhythmia should be obtained prior to instituting treatment. In patients with sustained arrhythmias, a 12-lead recording at rest is usually diagnostic, but a long continuous recording of the lead showing the clearest P wave (if present) should also be obtained. In patients with paroxysmal arrhythmias,

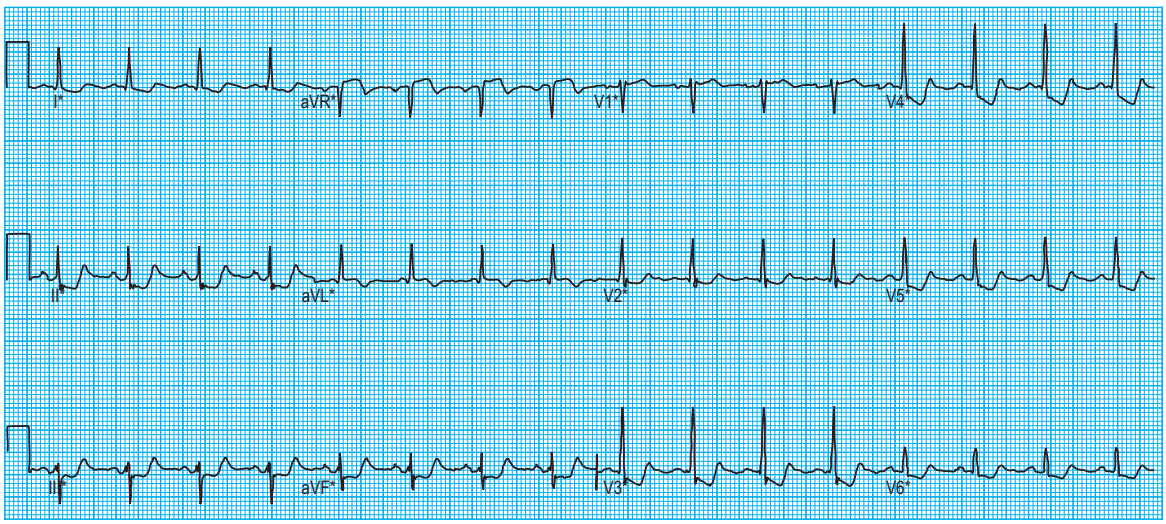



**Figure 13.16** Unstable angina or non-ST elevation myocardial infarction (depending on troponin release): 12-lead ECG showing planar/downsloping ST depression in the inferolateral territory.

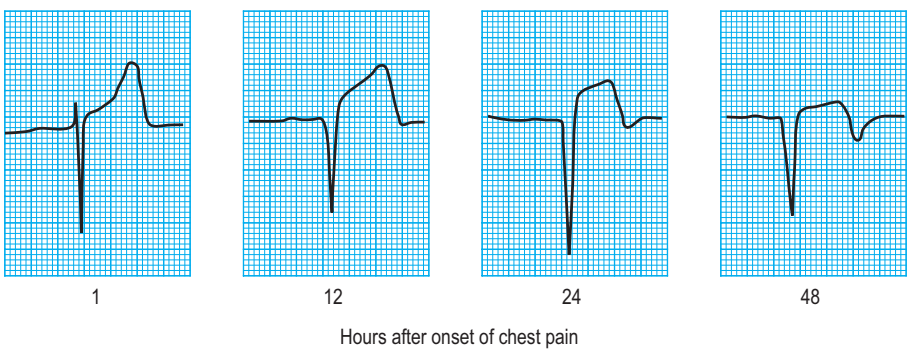



**Figure 13.17** Acute myocardial infarction: evolution of ECG changes. Elevation of the ST segment occurs during the first hour of chest pain. The Q wave develops during the subsequent 24 hours, usually persisting indefinitely. Within a day of the attack, the ST segment usually returns to the isoelectric line and T-wave inversion may occur.

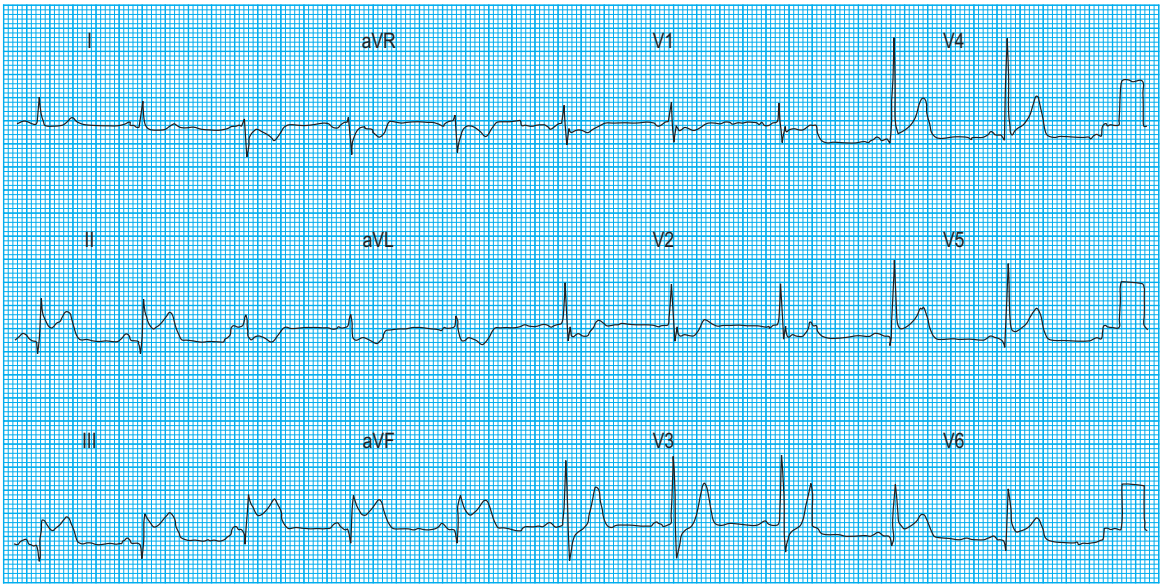



**Figure 13.18** Acute inferolateral infarction. ECG 2 hours after the onset of chest pain. Typical ST elevation in leads II, III and aVF is diagnostic of inferior myocardial infarction. ST elevation in leads $V_4$–$V_6$ indicates lateral extension. There is reciprocal ST depression in lead aVL. Prominent R waves associated with ST depression in leads $V_1$ and $V_2$ indicate posterior wall infarction. This pattern may reflect occlusion of the right coronary artery or a dominant circumflex coronary artery.

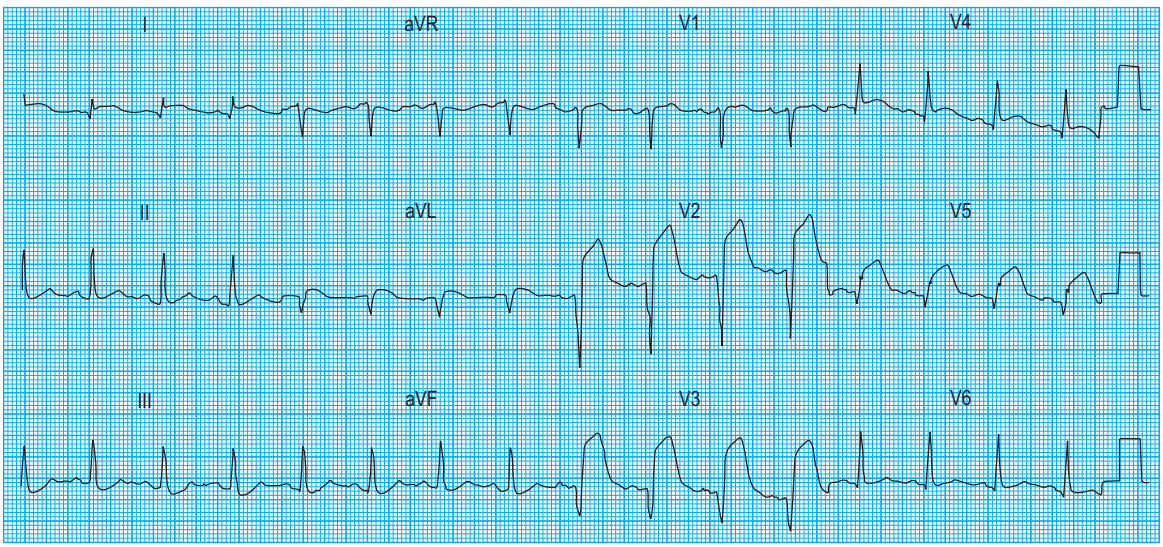



**Figure 13.19** Acute anterior infarction. ECG 1 hour after the onset of chest pain. Typical ST elevation in leads $V_2$–$V_5$ is diagnostic of anterior myocardial infarction. Additional ST elevation in standard leads I and aVL indicates lateral extension of the infarct. This pattern usually reflects proximal occlusion of the left anterior descending coronary artery.

the frequency and severity of symptoms determine which technique is used for electrocardiographic documentation.

## In-hospital ECG monitoring

Patients who have had out-of-hospital cardiac arrest or severe, arrhythmia-induced heart failure should undergo continuous ECG monitoring in hospital under the surveillance of staff trained in the recognition and treatment of arrhythmia. Patients with acute myocardial infarction should undergo ECG monitoring for 24 hours, after which time the risk of ventricular arrhythmia falls dramatically.

## Ambulatory (Holter) ECG monitoring

Patients with frequent palpitation or dizzy attacks are commonly investigated by means of an ambulatory 24-hour ECG. The availability of portable recorders allows this to be performed as an outpatient. Analysis

of the tape identifies any cardiac arrhythmias that occurred during the monitoring period (Fig. 13.20). A patient diary allows correlation between symptoms and heart rhythm.

## Patient-activated ECG recording

For patients with infrequent symptoms, the detection rate with 24-hour ambulatory monitoring is low and patient-activated recorders are more useful. The patient can keep the recorder for several weeks and activate it when symptoms occur. Rhythm strips are stored for later analysis.

## Implantable loop recording

Patients in whom there is clinical suspicion of serious arrhythmia but whose symptoms occur less than

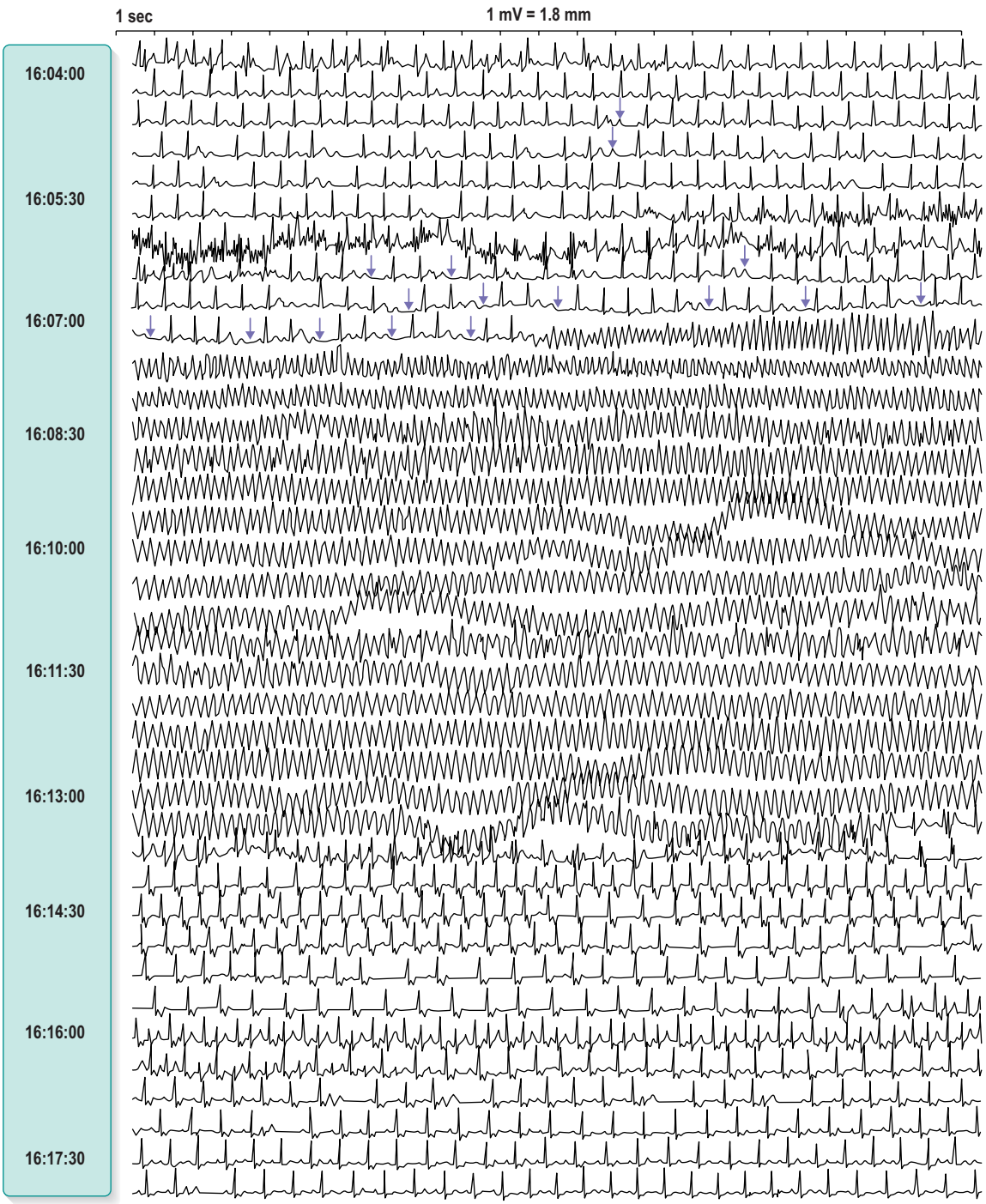



**Figure 13.20** Ventricular tachycardia: Holter recording. When tachycardias are paroxysmal in nature, continuous ECG monitoring is often necessary to document the arrhythmia. Here a Holter recording illustrates a long burst of rapid VT lasting a total of 6 minutes. Preceding the VT there is second-degree heart block (*arrows*).

once a month pose particular diagnostic difficulty. In this group, a miniaturized recording device can be implanted subcutaneously using local anaesthetic and interrogated electronically through the skin in the event of symptoms.

### Exercise testing

The ECG recorded during exercise may be helpful when there is a history of exertional palpitation. Arrhythmias provoked by ischaemia or increased sympathetic activity are more likely to be detected during exercise.

### Tilt testing

When malignant vasovagal syndrome is suspected, ECG and blood pressure recordings during tilting from supine to erect posture can be helpful. Abnormal bradycardia or hypotension sufficient to produce presyncope or syncope indicates a 'vasodepressor response' or 'cardioinhibitory response', respectively, and either is strongly suggestive of the diagnosis.

### Electrophysiological study

Electrophysiological study is a technique that requires cardiac catheterization with catheter-mounted electrodes. Premature stimuli are introduced into the atria or ventricles with a view to stimulating re-entry arrhythmias. In the normal heart, sustained arrhythmias are rarely provoked by premature stimuli. Thus, arrhythmia provocation is usually diagnostic, particularly when the arrhythmia reproduces symptoms. Electrophysiological study can identify accessory pathways and areas of focal atrial or ventricular ectopy as the prelude to radiofrequency ablation of the arrhythmia substrate.

## Diagnosis of atrial arrhythmias

The ECG in atrial arrhythmias (Fig. 13.21) shows a narrow and morphologically normal QRS complex when ventricular depolarization occurs by normal His-Purkinje pathways. Rate-related or pre-existing bundle branch block, however, results in broad ventricular complexes that are difficult to distinguish from ventricular tachycardia.

### Atrial ectopic beats

Atrial ectopic beasts rarely indicate heart disease. They often occur spontaneously, but may be provoked by toxic stimuli such as caffeine, alcohol and cigarette smoking. They are caused by the premature discharge of an atrial ectopic focus; an early and occasionally bizarre P wave is essential for the diagnosis. The premature impulse enters and depolarizes the sinus node such that a partially compensatory pause occurs before the next sinus beat during resetting of the sinus node.

### Atrial fibrillation

Prevalence of atrial fibrillation increases with age and is common in hypertensive heart disease, mitral valve disease, thyrotoxicosis and left ventricular failure. It can be precipitated by pneumonia and major surgery and also by various toxic stimuli, particularly alcohol. Atrial activity is chaotic and mechanically ineffective. P waves are therefore absent and replaced by irregular fibrillatory waves (rate 400–600/min). The long refractory period of the atrioventricular node ensures that only some of the atrial impulses are conducted to produce an irregular ventricular rate of 130–200 bpm. If the atrioventricular node is diseased, the ventricular rate is slower; however, in the presence of a rapidly conducting accessory pathway in WPW syndrome, dangerous ventricular rates above 300 bpm may occur.

### Atrial flutter

Atrial flutter is less common than atrial fibrillation, but it does occur under similar circumstances. Re-entry mechanisms produce an atrial rate close to 300 bpm. The normal atrioventricular node conducts with 2:1 block, giving a ventricular rate of 150 bpm. Higher degrees of block may reflect intrinsic disease of the atrioventricular node or the effects of nodal blocking drugs. The ECG characteristically shows sawtooth flutter waves, which are most clearly seen when the block is increased by carotid sinus pressure.

## Diagnosis of nodal arrhythmias

Often called supraventricular tachycardias (SVTs), nodal arrhythmias are usually paroxysmal without obvious cardiac or extrinsic causes. They are re-entry arrhythmias caused either by an abnormal pathway between the atrium and the atrioventricular node (atrionodal pathway) or by an accessory atrioventricular pathway (bundle of Kent), as seen in WPW syndrome. Like atrial arrhythmias, ventricular depolarization usually occurs by normal His-Purkinje pathways, producing a narrow QRS complex which confirms the supraventricular origin of the arrhythmia. Rate-related or pre-existing bundle branch block, however, produces broad ventricular complexes difficult to distinguish from ventricular tachycardia.

### Atrioventricular nodal re-entry tachycardia (AVNRT)

The abnormal atrionodal pathway provides the basis for a small re-entry circuit. In sinus rhythm, the ECG is usually normal, although occasionally the PR interval is short (Lown-Ganong-Levine syndrome). During tachycardia, the rate is 150–250 bpm (see Fig. 13.21). The arrhythmia is usually self-limiting. Sustained AVNRT will sometimes respond to carotid sinus pressure. If this fails, intravenous adenosine or verapamil usually are effective by blocking

the re-entry circuit within the atrioventricular node. Antitachycardia pacing or direct current cardioversion may also be used. Many patients are now being treated by catheter ablation to destroy the abnormal atrionodal pathway and avoid the need for long-term drug therapy (see below).

### Wolff-Parkinson-White syndrome

A congenital disorder that affects 0.12% of the population, Wolff-Parkinson-White syndrome, is caused by an accessory pathway (bundle of Kent) between the atria and the ventricles. During sinus rhythm, atrial impulses conduct more rapidly through the accessory pathway than the atrioventricular node, such that the initial phase of ventricular depolarization occurs early (pre-excitation) and spreads slowly through the ventricles by abnormal pathways. This produces a short PR interval and slurring of the initial QRS deflection (δ wave). The remainder of ventricular depolarization, however, is rapid because the delayed arrival of the impulse conducted through the atrioventricular node rapidly completes ventricular depolarization by normal His-Purkinje pathways (Fig. 13.22). Cardiac arrhythmias affect about 60% of patients with WPW syndrome and are usually re-entrant

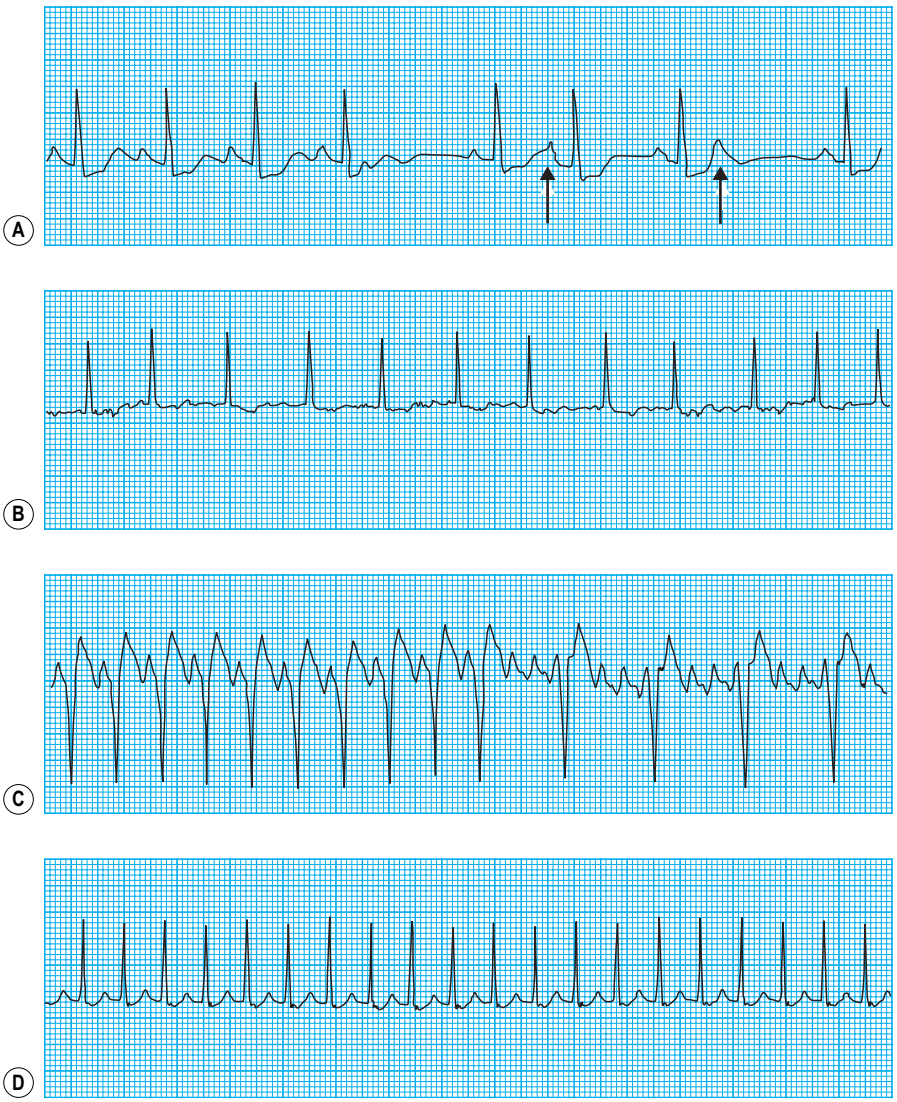



**Figure 13.21** Atrial arrhythmias. **(A)** Ectopic beats. After the fourth sinus beat there is a very early P wave which, finding the AV node refractory, is not conducted to the ventricle. This produces a pause before the next sinus beat, which itself is followed by a somewhat later atrial ectopic beat (*arrow*), which is conducted normally. This is followed by a sinus beat, following which the T wave is distorted by another early atrial ectopic (beat *arrowed*), which is also blocked. **(B)** Atrial fibrillation. Note the irregular fibrillatory waves and the irregular ventricular response. The ventricular rate is fairly slow because the patient was treated with a β-blocker. **(C)** A flutter. AV conduction with 2:1 block, giving a ventricular rate of about 150/min, which then gives way to 4:1 block. Sawtooth flutter waves at a rate of 300/min are clearly visible. **(D)** AV nodal re-entrant tachycardia (AVNRT). Often called supraventricular tachycardia (SVT), this arrhythmia causes a regular tachycardia, with a ventricular rate of about 180/min.

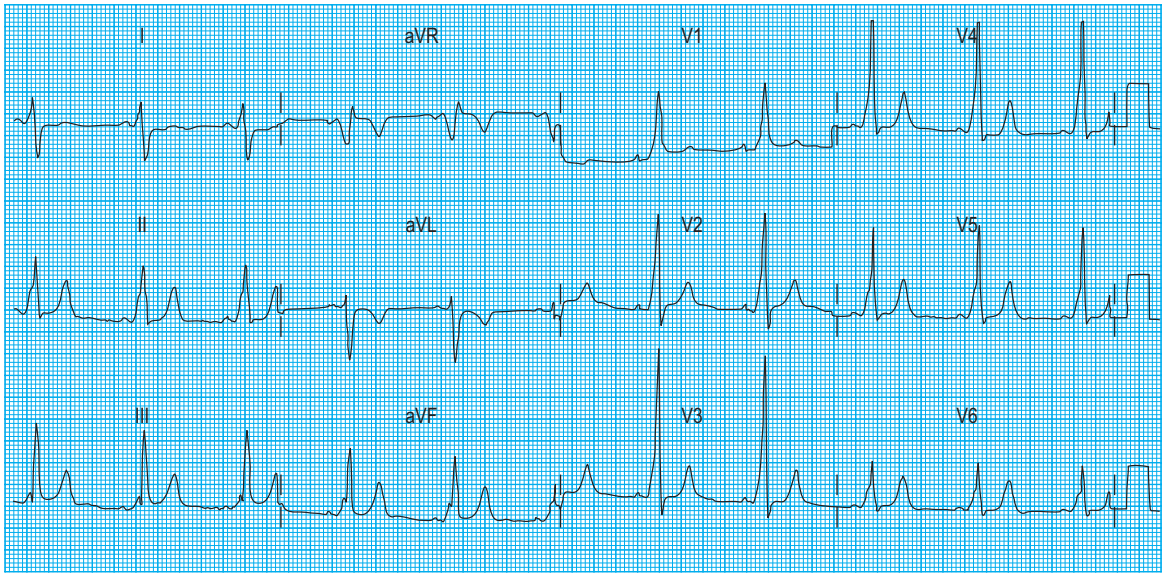



**Figure 13.22** WPW syndrome: 12-lead ECG. Ventricular pre-excitation is reflected on the surface ECG by a short PR interval and a slurred upstroke to the QRS complex (δ wave). The remainder of the QRS complex is normal because delayed arrival of the impulse conducted through the AV node rapidly completes ventricular depolarization through normal His-Purkinje pathways.

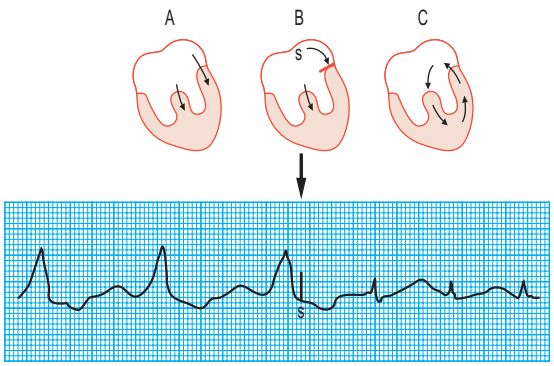



**Figure 13.23** WPW syndrome: re-entry tachycardia recorded at fast paper speed. Just after the third pre-excited (broad) complex, a premature atrial pacing stimulus (S) initiates an impulse that is blocked in the bundle of Kent, but conducted normally through the AV node, producing ventricular depolarization without pre-excitation. Thus, the QRS complex is narrow and lacks a δ wave. The impulse is conducted retrogradely through the bundle of Kent, re-enters the proximal conducting system and completes the re-entry circuit, initiating a self-sustaining orthodromic re-entry tachycardia (last three complexes).

(rate 150–250 bpm) triggered by an atrial premature beat. In most patients, the re-entry arrhythmia is 'orthodromic', with anterograde conduction through the atrioventricular node and retrograde conduction through the accessory pathway (atrioventricular re-entry tachycardia (AVRT), Fig. 13.23). This results in a narrow complex tachycardia (without pre-excitation) that is indistinguishable from AVNRT. Occasionally, the re-entry circuit is in the opposite direction ('antidromic'), producing a very broad, pre-excited tachycardia.

Patients with WPW syndrome are more susceptible than the general population to atrial fibrillation. If the accessory pathway is able to conduct the fibrillatory impulses rapidly to the ventricles, it may result in ventricular fibrillation and sudden death. Digoxin (and to a lesser extent verapamil) should be avoided because it shortens the refractory period of the accessory pathway and can heighten the risk. Patients with dangerous accessory pathways of this type require catheter ablation of the pathway. Ablation therapy also cures AVNRT and is the treatment of choice in patients with frequent attacks.

## Diagnosis of ventricular arrhythmias

### Ventricular premature beats

Ventricular premature beats may occur in normal individuals, either spontaneously or in response to toxic stimuli such as caffeine or sympathomimetic drugs. They are caused by the premature discharge of a ventricular ectopic focus that produces an early and broad QRS complex (Fig. 13.24). The premature impulse may be conducted backwards into the atria, producing a retrograde P wave, but penetration of the sinus node is rare. Thus, resetting of the sinus node does not usually occur and there is a compensatory pause before the next sinus beat.

### Ventricular tachycardia

Ventricular tachycardia is defined as three or more consecutive ventricular beats at a rate above 120 per minute. Ventricular depolarization inevitably occurs slowly by abnormal pathways, producing a broad QRS complex. This distinguishes it from most

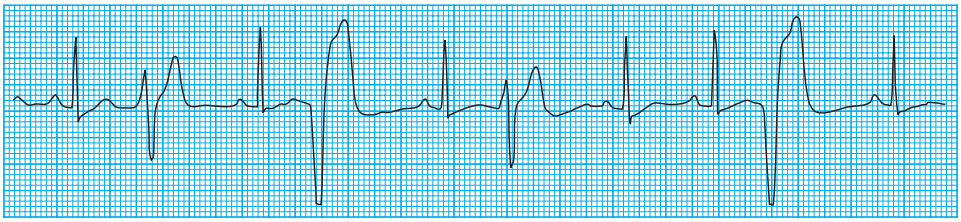


**Figure 13.24** Multifocal, ventricular ectopic beats. Frequent broad complex ectopic beats are seen early after the sinus beats. Note, however, that the ectopic beats have two different morphologies, indicating that they arise from different foci. Note also that the coupling interval (interval between QRS complex and ectopic beat) is identical for beats arising from any particular focus.

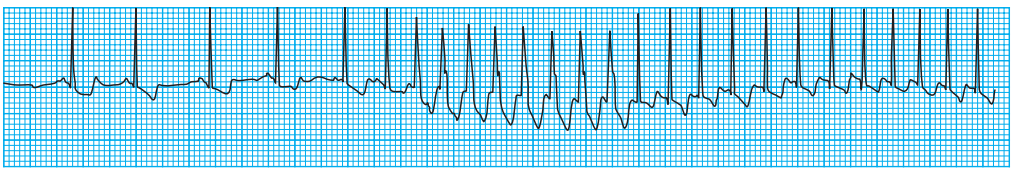


**Figure 13.25** Paroxysmal AV nodal re-entrant tachycardia. This Holter recording shows sinus rhythm giving way to a broad complex tachycardia. However, this is clearly the result of temporary bundle branch block because it converts spontaneously to a narrow complex tachycardia, confirming that the arrhythmia is junctional, not ventricular, in origin.

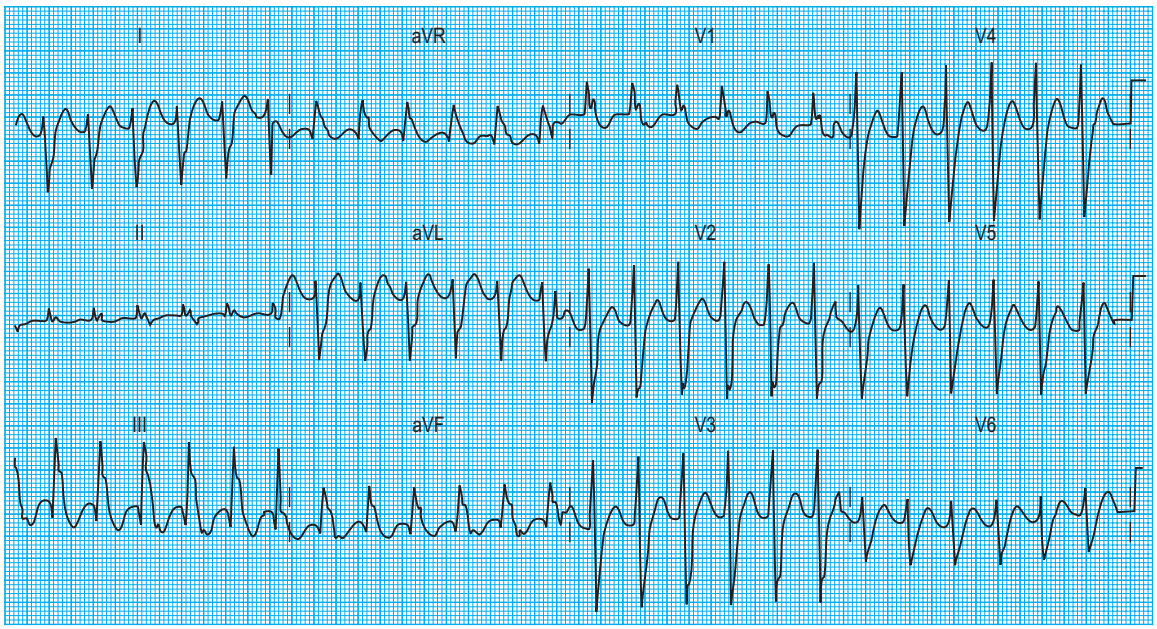



**Figure 13.26** Ventricular tachycardia: 12-lead ECG. The recording shows a broad complex tachycardia. The following features suggest or confirm the ventricular origin of the tachycardia: very broad QRS complex (>140 ms); extreme right axis deviation; atrioventricular dissociation: note the dissociated P waves in lead $V_1$; the 'rSR' complex in $V_1$.

atrial and junctional tachycardias that have a narrow QRS complex, although differential diagnosis may be more difficult for atrial or junctional tachycardias with a broad QRS complex caused by rate-related or pre-existing bundle branch block (Fig. 13.25). Nevertheless, ventricular tachycardia usually can be identified by careful scrutiny of the 12-lead ECG (Fig. 13.26). Support for the diagnosis is provided by a very broad QRS complex (>140 ms), extreme left or right axis deviation, concordance of the QRS deflections in $V_1$–$V_6$ (either all positive or all negative) and configurational features of the QRS complex, including an 'rSR' complex in $V_1$ and a QS complex in $V_6$. Confirmation of the diagnosis is provided by any evidence of AV dissociation: either P waves, at a slower rate than the QRS complexes, 'marching through' the tachycardia (Fig. 13.27A), or ventricular capture and/or fusion beats, in which the dissociated atrial rhythm penetrates the ventricle by conduction through the AV node

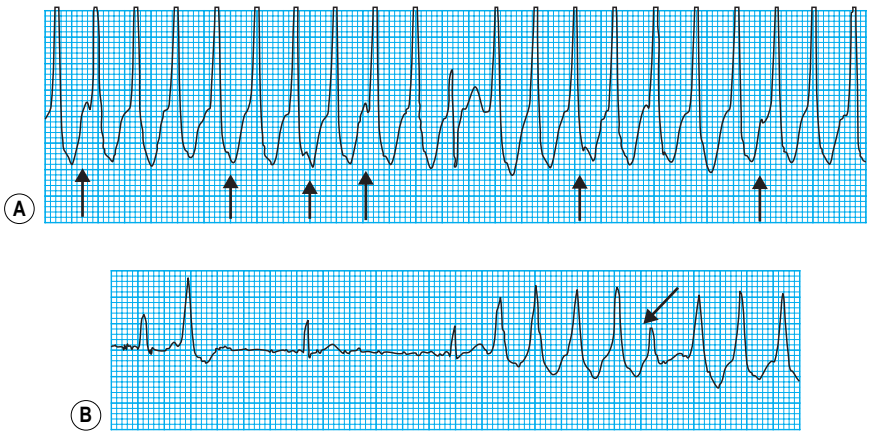



**Figure 13.27** **(A)** Ventricular tachycardia: AV dissociation. P waves (*arrows*) can be seen 'marching through' the tachycardia, confirming its ventricular origin. The tachycardia is interrupted by a narrow capture beat. **(B)** Ventricular tachycardia (VT): fusion. In this example, VT is initiated by a very early ventricular ectopic beat (morphologically similar to the previous isolated ectopic beat) and is interrupted by a fusion beat (*arrow*), confirming the ventricular origin of the tachycardia.

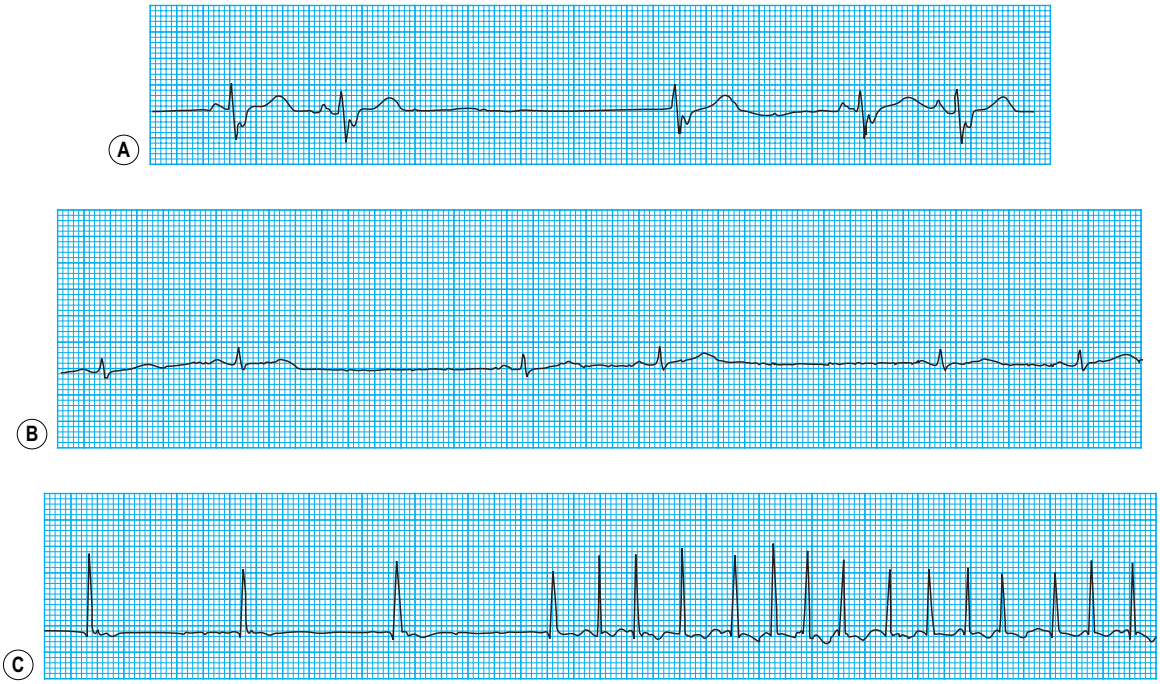



**Figure 13.28** Sinoatrial disease. **(A)** Sinus arrest with late junctional escape. After the second sinus beat there is a long pause, interrupted by a single junctional escape beat, before sinus rhythm is re-established. **(B)** Sinoatrial block. Pauses after the second and fourth complexes are the result of sinoatrial block, which has prevented sinus impulses from depolarizing the atrium. No P waves are seen but, because the sinus discharge continues uninterrupted, the pauses are each a precise multiple of the preceding PP interval. Sinoatrial block is probably rare. **(C)** Bradycardia-tachycardia syndrome. A slow junctional rhythm gives way to rapid atrial fibrillation.

and interrupts the tachycardia, producing a normal ventricular complex (capture, Fig. 13.27A) or, more commonly, a broad hybrid complex (fusion) that is part sinus and part ventricular in origin (Fig. 13.27B). *Torsades de pointes*, a broad complex tachycardia with changing wavefronts, also provides unequivocal evidence of ventricular tachycardia and is particularly characteristic of the arrhythmia that complicates long QT syndrome, often resulting in sudden death. The syndrome may be inherited as an autosomal-dominant (Romano-Ward syndrome) or as an autosomal-recessive (Lange-Nielsen syndrome) trait, when it is associated with congenital deafness. Another important cause of *Torsade de pointes* is the use of drugs (antiarrhythmics and others, such as anti-psychotic therapy) that cause prolongation of the QT interval.

## Ventricular fibrillation

Ventricular fibrillation occurs most commonly in severe myocardial ischaemia, either with or without frank infarction. It is a completely disorganized

arrhythmia characterized by irregular fibrillatory waves with no discernible QRS complexes. There is no effective cardiac output and death is inevitable unless resuscitation with direct current defibrillation is instituted rapidly.

## Diagnosis of sinoatrial disease

Sinus node discharge is not itself visible on the surface ECG, but the atrial depolarization it triggers produces the P wave. The spontaneous discharge of the normal sinus node is influenced by a variety of neurohumoral factors, particularly vagal and sympathetic activity, which respectively slow and quicken the heart rate. In sinoatrial disease (Fig. 13.28 and Box 13.16), sinus node discharge may be abnormally slow, blocked (with failure to activate atrial depolarization) or absent altogether. Under these circumstances, the sinus rate may be very slow, the atrium may fibrillate, or pacemaker function may be assumed by foci lower in the atrium, the atrioventricular node or the His-Purkinje conducting tissue in the ventricles. The intrinsic rate of these 'escape' pacemaker foci is slower than the normal sinus rate.

### Sinus bradycardia (<50 bpm)

Sinus bradycardia is physiological during sleep and in trained athletes, but in other circumstances often reflects sinoatrial disease, particularly when the heart rate fails to increase normally with exercise.

**Box 13.16** Sinoatrial disease

**Typical patient**

- Elderly, often with no previous cardiac history

**Causes**

- Acute: myocardial infarction, coronary artery disease, drugs (e.g. β-blockers, digoxin), hypothermia, atrial surgery
- Chronic: idiopathic fibrotic disease, congenital heart disease, ischaemic heart disease, amyloid

**Major symptoms**

- Intermittent syncopal or presyncopal attacks
- Patients may also complain of exertional fatigue (chronotropic incompetence) or palpitations (tachycardia-bradycardia syndrome).

**Major signs**

- Often none
- Sometimes sinus bradycardia or slow atrial fibrillation

**Diagnosis**

- ECG: often normal. May show sinus bradycardia or slow atrial fibrillation
- Ambulatory ECG: 24-hour Holter recording may show pauses diagnostic of sinoatrial disease.

**Comments**

- Documentation of the sinus pauses (or very slow atrial fibrillation) during an attack of symptoms provides the most robust diagnostic information.

### Sinoatrial block

If the sinus impulse is blocked and fails to trigger atrial depolarization, a pause occurs in the ECG. No P wave is seen during the pause owing to the absence of atrial depolarization. The electrically 'silent' sinus discharge, however, continues uninterrupted. Thus, the pause is always a precise multiple of preceding PP intervals. Sinoatrial block that cannot be abolished by atropine-induced vagal inhibition usually indicates sinoatrial disease, particularly with pauses longer than 2 seconds.

### Sinus arrest

Failure of sinus node discharge produces a pause on the ECG that bears no relation to the preceding PP interval. Pauses longer than 2 seconds are usually pathological. Prolonged pauses are often terminated by an escape beat from a 'junctional' focus in the bundle of His.

### Bradycardia-tachycardia syndrome

In the bradycardia-tachycardia syndrome, atrial bradycardias are interspersed by paroxysmal tachyarrhythmias, usually atrial fibrillation. Nevertheless, it is the bradycardia that usually causes symptoms, particularly dizzy attacks and blackouts.

## Diagnosis of atrioventricular block

In atrioventricular block (Fig. 13.29), conduction is delayed or completely interrupted, either in the atrioventricular node or in the bundle branches (Box 13.17). When conduction is merely delayed (e.g. first-degree atrioventricular block, bundle branch block), the heart rate is unaffected. When conduction is completely interrupted, however, the heart rate may slow sufficiently to produce symptoms. In second-degree atrioventricular block, failure of conduction, by definition, is intermittent; if sufficient sinus impulses are conducted to maintain an adequate ventricular rate, symptoms may be avoided. In third-degree atrioventricular block there is complete failure of conduction and continuing ventricular activity depends on the emergence of an escape rhythm. If the block is within the atrioventricular node, the escape rhythm usually arises from a focus just below the node in the bundle of His (junctional escape), and is often fast enough to prevent symptoms. If both bundle branches are blocked, however, the escape rhythm must arise from a focus lower in the ventricles. Ventricular escape rhythms of this type are nearly always associated with symptoms because they are not only very slow but also unreliable, and may stop altogether, producing prolonged asystole.

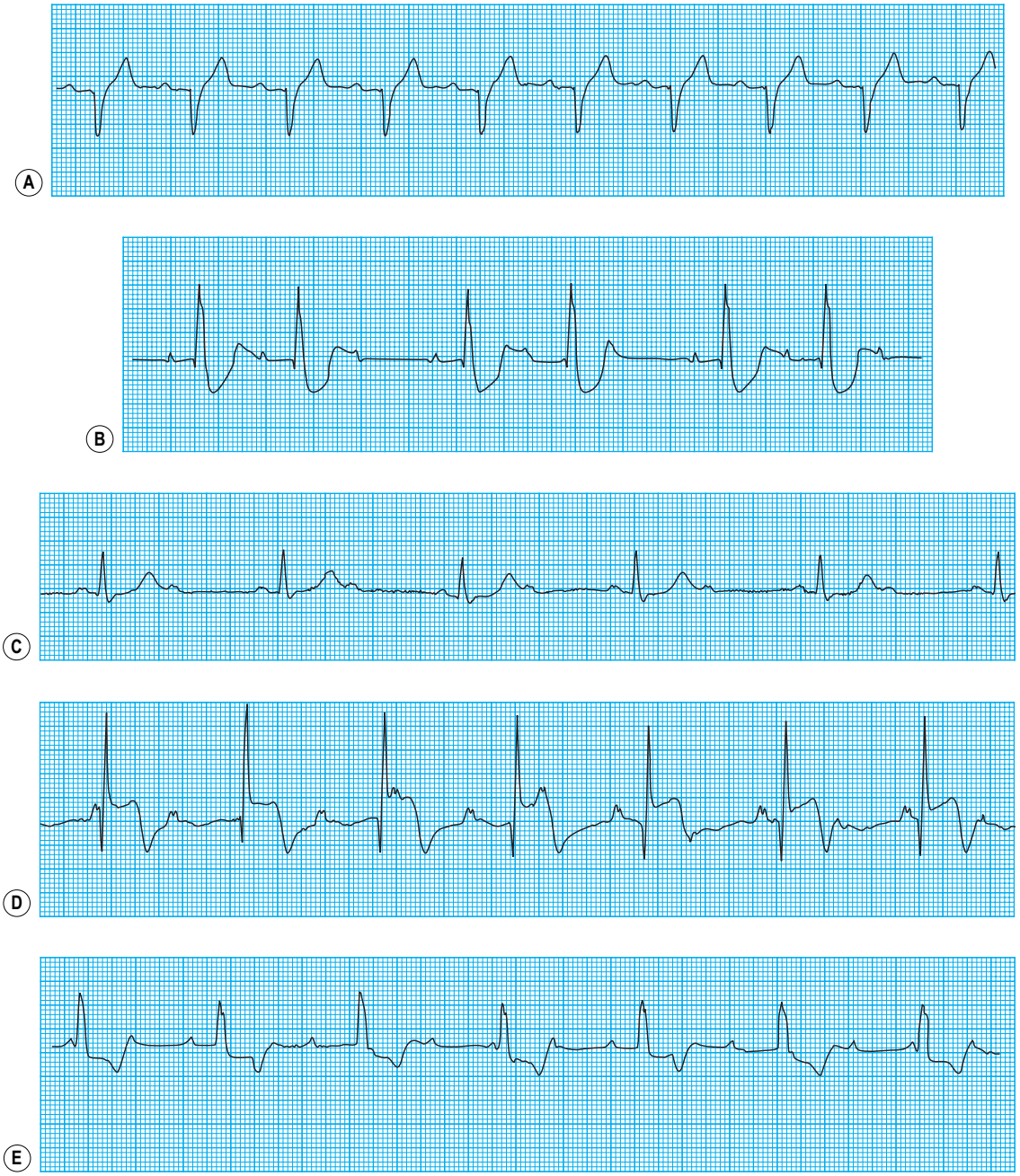



**Figure 13.29** Atrioventricular conducting tissue disease. **(A)** 1° AV block. Delayed AV conduction causes a prolonged PR interval (>0.20 s). **(B)** 2° AV block, Wenckebach type. This, also called Mobitz type I block, occurs within the AV node. Three Wenckebach cycles are shown. Successive sinus beats find the AV node increasingly refractory until failure of conduction occurs. This delay permits recovery of nodal function and the process repeats itself. **(C)** 2° AV block at bundle branch level (Mobitz type II). This is standard lead I. Note that the PR interval of conducted beats is normal, but the QRS complex shows right bundle branch block. Intermittent block in the left bundle results in failure of conduction of alternate P waves. **(D)** 3° (complete) AV block at level of AV node. In this patient with acute inferior myocardial infarction there is complete failure of AV conduction, as reflected by the dissociated atrial and ventricular rhythms. Note the regular P waves and the regular slower QRS complexes occurring independently of one another. Because block is at the level of the AV node, a junctional escape rhythm has taken over with a narrow QRS complex. **(E)** 3° (complete) AV block at bundle branch level. The atrial and ventricular rhythms are dissociated because none of the atrial impulses is conducted. The ECG shows regular P waves and regular but slower QRS complexes. Because the escape rhythm is ventricular in origin, the QRS complexes are broad and the rate is slow.

**Box 13.17** Causes of atrioventricular heart block

**Acute**

- Myocardial infarction
- Drugs (e.g. β-blockers, verapamil, digoxin, adenosine)
- Surgical or catheter ablation of bundle of His
- Endocarditis complicated by aortic root abscess

**Chronic**

- Idiopathic fibrosis of both bundle branches
- Ischaemic heart disease
- Congenital heart disease
- Calcific aortic valve disease
- Chagas' disease
- Infiltrative disease (amyloid, haemochromatosis)
- Granulomatous disease (sarcoid, tuberculosis)

### First-degree atrioventricular block

Delayed atrioventricular conduction causes prolongation of the PR interval (>0.20 s). Ventricular depolarization occurs rapidly by normal His-Purkinje pathways and the QRS complex is usually narrow.

### Second-degree atrioventricular block: Mobitz type I (Wenckebach)

Second-degree atrioventricular block (Mobitz type I) commonly occurs in inferior myocardial infarction. Successive sinus beats find the atrioventricular node increasingly refractory until failure of conduction occurs. The delay permits recovery of nodal function, and the process may then repeat itself. The ECG shows progressive prolongation of the PR interval, culminating in a dropped beat. Block is within the atrioventricular node itself and ventricular depolarization occurs rapidly by normal pathways. Thus, the QRS complex is usually narrow.

### Second-degree atrioventricular block: Mobitz type II

Mobitz type II second-degree atrioventricular block indicates advanced conducting tissue disease affecting the bundle branches. The ECG typically shows a normal PR interval with bundle branch block in conducted beats, and intermittent block in the other bundle branch resulting in complete failure of atrioventricular conduction and dropped beats.

### Third-degree (complete) atrioventricular block

The atrial and ventricular rhythms are 'dissociated' in third-degree atrioventricular block because none of the atrial impulses are conducted. Thus, the ECG shows regular P waves (unless the atrium is fibrillating) and regular but slower QRS complexes occurring independently of each other. When block is within the atrioventricular node (e.g. inferior myocardial infarction, congenital atrioventricular block), a junctional escape rhythm with a reliable rate (40–60 bpm) takes over (see Fig. 13.29). Ventricular depolarization occurs rapidly by normal pathways, producing a narrow QRS complex. However, when the block is within the bundle branches (e.g. idiopathic fibrosis), there is always extensive conducting tissue disease. The ventricular escape rhythm is slow and unreliable, with a broad QRS complex (see Fig. 13.29).

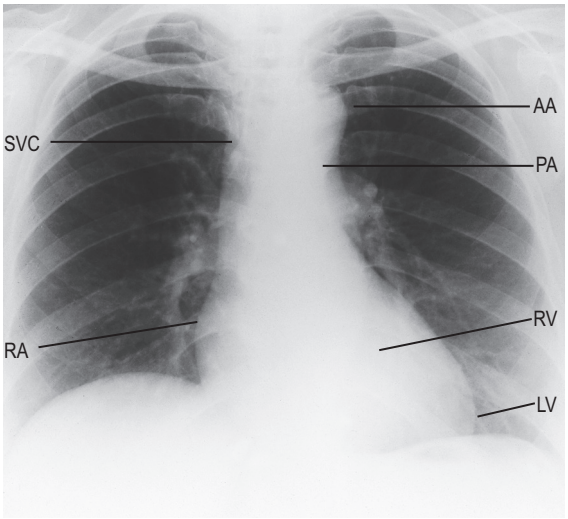



**Figure 13.30** Normal chest X-ray: posteroanterior projection. Note the heart is not enlarged (cardiothoracic ratio <50%) and the lung fields are clear. AA, aortic arch; LV, left ventricle; PA, pulmonary artery; RA, right atrium; RV, right ventricle; SVC, superior vena cava.

### Right bundle branch block

A right bundle branch block may be a congenital defect, but more commonly, it is the result of organic conducting tissue disease. Right ventricular depolarization is delayed, resulting in a broad QRS complex with an 'rSR' pattern in lead $V_1$ and prominent S waves in leads I and $V_6$.

### Left bundle branch block

Left bundle branch block always indicates organic conducting tissue disease. The entire sequence of ventricular depolarization is abnormal, resulting in a broad QRS complex with large slurred or notched R waves in leads I and $V_6$.

## The chest X-ray

Good-quality posteroanterior (PA) and lateral chest X-rays are always helpful in the assessment of the cardiac patient (Fig. 13.30).

## Cardiac silhouette

Although the PA chest X-ray exhibits a wide range of normality, the maximum diameter of the heart should not be more than 50% of the widest diameter of the thorax. Cardiac enlargement is caused either



**Figure 13.31** Pericardial effusion with tamponade: chest X-ray. There is a left hilar mass caused by carcinoma. Pericardial infiltration has produced effusion and tamponade, evidenced by the severely enlarged and globular cardiac silhouette. Malignant disease is now the most common cause of tamponade in most developed countries.

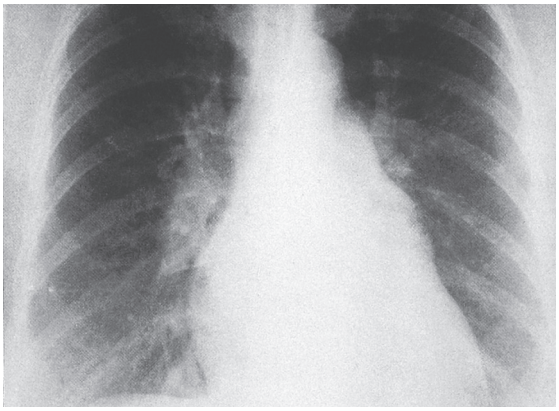


**Figure 13.32** Left atrial dilatation. This is a penetrated PA chest X-ray in a patient with mitral stenosis. The dilated left atrium causes a bulge on the left heart border below the pulmonary artery which is also dilated, widening of the carina and the double-density sign at the right heart border.

by dilatation of the cardiac chambers or by pericardial effusion (Fig. 13.31). Myocardial hypertrophy only affects heart size if very severe.

### Ventricular dilatation

The PA chest X-ray does not reliably distinguish left from right ventricular dilatation. For this, the lateral chest X-ray is more helpful. Dilatation of the posteriorly located left ventricle encroaches on the retrocardiac space, whereas dilatation of the anteriorly located right ventricle encroaches on the retrosternal space.

### Atrial dilatation

Right atrial dilatation is usually caused by right ventricular failure, but it does occur as an isolated finding in tricuspid stenosis and Ebstein's anomaly. It produces cardiac enlargement without specific radiographic signs.

Left atrial dilatation occurs in left ventricular failure and mitral valve disease (Fig. 13.32). Radiographic signs are:

- Flattening and later bulging of the border on the left side of the heart below the main pulmonary artery
- Elevation of the left main bronchus, with widening of the carina
- Appearance of the medial border of the left atrium behind the right side of the heart (double-density sign)

### Vascular dilatation

Aortic dilatation caused by an aneurysm or dissection may produce widening of the entire upper mediastinum. Localized dilatation of the proximal aorta, which occurs in aortic valve disease, produces a prominence in the right upper mediastinum (Fig. 13.33). Dilatation of the main pulmonary artery occurs in pulmonary hypertension and pulmonary stenosis and produces a prominence below the aortic knuckle (Fig. 13.34).

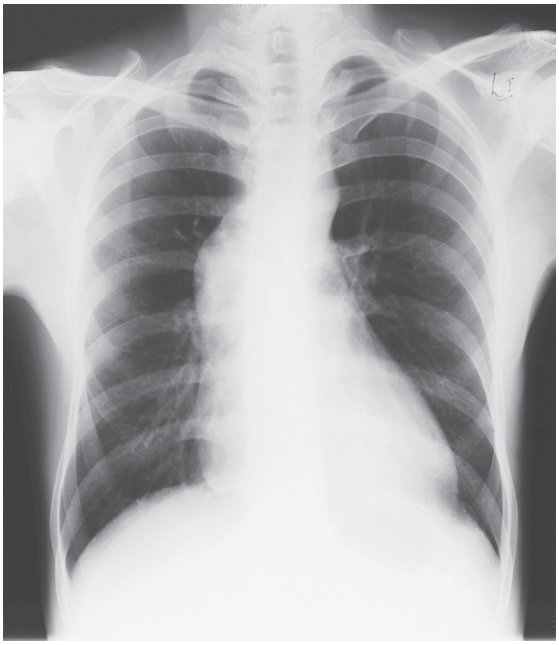


**Figure 13.33** Chest X-ray in a patient with Marfan's syndrome. Note the dilatation of the ascending aorta.

### Intracardiac calcification

Because the radiodensity of cardiac tissue is similar to that of blood, intracardiac structures can rarely be identified unless they are calcified. Valvular, pericardial or myocardial calcification may occur, which usually indicates important disease of these structures. Calcification is best appreciated on the deeply penetrated lateral chest X-ray.

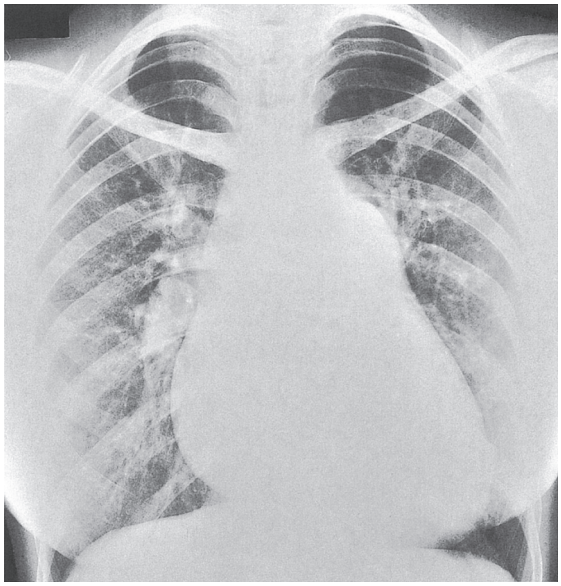


**Figure 13.34** Atrial septal defect: chest X-ray. Note the prominent proximal pulmonary arteries and the pulmonary plethora reflecting increased pulmonary flow.

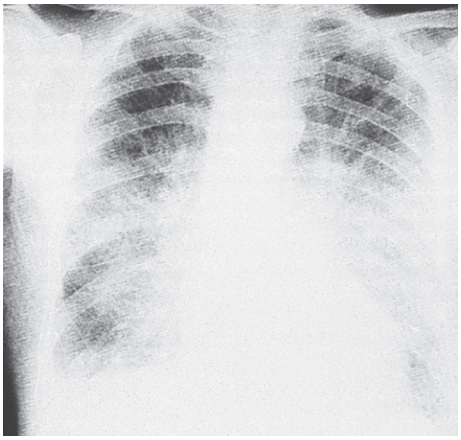


**Figure 13.35** Chest X-ray in acute left ventricular failure: the patient had severe pulmonary oedema caused by acute myocardial infarction. The heart is not yet enlarged, but there is prominent alveolar pulmonary oedema in a perihilar ('bat's-wing') distribution. Note the bilateral pleural effusions.

### Lung fields

Common lung field abnormalities in cardiovascular disease are caused either by altered pulmonary flow or by increased left atrial pressure.

### Altered pulmonary flow

Increments in pulmonary flow sufficient to cause radiographic abnormalities are caused by left-to-right intracardiac shunts (e.g. atrial septal defect; see Fig. 13.34, ventricular septal defect, patent ductus arteriosus). Prominence of the vascular markings gives the lung fields a plethoric appearance. Reductions in pulmonary flow, on the other hand, cause reduced vascular markings. This may be regional (e.g. pulmonary embolism) or global (e.g. severe pulmonary hypertension).

### Increased left atrial pressure

Increased left atrial pressure, which occurs in mitral stenosis and left ventricular failure, produces corresponding rises in pulmonary venous and pulmonary capillary pressures. Prominence of the upper lobe veins is an early radiographic finding. As the left atrial and pulmonary capillary pressures rise above 18 mmHg, transudation into the lung produces interstitial pulmonary oedema, characterized by prominence of the interlobular septa, particularly at the lung bases (Kerley B lines). Further elevation of pressure leads to alveolar pulmonary oedema, characterized by perihilar 'bat's-wing' shadowing (Fig. 13.35).

### Other lung field abnormalities

#### *Pulmonary infarction*

Localized and typically wedge-shaped areas of consolidation are occasionally seen in pulmonary embolic disease, although more often the bronchial circulation protects against ischaemic damage.

#### *Pneumonic consolidation and abscess*

In patients with right-sided endocarditis, infected pulmonary emboli commonly cause septic foci within the lung fields.

#### *Interstitial lung disease*

In longstanding pulmonary hypertension complicating rheumatic mitral valve disease, hemosiderosis (stippled shadowing throughout the lung fields) was once a common X-ray finding. It is now rarely seen.

## Bony abnormalities

Bony abnormalities are unusual in cardiovascular disease, apart from coarctation of the aorta and thoracic outlet syndromes. In coarctation, dilated bronchial collateral vessels erode the inferior aspect of the ribs to produce notches, although they are rarely present before adolescence. Cervical ribs may compress the neurovascular bundle in the thoracic outlet, and special thoracic outlet views are necessary for radiographic diagnosis.

## Echocardiography

Echocardiography is one of the most versatile non-invasive imaging techniques used in clinical cardiology. Because it does not use ionizing radiation, it is free of risk and can be employed safely throughout pregnancy. Transthoracic imaging with the transducer applied to the chest wall is usually satisfactory, but better quality information is obtained via the transoesophageal approach in which the transducer is mounted on a probe and positioned

in the oesophagus, directly behind the heart. This provides higher resolution images because there are no intervening ribs or lung tissue and the probe is closely applied to the posterior aspect of the heart. It is particularly useful for imaging the left atrium, the aorta, the interatrial septum and prosthetic heart valves.

## Principles

### Physics

A transducer containing a piezoelectric element converts electrical energy into an ultrasound beam that is directed towards the heart. The beam is reflected when it strikes an interface between tissues of different densities. The reflected ultrasound, or echo, is converted back to electrical energy by the piezoelectric element, which permits the construction of an image using two basic units of information:

1. The intensity of the echoes, which defines the density difference at tissue interfaces within the heart.
2. The time taken for echoes to arrive back at the transducer, which defines the distance of the cardiac structures from the transducer.

Density differences within the heart are greatest between the blood-filled chambers and the myocardial and valvular tissues, all of which are clearly visible on the echocardiogram. Because the depth of the myocardial and valvular tissues with respect to the transducer changes constantly throughout the cardiac cycle, the time taken for echo reflection changes accordingly. Thus, real-time imaging throughout the cardiac cycle provides a dynamic record of cardiac function.

### M-mode echocardiogram

The M-mode echocardiogram provides a unidimensional view through the heart. Continuous recording on photographic paper provides an additional time dimension, thereby permitting appreciation of the dynamic component of the cardiac image. By convention, cardiac structures closest to the transducer are displayed at the top of the record and more distant structures are displayed below. Thus, on the transthoracic M-mode echocardiogram, anteriorly located ('right-sided') structures lie above the posteriorly located ('left-sided') structures, but on the transoesophageal echocardiogram, the display is reversed. M-mode is particularly useful for measuring chamber dimensions and left ventricular wall thickness, and for timing events within the cardiac cycle (Fig. 13.36A).

### Two-dimensional (2D) echocardiogram

A two-dimensional echocardiogram provides more detailed information about morphology than does the M-mode recording. By projecting a fan of echoes in an arc of up to 80°, a 2D 'slice' through the heart can be obtained, the precise view depending on the location and angulation of the transducer (Fig. 13.36B).

## Clinical applications

### Congenital heart disease

Echocardiography, particularly the 2D technique, has revolutionized the diagnosis of congenital heart disease, in most cases, obviating the need for invasive investigation by cardiac catheterization (Fig. 13.37). The relationships of the cardiac chambers and their connections with the great vessels are readily determined. Valvular abnormalities and septal defects can also be recognized. Recent technology has permitted in utero fetal imaging for the antenatal diagnosis of cardiac defects.

### Myocardial disease

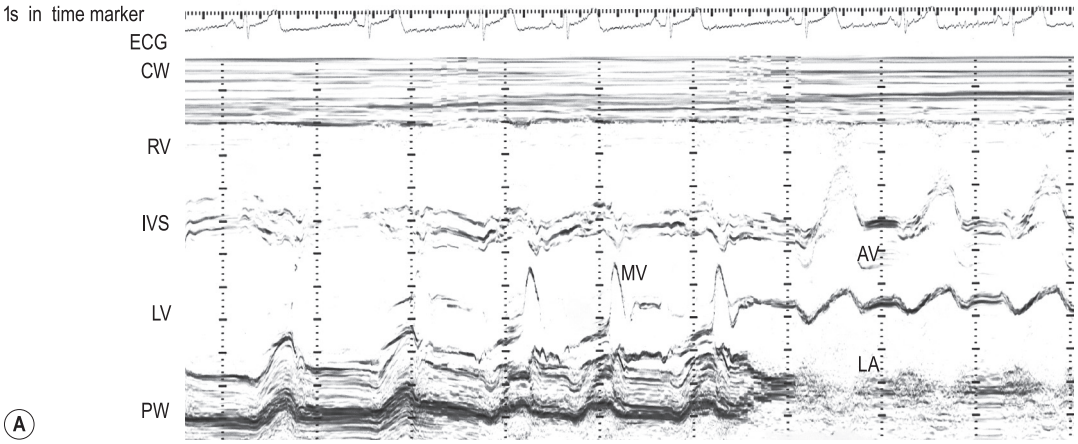
Echocardiography permits accurate assessment of cardiac dilatation, hypertrophy and contractile function. Dilated cardiomyopathy produces ventricular dilatation with global contractile impairment (Fig. 13.38). This must be distinguished from the regional contractile impairment that follows myocardial infarction in patients with coronary artery disease (Fig. 13.39). Hypertrophic cardiomyopathy is characterized by thickening (hypertrophy) of the left ventricular myocardium, usually with disproportionate involvement of the interventricular septum (asymmetric septal hypertrophy). In aortic and hypertensive heart disease, on the other hand, left ventricular hypertrophy is usually symmetrical (Fig. 13.40).

### Valvular disease

Echocardiography is of particular value for identifying both structural and dynamic valvular abnormalities and any associated chamber dilatation or hypertrophy (Boxes 13.18–13.21). The severity of valvular involvement in congenital, rheumatic, degenerative and infective disease may thus be defined; the technique is diagnostic for bicuspid aortic valve and mitral valve prolapse, and readily identifies valve thickening and calcification in rheumatic and calcific disease (Figs 13.40 and 13.41). Vegetations in infective endocarditis usually can be visualized if they are large enough (>3 mm, Fig. 13.42). The transoesophageal approach is usually necessary for endocarditis involving prosthetic heart valves.

### Pericardial disease

Although the echocardiogram is of little value in constrictive pericarditis, it is the most sensitive technique available for the diagnosis of pericardial effusion (Fig. 13.43). The effusion appears as an echo-free space distributed around the ventricles,



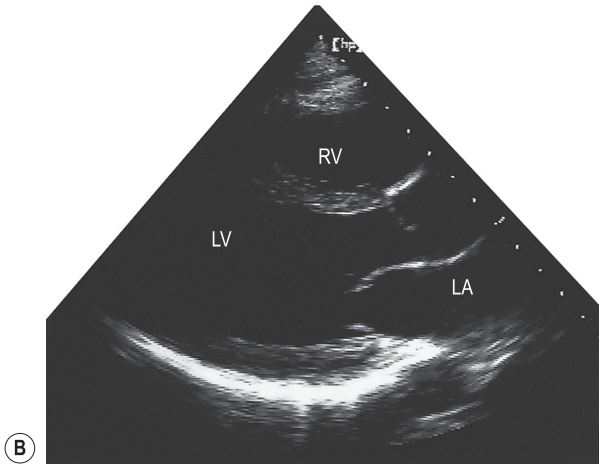




**Figure 13.36** **(A)** M-mode echocardiography. The figure shows a sweep as the transducer is angulated from the left ventricle to the aortic root. **(B)** Transthoracic 2D echocardiography. Parasternal long-axis and apical four-chamber views are shown. The dots are a 1-cm scale. AV, aortic valve; CW, chest wall; IVS, interventricular septum; LA, left atrium; LV, left ventricle; MV, mitral valve; PW, posterior LV wall; RV, right ventricle.

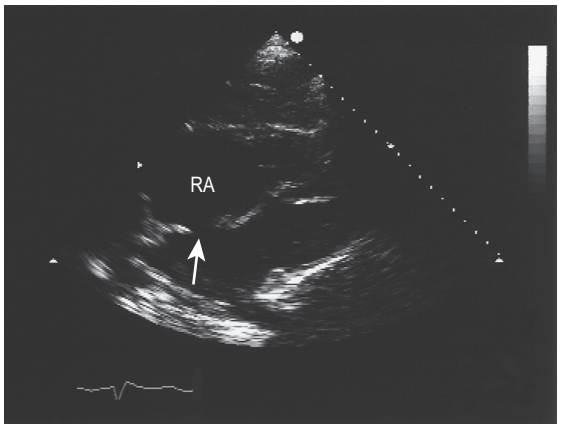



**Figure 13.37** Atrial septal defect: 2D echocardiogram (subcostal view). In this view, good views of the interatrial septum can usually be obtained without resorting to transoesophageal echocardiography. Note the atrial septal defect (*arrow*) and the dilatation of the right atrium (RA).

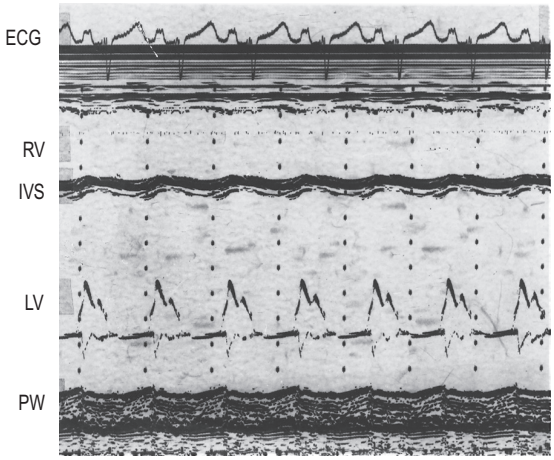



**Figure 13.38** Dilated cardiomyopathy: echocardiogram. This M-mode study shows severe dilatation of the left ventricular cavity and severe global contractile impairment. The patient later underwent successful heart transplantation.

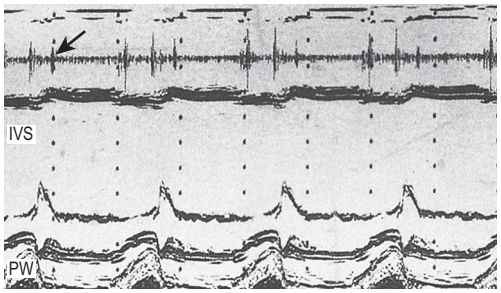



**Figure 13.39** Heart failure: echocardiogram. This M-mode study shows considerable dilatation of the left ventricle. Note that the interventricular septum (IVS) is almost akinetic, but the posterior wall (PW) is contracting normally. Regional contractile impairment of this type indicates coronary heart disease. The phonocardiogram recorded simultaneously shows normal first and second heart sounds and also a third heart sound (*arrow*).

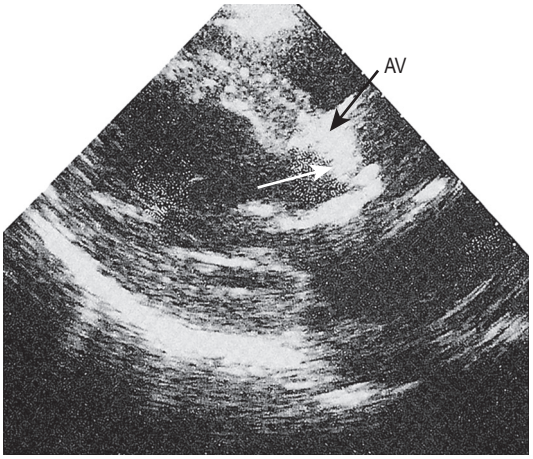



**Figure 13.40** Aortic stenosis and left ventricular hypertrophy: 2D echocardiogram (long-axis view). The aortic valve (AV) is grossly thickened and calcified. Concentric left ventricular hypertrophy is present (*arrow*).

**Box 13.18** Aortic stenosis

**Typical patient**

- Middle-aged (congenitally bicuspid valve) or elderly (degenerative calcific disease) man or woman

**Major symptoms**

- Exertional shortness of breath is usual presenting symptom.
- Angina may also occur and, in advanced cases, syncopal attacks or sudden death.

**Major signs**

- Carotid pulse: slow upstroke with plateau
- Auscultation: fourth heart sound at cardiac apex; ejection systolic murmur at base of heart, radiating to neck. The murmur may be preceded by an ejection click if the valve is mobile and not heavily calcified.

**Diagnosis**

- ECG: left ventricular hypertrophy
- Chest X-ray: dilatation of ascending aorta
- Echocardiogram: calcified immobile aortic valve with left ventricular hypertrophy. Doppler studies permit quantification of the severity of stenosis.

**Additional investigations**

- Cardiac catheterization is necessary to evaluate coronary arteries in patients being considered for aortic valve replacement surgery.

**Comments**

- Aortic stenosis is now the most commonly acquired valve lesion in developed countries.
- The use of percutaneous implanted aortic valves (TAVI) has significantly increased the number of patients with aortic stenosis who can now be treated.

**Box 13.19** Aortic regurgitation

**Typical patient**

- Young men (Marfan's syndrome, etc.) or older patients (longstanding hypertension) with dilating disease of the aortic root

**Major symptoms**

- Exertional shortness of breath is usual presenting symptom.
- Angina may also occur.

**Major signs**

- Carotid pulse: sharp upstroke with early diastolic collapse
- Blood pressure: systolic hypertension with wide pulse pressure
- Auscultation: early diastolic murmur at left sternal edge. Third heart sound at cardiac apex in severe cases. Mid-diastolic murmur (Austin Flint) may be heard at apex owing to preclosure of mitral valve by regurgitant jet.

**Diagnosis**

- ECG: left ventricular hypertrophy
- Chest X-ray: cardiac enlargement with dilatation of ascending aorta
- Echocardiogram: often normal valve with dilated aortic root. Doppler studies confirm regurgitant jet.

**Additional investigations**

- Cardiac catheterization is necessary to evaluate coronary arteries in older patients (age >50) being considered for aortic valve replacement surgery.

**Comments**

- The timing of valve replacement surgery is difficult, but should anticipate irreversible left ventricular contractile failure.

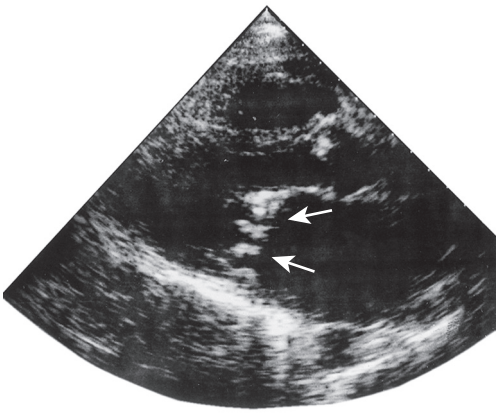


**Figure 13.41** Mitral stenosis: 2D echocardiogram (parasternal long-axis view). The mitral valve leaflets are densely thickened (*arrows*) and the left atrium is severely dilated.

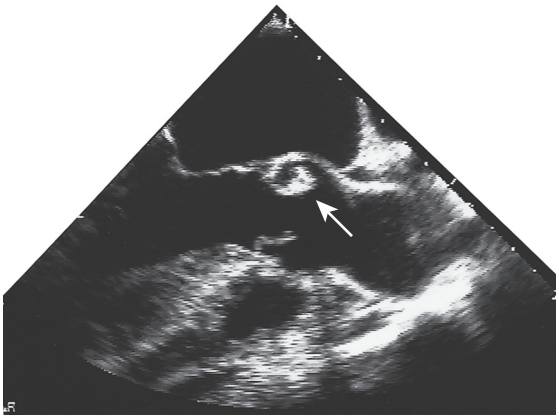


**Figure 13.42** Infective endocarditis: transoesophageal echocardiogram. A vegetation (*arrow*) is adherent to the aortic valve leaflet.

**Box 13.20** Mitral stenosis

**Typical patient**

- Young to middle-aged woman with a history of rheumatic fever in childhood.

**Major symptoms**

- Exertional shortness of breath with orthopnoea in advanced cases.
- Palpitations commonly signal the development of atrial fibrillation, which puts the patient at serious risk of peripheral embolism and stroke.

**Major signs**

- Pulse: atrial fibrillation in many cases.
- Auscultation: loud S1 with early diastolic opening snap, followed by low-pitched mid-diastolic murmur best heard at cardiac apex. If the patient is in sinus rhythm, there is presystolic accentuation of the murmur.

**Diagnosis**

- ECG: atrial fibrillation usually.
- Chest X-ray: signs of left atrial enlargement (flat left heart border, widening of carinal angle and double-density sign at right heart border). Pulmonary congestion
- Echocardiogram: rheumatic mitral valve and left atrial dilatation. Doppler studies permit quantification of the severity of stenosis.

**Additional investigations**

- Cardiac catheterization is necessary to evaluate coronary arteries in patients aged >50 being considered for mitral valve replacement surgery.

**Comments**

- In patients with atrial fibrillation, anticoagulation with warfarin is mandatory to protect against stroke.

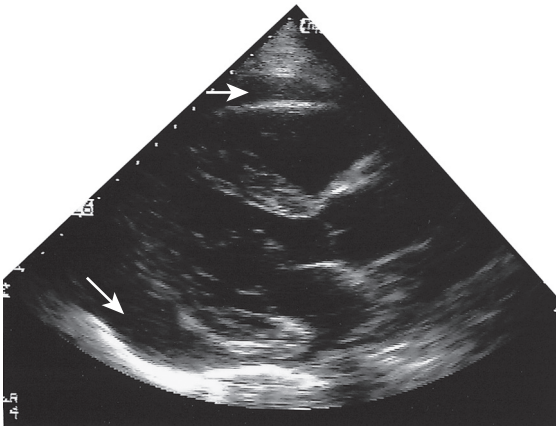


**Figure 13.43** Pericardial effusion: 2D echocardiogram (parasternal long-axis view). Note the echo-free space (*arrows*) around the heart, but not behind the left atrium.

but usually avoiding the potential space behind the left atrium.

### Other clinical applications

Intracardiac tumours, particularly myxomas (see Fig. 13.1) and thrombi (Fig. 13.44), are readily visualized by echocardiography. The transoesophageal technique is more sensitive for identifying thrombus in the left atrial appendage and is also helpful for diagnosing aortic disease, such as aneurysm and dissection, because it provides better views of the thoracic aorta than is possible with conventional 2D echocardiography (Fig. 13.45).

## Stress echocardiography

Stress echocardiography is increasingly being used for the diagnosis of myocardial ischaemia in suspected coronary disease. Left ventricular imaging during increasing dobutamine infusion or exercise

**Box 13.21** Mitral regurgitation

**Typical patient**

- Mitral valve prolapse (floppy mitral valve) causes regurgitation of variable severity, and more commonly affects women at almost any age.
- Patients with subvalvular disease (papillary muscle dysfunction or chordal rupture) are usually elderly men or women.

**Major symptoms**

- Exertional shortness of breath with orthopnoea in advanced cases.
- Palpitations commonly signal the development of atrial fibrillation, which puts the patient at serious risk of peripheral embolism and stroke.

**Major signs**

- Pulse: often sinus rhythm, but may be atrial fibrillation
- Auscultation: pansystolic murmur at cardiac apex, radiating to axilla. Often associated with third heart sound.

**Diagnosis**

- ECG: atrial fibrillation, but may be normal.
- Chest X-ray: cardiac enlargement with variable signs of left atrial enlargement, though these are usually less marked than in mitral stenosis. Pulmonary congestion in severe cases.
- Echocardiogram: prolapsing (floppy) mitral valve may be seen; in subvalvular disease the valve often appears normal. Left ventricular and left atrial dilatation. Doppler studies confirm regurgitant jet.

**Additional investigations**

- Transoesophageal echo is sometimes necessary to clarify the severity and mechanism of regurgitation.
- Cardiac catheterization is necessary to evaluate coronary arteries in patients aged >50 being considered for mitral valve replacement surgery.

**Comments**

- In patients with atrial fibrillation, anticoagulation with warfarin is mandatory to protect against stroke.

permits assessment of regional wall motion in response to stress. Decreasing systolic wall motion or wall thickening indicates ischaemia and the need for further investigation. Stress echocardiography is also used to identify myocardial viability in patients with impaired cardiac function. Improvement in regional hypokinesia in response to dobutamine may indicate 'hibernating' (and hence potentially salvageable) myocardium likely to respond favourably to revascularization by angioplasty or bypass surgery.

## Doppler echocardiography

Doppler echocardiography permits evaluation of the direction and velocity of blood flow within the heart and great vessels. It is widely used for measuring the severity of valvular stenosis and identifying valvular regurgitation and intracardiac shunts through septal defects.

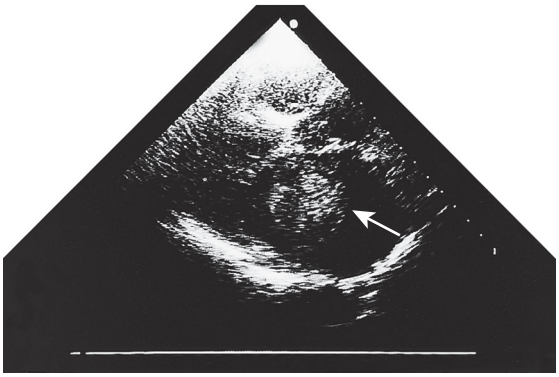


**Figure 13.44** Mitral stenosis: 2D echocardiogram (long-axis view). The left atrium is severely dilated and a large thrombus (*arrow*) is visible, emphasizing the importance of anticoagulation in patients with mitral valve disease and atrial fibrillation.

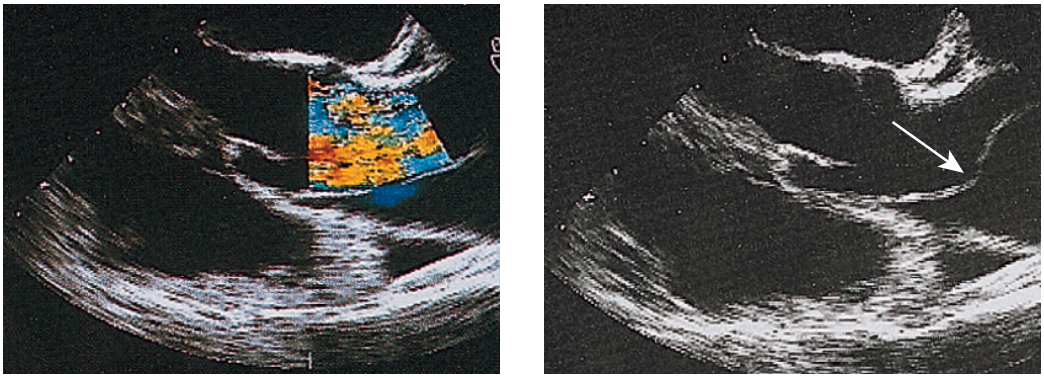


**Figure 13.45** Aortic dissection: transoesophageal echocardiogram. Right panel: this long-axis transoesophageal echocardiogram reveals the dilated aortic root and an S-shaped flap (*arrow*) traversing the lumen. Left panel: same patient with colour Doppler superimposed to show flow in the true lumen.

## Principles

### Physics

According to the Doppler principle, when an ultrasound beam is directed towards the bloodstream, the frequency of the sound waves reflected from the blood cells is altered. The frequency shift or Doppler effect is related to the direction and velocity of flow. If continuous-wave Doppler is used, blood flow at any point along the path of the ultrasound beam is detected, such that a 'clean' Doppler signal from the area of interest may be difficult to obtain. Pulsed Doppler, however, has a range-gating facility that permits frequency sampling from any specific point within the heart, preselected on the echocardiogram. This lends greater precision to the technique. Nevertheless, pulsed Doppler is less able than continuous-wave Doppler to quantify very high-velocity jets, such as those that occur in aortic stenosis.

### Colour-flow mapping

Colour-flow mapping has been a major technological advance. Instead of the unidirectional ultrasound beam used in continuous-wave and pulsed Doppler imaging, the beam is rotated through an arc. Frequency sampling throughout the arc permits the construction of a colour-coded map, red indicating flow towards and blue away from the transducer. Colour-flow data can be superimposed on the standard 2D echocardiogram to identify precisely the patterns of flow within the four chambers of the heart. This simplifies the interpretation of Doppler imaging and provides more useful qualitative data, although it is less useful for quantitative assessment of valve gradients, which requires the precision of conventional Doppler technique.

## Clinical applications

In paediatric cardiology, the combination of 2D echocardiography and colour-flow Doppler mapping has made possible the 'non-invasive' diagnosis of a large majority of congenital defects, often without the need for cardiac catheterization. These techniques have also revolutionized the diagnosis of valvular disease in all age groups. In valvular regurgitation, the retrograde flow that occurs after valve closure is readily detected by Doppler echocardiography, although only an approximate estimate of its severity is possible (Fig. 13.46). In valvular stenosis, the peak velocity (as opposed to the volume) of flow across the valve is directly related to the degree of stenosis. Thus, measurement of Doppler flow velocity (ideally by continuous wave) permits quantification of the stenosis by the application of the Bernoulli equation:

$$\text{pressure gradient} = 4 \times \text{velocity}^2$$

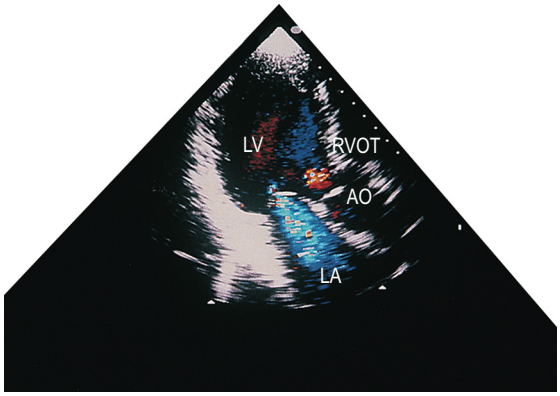



**Figure 13.46** Mitral regurgitation: colour-flow Doppler. This is an apical long-axis view of the heart showing a large jet of mitral regurgitation (*blue*) occupying most of the left atrial cavity (LA).

## New developments

In recent years, real time three-dimensional imaging has emerged as a powerful technique—particularly in the assessment of valvular disease and in guiding percutaneous interventions such as minimally invasive transcatheter aortic valve implantation (TAVI).

Doppler imaging techniques can also be applied to the myocardium itself (tissue Doppler) to provide measures of myocardial strain. Abnormalities in strain often predate clinical symptoms and have been used to identify and treat patients at risk of developing cardiac disease.

# Cardiovascular radionuclide imaging

## Principles

All radionuclide techniques require the internal administration of a radioisotope; the distribution of radioactivity in the area of interest is then imaged with a gamma camera. Ideally, the isotope should be distributed homogeneously in that part of the cardiovascular system under investigation: thus, isotopes that remain in the intravascular space during imaging are used for radionuclide angiography. In myocardial perfusion scintigraphy, however, isotopes taken up by the myocardium are required. Because of their potential toxicity, isotopes with a short half-life are usually used.

## Clinical applications

### Radionuclide ventriculography

Radionuclide ventriculography is used for assessment of ventricular function. Red cells labelled with technetium-99m ($^{99m}$Tc) are allowed to equilibrate in the blood pool and the heart is then imaged under the gamma camera. The waxing and waning of radioactivity within the ventricular chambers

during diastole and systole, respectively, permits the construction of a dynamic ventriculogram. Left ventricular contractile function can be evaluated quantitatively by calculating the ejection fraction, or qualitatively by observing wall movement. In clinical practice, this has now been replaced by other techniques, such as echocardiography and magnetic resonance imaging, for the assessment of left ventricular function.

## Myocardial perfusion scintigraphy

Myocardial perfusion scintigraphy is used for the diagnosis and assessment of coronary artery disease (Fig. 13.47). The patient is 'stressed' in an attempt to provoke myocardial ischaemia, either by a standardized exercise test, an intravenous dobutamine infusion or an intravenous adenosine infusion. Isotope is injected intravenously at peak stress and the heart is imaged under a gamma camera. Thallium-201 ($^{201}$Tl) has now given way to $^{99m}$Tc-labelled methoxy-isobutyl- isonitrile (MIBI) which provides better image quality. Isotope is distributed homogeneously in normally perfused myocardium, ischaemic or infarcted areas appearing as scintigraphic defects. If $^{201}$Tl is used, repeat imaging after 2–4 hours' rest permits the reassessment of scintigraphic defects; those that disappear (reversible defects) indicate areas of stress-induced ischaemia, whereas those that persist (fixed defects) indicate infarcted myocardium. If $^{99m}$Tc-labelled MIBI is used, resting images for assessment of reversibility require a separate injection of isotope 24 hours after (or before) the stress images.

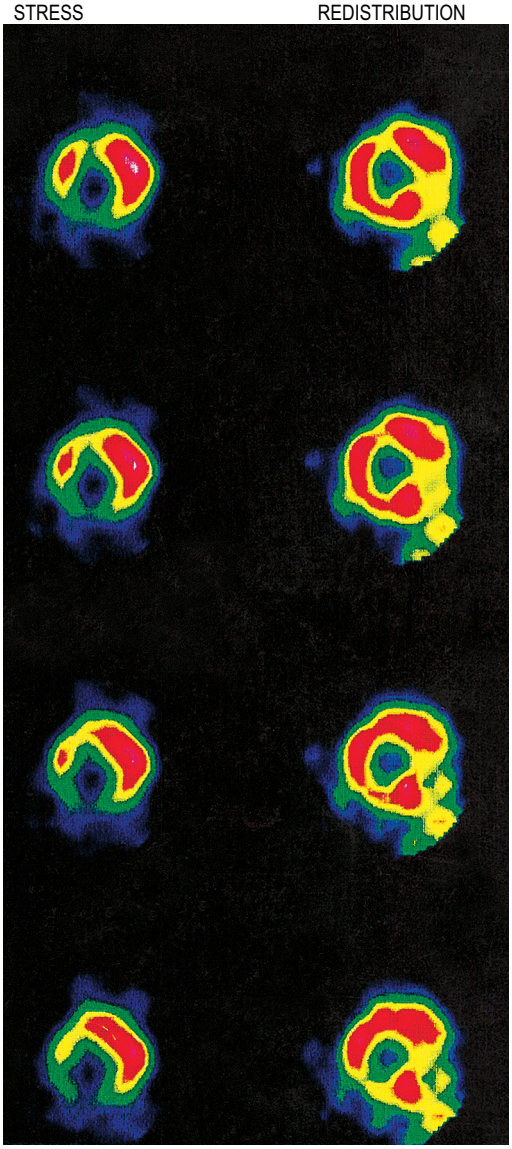



**Figure 13.47** Isotope perfusion scan. These are tomographic slices across the short axis of the left ventricle. A posterior wall defect is seen during stress, but it largely disappears during rest as isotope 'redistributes' into the ischaemic area. A smaller fixed defect is seen in the anterior wall, indicating infarction in that territory.

## Positron emission tomography

Positron emission tomography (PET) scanning is used to determine myocardial 'viability' in patients with heart failure (see stress echocardiography, above). Simultaneous assessment of myocardial perfusion using $^{13}$N ammonia and glucose uptake using a glucose analogue permits the identification of viable but dysfunctional myocardium, in which perfusion is impaired but metabolic activity in terms of glucose uptake remains normal. This perfusion-metabolic 'mismatch' indicates viable muscle likely to respond favourably to revascularization by angioplasty or bypass surgery.

New PET tracers, such as rubidium, can be used to assess myocardial perfusion. A specialist cyclotron is not required for this tracer, reducing the costs involved.

## Pulmonary scintigraphy (radioisotope imaging)

Pulmonary scintigraphy is used for the diagnosis of pulmonary embolism (Fig. 13.48). $^{99m}$Tc-labelled microspheres injected intravenously become trapped within the pulmonary capillaries. The normal pulmonary perfusion scintigram shows a homogeneous distribution of radioactivity throughout both lung fields. Pulmonary embolism causes regional impairment of pulmonary flow, which results in a perfusion defect on the scintigram; however, the appearance is non-specific and occurs in many other pulmonary disorders, particularly chronic obstructive pulmonary disease. Specificity is enhanced by simultaneous ventilation scintigraphy (a ventilation/perfusion or V/Q scan). Inhaled xenon-133 ($^{133}$Xe) is distributed homogeneously throughout the normal lung, and in pulmonary embolism (unlike other pulmonary disorders), distribution remains homogeneous. Thus, a scintigraphic perfusion defect not 'matched' by a ventilation defect is highly specific for pulmonary embolism. CT pulmonary angiography is now the

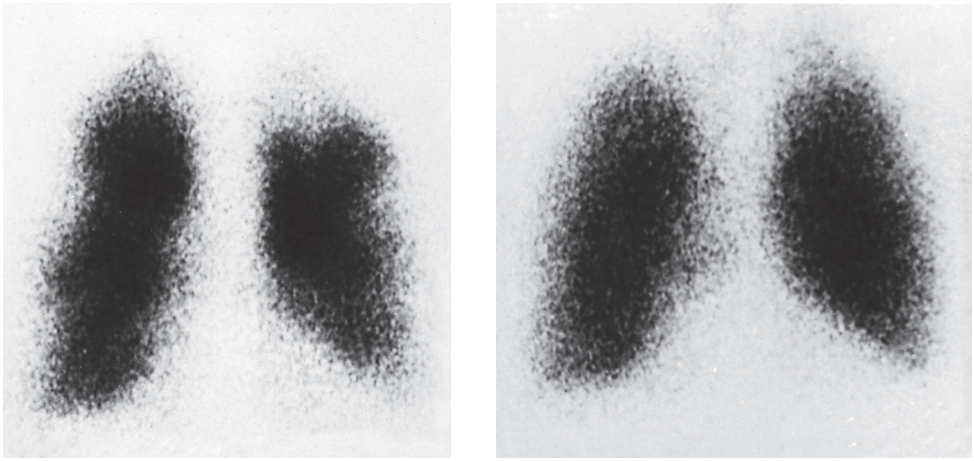


**Figure 13.48** Ventilation (right) and perfusion (left) lung scans in pulmonary embolism. Contrast the homogeneous distribution of isotope in the ventilation scan with the regional defects in the perfusion scan.

diagnostic test of choice for pulmonary embolism, but V/Q scanning is still valuable in patients with suspected pulmonary embolism who have severe renal impairment in whom X-ray contrast is contraindicated.

## Computed tomography

### Principles

Computed tomography measures the attenuation of X-rays after they have traversed body tissues. Attenuation is greatest for tissues such as bone, which are relatively radio-opaque, and least for tissues such as lung or fat, which are relatively radiolucent. From X-ray attenuation measurements, taken as a sensor rotates around the chest, cross-sectional images are constructed. Image resolution is excellent, and contrast injection into a peripheral vein opacifies the blood pool allowing assessment of the coronary arteries and intracardiac structures. In the past, the clinical application of cardiac CT was limited by image acquisition times of up to 5 seconds, during which the constant motion of the heart degraded the image, and by high radiation doses. The current generation of ultrafast CT scanners with image acquisition times of less than 1 second provides high-resolution cardiac images in both static and video mode, whereas radiation doses have reduced substantially.

### Clinical applications

Computed tomography (with contrast enhancement) diagnoses pulmonary embolism and aortic dissection (Fig. 13.49) with a sensitivity of about 95%. It is also used for accurate assessment of pericardial thickness in constrictive disease (Fig. 13.50) and in the diagnosis of cardiac tumours. Recently, multislice CT has emerged as a useful investigative tool for the diagnosis of coronary artery disease. The quantification of coronary calcification has found application as a marker of coronary

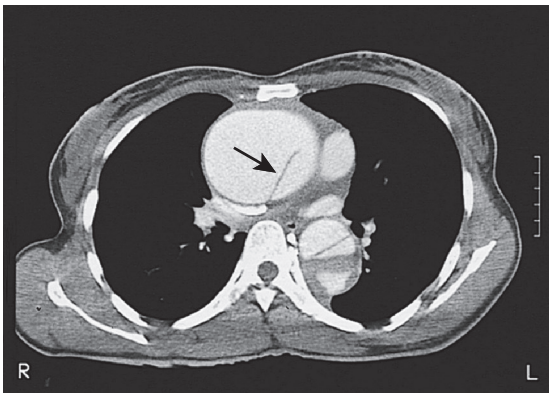



**Figure 13.49** Aortic dissection: CT scan (Marfan's syndrome). The ascending aorta is dilated and the intimal flap (*arrow*) clearly visible. This flap extends around the arch (not seen here) into the descending aorta, where again it is clearly visible.

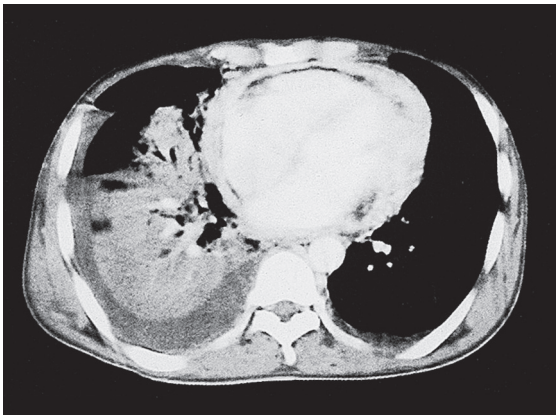


**Figure 13.50** Non-calcific pericardial constriction: CT scan. There is consolidation in the right lung and severe pericardial thickening. The patient had pulmonary tuberculosis with pericardial involvement and presented with fever and signs of constriction. Antituberculous therapy caused regression of all symptoms and signs, although the patient is at major risk of developing constriction later as the pericardium becomes fibrotic and calcified.

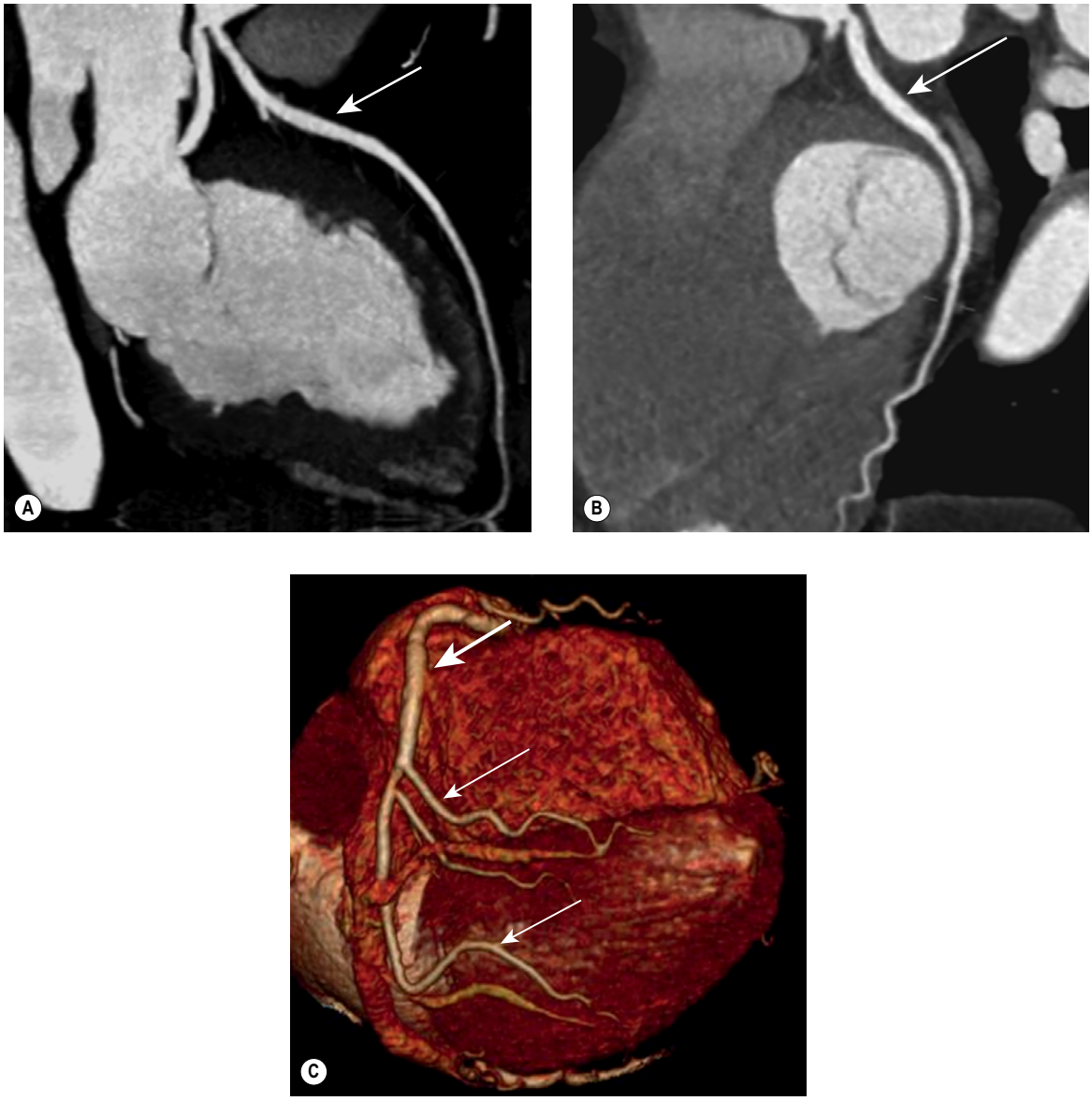



**Figure 13.51** CT coronary angiogram: normal study in a 42-year-old man with atypical chest pain and a family history of premature coronary artery disease. **(A)** Multiplanar reformatted image of the left anterior descending artery (*arrow*). **(B)** Multiplanar reformatted image of the circumflex artery (*arrow*). **(C)** 3D volume rendered image of the right coronary artery (*large arrow*), its posterior descending branch (*upper small arrow*) and posterior left ventricular branch (*lower small arrow*).

risk in asymptomatic or low-risk individuals; a zero calcium score has a high accuracy rate for the exclusion of coronary artery disease. CT coronary angiography is able to diagnose normal coronary arteries or mild plaque disease extremely accurately (Fig. 13.51), but it cannot distinguish reliably between moderate and severe stenoses or determine the degree of stenosis in heavily calcified coronary segments. CT coronary angiography is therefore best used in the assessment of low- to medium-risk patients to 'rule out' significant obstructive coronary artery disease. Additional applications of CT include the evaluation of graft patency following coronary bypass surgery and analysis of ventricular wall motion.

## Magnetic resonance imaging

### Principles

Magnetic resonance imaging (MRI) utilizes the fact that certain nuclei with an intrinsic spin generate magnetic fields and behave like tiny bar magnets. Placed in a magnetic field, these nuclei align and adopt a resonant frequency that is unique to that nucleus and the strength of the magnetic field. If the nuclei are exposed to pulsed radiowaves of that frequency, they resonate and release energy, which allows their location to be determined.

For imaging purposes, the patient lies in a strong magnetic field that is artificially graded. The hydrogen protons of fat and water are imaged and,

on exposure to pulsed radiowaves, they resonate at different frequencies in different parts of the imaging zone. Analysis of the emitted frequencies permits the construction of tomographic and 3D images of the heart. If data acquisition is gated to a specific part of the cardiac cycle, motion artefact is eliminated and excellent image resolution can be obtained. Different scanning parameters and patterns of gadolinium enhancement after its intravenous administration allow precise myocardial tissue characterization and detection of myocardial ischaemia.

### Clinical applications

Cardiac MRI is now widely used for the assessment of cardiac structure and ventricular function. It can differentiate between myocardial infarction (Fig. 13.52), oedema, fibrosis and fat. MRI is well validated for the assessment of myocardial viability and ischaemia. Therefore, it is used in the diagnosis of coronary artery disease, myocarditis (Fig. 13.53) and cardiomyopathies. It is also used to guide therapy in patients with documented coronary artery disease, particularly those with chronic coronary occlusions or multivessel disease, the presence of viability or ischaemia, respectively, providing justification for revascularization. Cardiac MRI does not involve the use of ionizing radiation so it can be used safely for repeated follow up assessments. MRI provides an accurate assessment of valve regurgitant fraction and it is also highly sensitive for the diagnosis of aortic dissection (Fig. 13.54), intracardiac tumours and thrombi.

## Cardiac catheterization

Catheters introduced into an artery or vein may be directed into the left or right sides of the heart, respectively. Arterial access is gained percutaneously from the femoral or radial artery, or by surgical cutdown to the brachial artery. Venous access is usually from the femoral vein. Originally developed for diagnostic purposes, catheter techniques have now found widespread application in the interventional management of cardiovascular disease.

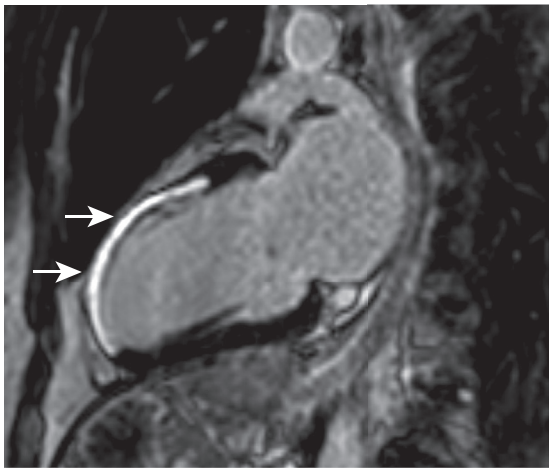


**Figure 13.52** Anteroapical myocardial infarction: MRI scan. Delayed gadolinium enhancement (*white arrows*) of the full thickness of the anteroapical left ventricular wall indicating infarcted, non-viable tissue.

## Cardiac angiography

Coronary angiography uses relatively small volumes of contrast (5–8 ml) injected manually, but other angiographic procedures require larger amounts (up to 40 ml), introduced by power injection. Digital subtraction techniques permit a reduction in contrast volume but at present have only a limited role in cardiovascular angiographic diagnosis (see below). The current generation of angiographic laboratories uses digital technology to provide high-quality dynamic images of ventricular wall movement, blood flow and intravascular anatomy.

### Aortic root angiography

Injection of contrast into the aortic root demonstrates the vascular anatomy in suspected aneurysm or dissection, and also permits evaluation of aortic valve function (Fig. 13.55). The normal aortic valve prevents diastolic backflow of contrast, but in aortic regurgitation, variable opacification of the left ventricle occurs, depending on the severity of the valve lesion.

### Left ventricular angiography

Contrast injection into the left ventricle defines ventricular anatomy and wall motion, and also permits evaluation of mitral valve function. Dilatation of the ventricle and contractile dysfunction occurs in left ventricular failure. An akinetic segment denotes previous myocardial infarction. Exaggerated contractile function with systolic obliteration of the cavity occurs in hypertrophic cardiomyopathy. Filling defects within the ventricular lumen may indicate thrombus or neoplasm. The normal mitral valve prevents systolic backflow of contrast into the left atrium, but in mitral regurgitation, variable atrial

**Box 13.22** Indications for coronary angiography

- ST elevation myocardial infarction
- High-risk non-ST elevation acute coronary syndromes
- Severe angina unresponsive to medical treatment
- Angina or a positive exercise test following myocardial infarction
- Cardiac arrhythmias with clinical suspicion of underlying coronary artery disease
- Preoperatively in patients requiring valve surgery when advanced age (>50) or angina suggest a high probability of coronary artery disease

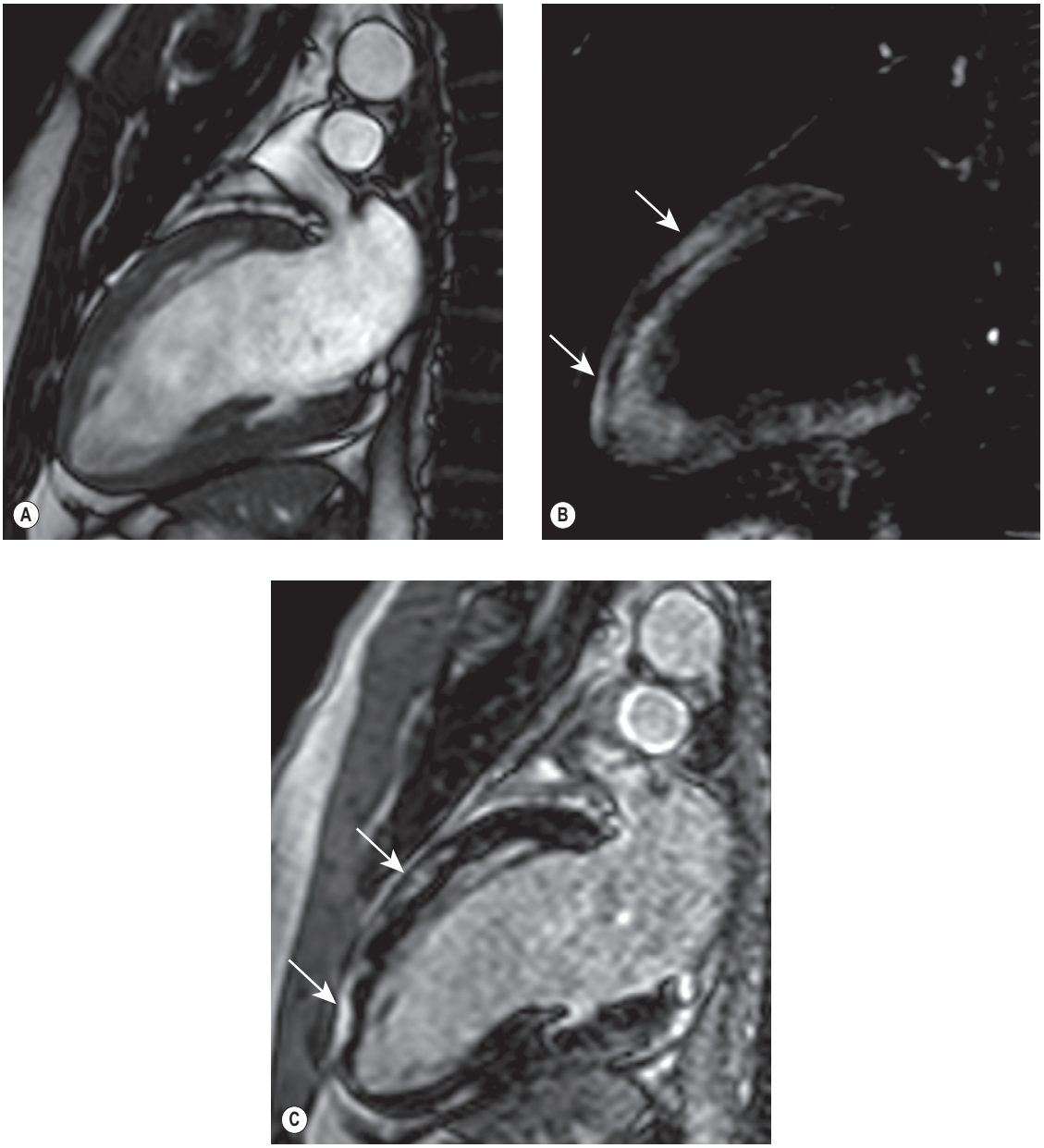



**Figure 13.53** Myocarditis: MRI scan. **(A)** Still image from a cine loop of a two-chamber view of the left ventricle. Measurements of systolic function, chamber size and myocardial mass can be made from the cine loops. **(B)** Two-chamber view from the same patient with imaging for water content (T2 weighted) showing regions of myocardial oedema (*arrows*). **(C)** Late enhancement imaging following gadolinium infusion allows detection of areas of increased interstitial expansion from scar or oedema (*arrows*). The subepicardial pattern indicates myocarditis.

opacification occurs, depending on the severity of the valve lesion (Fig. 13.56).

## Coronary angiography

Coronary angiography is the most reliable technique for diagnostic imaging of the coronary arteries and assessment of lesion severity. Indications are summarized in Box 13.22. The technique requires selective injection of contrast into the left and right coronary arteries to opacify the lumen (Fig. 13.57), and multiple views in different projections are necessary for a complete study. Intraluminal filling defects or occlusions indicate coronary artery disease, which is nearly always caused by atherosclerosis. In stable patients, coronary angiography often reveals a stenosis or stenoses of intermediate severity. Additional information regarding the functional significance and anatomic severity of the disease is required before treatment decisions can be made. This can be obtained either from non-invasive myocardial perfusion imaging that looks for ischaemia in the territory of the affected artery (see MRI perfusion imaging) or from catheter laboratory-based techniques.

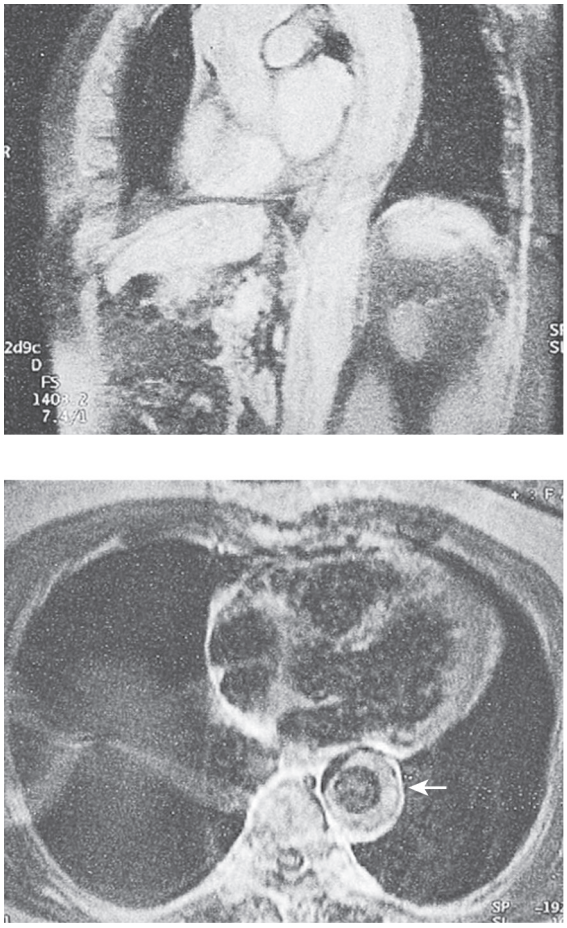


**Figure 13.54** Aortic dissection: MRI scan. The upper panel (coronal section) reveals an extensive aortic dissection extending from the aortic root through the arch and into the descending aorta. The lower panel reveals a transverse view through the heart and descending aorta. Note the thrombus in the false lumen surrounding the true lumen in the descending aorta (*arrow*).

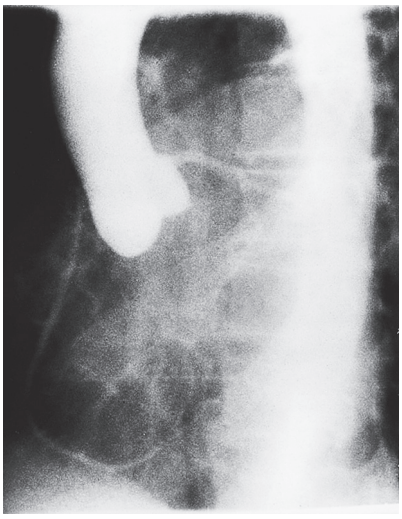


**Figure 13.55** Aortic root angiogram: normal study. Contrast injection provides an X-ray image of the ascending aorta. The coronary arteries arising from the sinuses of Valsalva are clearly seen.

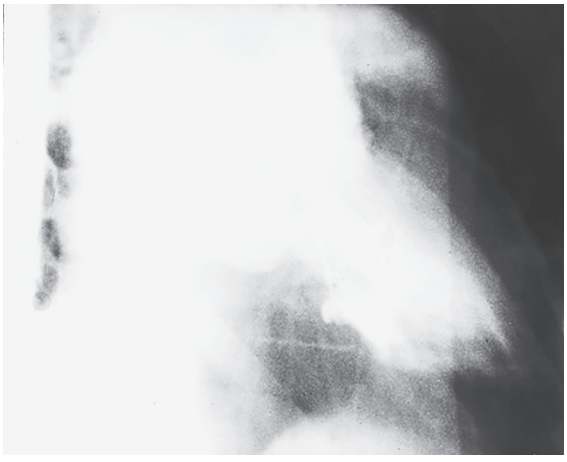


**Figure 13.56** Left ventricular angiogram showing mitral regurgitation. Contrast injection into the left ventricle has resulted in prompt opacification of the left atrium owing to backflow across the diseased mitral valve.

## Pressure wire

Pressure wire, a relatively new catheter laboratory-based technology, is increasingly used to assess functional significance and guide management of coronary stenoses. The wire has a pressure sensor 3 cm from its distal tip. The pressure sensor is calibrated and then equalized to the pressure at the guide catheter tip in the left main stem, before being passed down the artery so that the pressure sensor is distal to the stenosis to be interrogated. Adenosine is administered by intravenous infusion to induce maximal hyperaemia in the myocardial capillary bed and simulate the physiological state during exercise. The fractional flow reserve (FFR) is calculated automatically by computer software from the pressure measured distal to the stenosis referenced against the pressure measured proximal to the stenosis at the guide catheter tip. An FFR <0.75 signifies a stenosis that is likely to cause myocardial ischaemia and the stenosis is deemed functionally significant. Clinical studies have shown that pressure wire-guided management is superior to angiographic-driven decision-making, predominantly through avoiding procedure-related myocardial infarction in patients who undergo revascularization.

## Intravascular ultrasound

An ultrasound transducer mounted at the tip of a coronary catheter is used to provide cross-sectional images of the artery. The technique permits accurate measurement of the luminal area and provides information about plaque composition and structure that cannot be obtained from angiographic images (Fig. 13.58). Intravascular ultrasound has a clinical

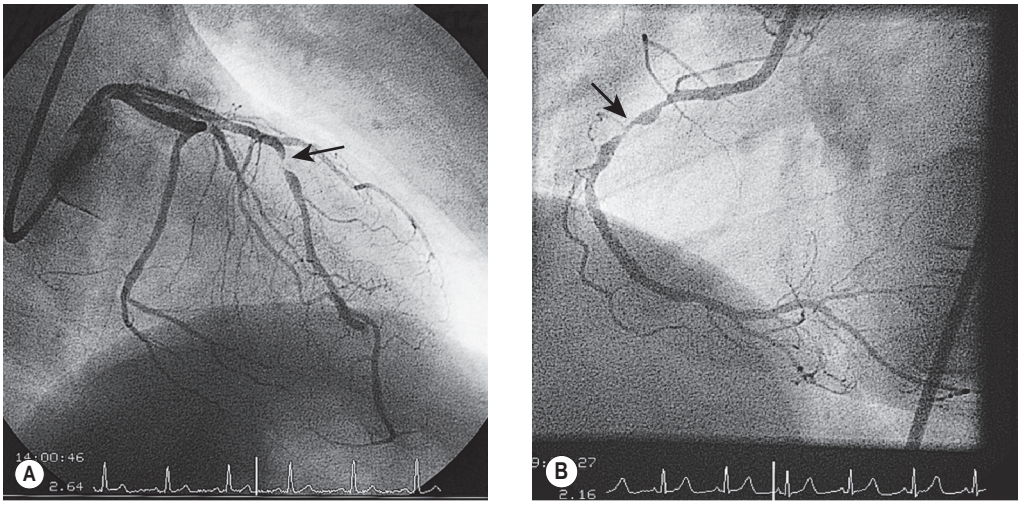



**Figure 13.57** Coronary angiograms. **(A)** Left anterior descending disease (*arrow*). This tight stenosis threatens the coronary supply to the anterior wall of the left ventricle. **(B)** Right coronary artery disease (*arrow*). Serial stenoses in this dominant right coronary artery threaten the supply to the inferior wall of the heart.

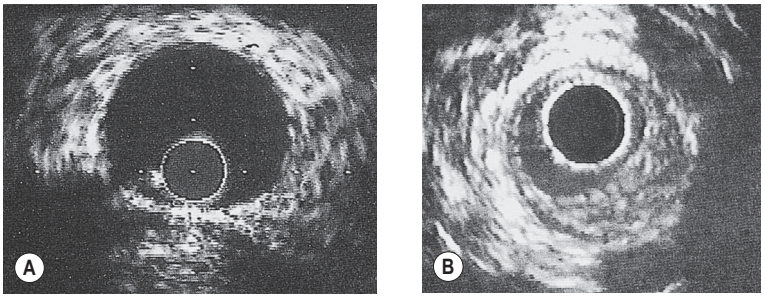



**Figure 13.58** Intravascular ultrasound. **(A)** Non-obstructive, calcified plaque between the 5 and 7 o'clock position in a distal segment of the right coronary artery. The ultrasound's transducer is clearly visible at the intersection of the horizontal and vertical centimetre scales. **(B)** Coronary artery disease: a large semilunar, non-calcified coronary plaque that produced an intermediate severity stenosis on coronary angiography is shown extending from the 1 o'clock to the 7 o'clock position, severely reducing the vessel lumen.

role in quantifying the severity of intermediate stenoses identified on coronary angiography, optimizing stent deployment, assessing in-stent restenosis and stent thrombosis and in clarifying the cause of equivocal angiographic appearances owing to possible thrombus, dissection, calcium or plaque.

## Pulmonary angiography

Injection of contrast medium into the main pulmonary artery opacifies the arterial branches throughout both lung fields. The normal flow distribution is homogeneous. Vascular occlusions with regional perfusion defects usually indicate pulmonary thromboembolism (particularly when intraluminal filling defects are present), but may also occur in advanced emphysema. The high diagnostic accuracy of CT pulmonary angiography means that invasive pulmonary angiography is now rarely performed.

## Intracardiac pressure measurement

Cardiac catheterization for measurement of blood flow and pressure within the heart and great vessels is widely used both for diagnostic purposes and to guide treatment. The fluid-filled catheter is attached to a pressure transducer, which converts the pressure waves into electrical signals. For measurement of right-sided pressures, the catheter is directed by the venous route into the right atrium and then advanced through the right ventricle into the pulmonary artery. For measurement of left-sided pressures, the catheter is directed by the arterial route into the ascending aorta and advanced retrogradely through the aortic valve into the left ventricle. Because access to the left atrium is technically difficult, left atrial pressure is usually measured indirectly using the pulmonary artery wedge pressure.

The pulmonary artery wedge pressure is obtained during right-heart catheterization by advancing the catheter distally into the pulmonary arterial tree until the tip wedges in a small branch. Alternatively, a catheter with a preterminal balloon (Swan-Ganz catheter) may be used. Inflation of the balloon in the pulmonary artery causes the catheter tip to be carried with blood flow into a more distal branch, which becomes occluded by the balloon. Regardless of which method is used, the wedge pressure recorded

at the catheter tip is a more or less accurate measure of the left atrial pressure transmitted retrogradely through the pulmonary veins and capillaries.

## Haemodynamic evaluation of valvular stenosis

In the normal heart, there is no pressure gradient across an open valve. Such a gradient usually indicates valvular stenosis (Fig. 13.59) and, as stenosis worsens, the pressure gradient increases. This therefore provides a useful index of the severity of stenosis. However, it must be recognized that the pressure gradient is influenced by the flow through the valve. For example, if cardiac output is very low, the gradient may be small despite the presence of

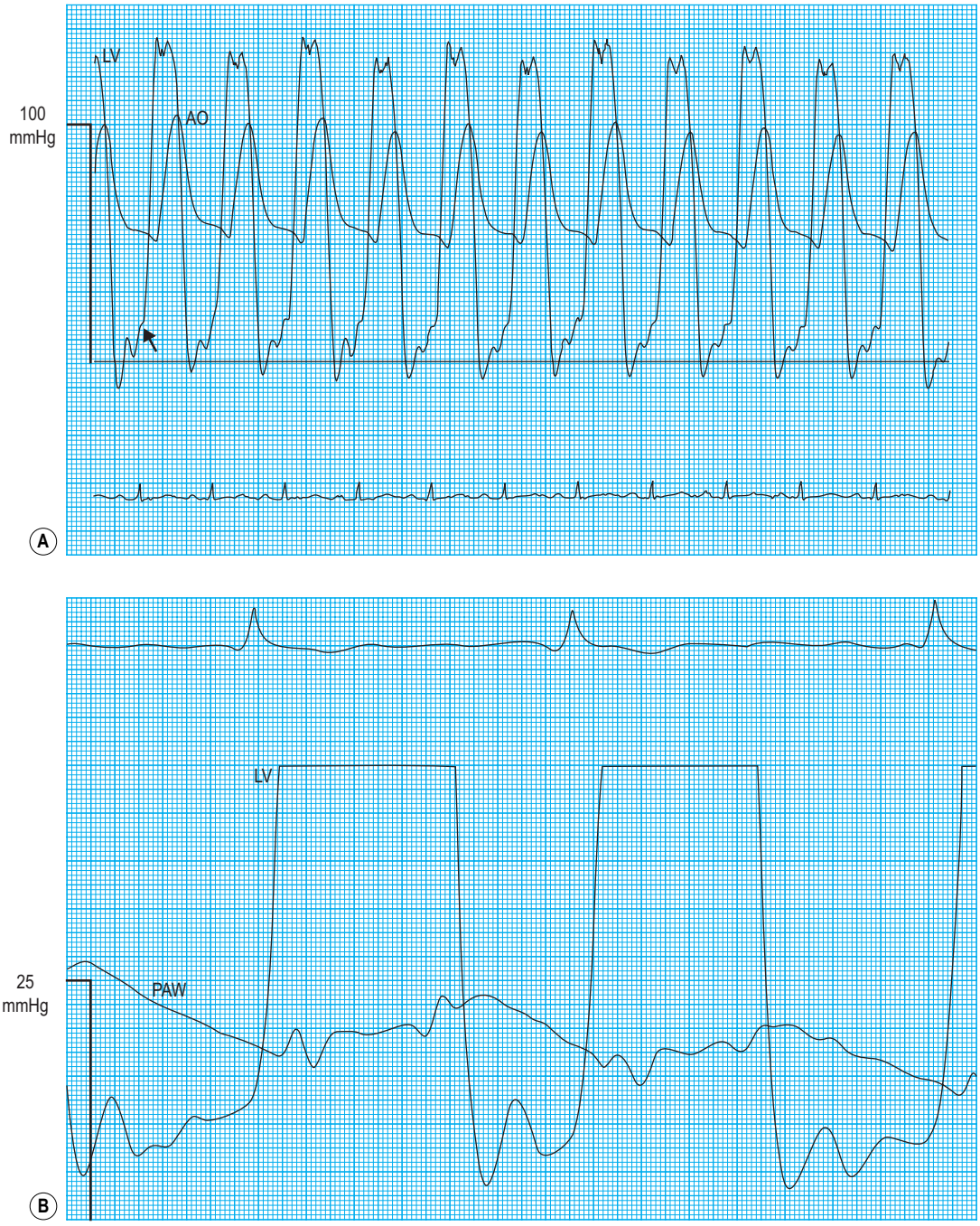



**Figure 13.59** Valvular stenosis: pressure signals. **(A)** Aortic stenosis: simultaneous left ventricular (LV) and aortic (AO) pressure signals. In the normal heart, the pressure signals essentially should be similar throughout systole. Here there is a peak systolic gradient of about 35 mmHg across the aortic valve. Note the prominent 'a' wave (*arrow*), reflecting the major contribution that atrial systole makes to filling of the hypertrophied, non-compliant ventricle. Note also the pulsus alternans indicating left ventricular failure. **(B)** Mitral stenosis: simultaneous recordings of the pulmonary artery wedge (PAW) and left ventricular (LV) pressure signals. In the normal heart, the pressure signals should be superimposed throughout diastole. Here there is a pressure gradient >10 mmHg, indicating severe mitral stenosis. Note that the patient is in atrial fibrillation and the pressure gradient varies inversely with the RR interval, tending to increase as the RR interval shortens.

severe stenosis. This applies particularly to the aortic valve because flow velocity is normally high. In patients with poor left ventricular function, the 2D appearances of the valve on echocardiography, the degree of calcification on CT and estimates of valve area should be considered before concluding that significant aortic stenosis is not present.

## Haemodynamic evaluation of intracardiac shunts

Left-to-right intracardiac shunts through atrial or ventricular septal defects introduce 'arterialized' blood into the right side of the heart. This results in an abrupt increase or step-up in the oxygen saturation of venous blood at the level of the shunt, which can be detected by right heart catheterization. Thus, by drawing serial blood samples for oxygen saturation from the pulmonary artery, right ventricle, right atrium and vena cava, the shunt may be localized to the site at which the step-up in oxygen saturation occurs. The magnitude of the step-up is related to the size of the shunt, but precise quantification of the shunt requires measurement of pulmonary and systemic blood flow. The extent to which the pulmonary-systemic flow ratio exceeds 1 is a measure of the size of the shunt.

## Haemodynamic evaluation of constriction and tamponade

In constrictive pericarditis and tamponade (Boxes 13.23 and 13.24), diastolic relaxation of the ventricles is impeded, preventing adequate filling. Compensatory increments in atrial pressures occur to help maintain ventricular filling, and because these disorders usually affect both ventricles equally, the filling pressures also equilibrate. Thus, simultaneous left- and right-sided recordings in constrictive pericarditis and tamponade show characteristic elevation and equalization of the filling pressures (atrial or ventricular end-diastolic), with loss of the normal differential (Fig. 13.60). Similar physiology characterizes restrictive cardiomyopathy, in which infiltrative disease (usually amyloid in the UK) impedes relaxation of the ventricles.

## Measurement of cardiac output

Measurement of cardiac output usually is calculated by thermodilution using a Swan-Ganz catheter with a right atrial portal and a terminal thermistor positioned in the pulmonary artery. A known volume of cold saline (usually 10 ml) is injected into the right atrium and the temperature reduction in the pulmonary artery is recorded at the thermistor. The contour of the cooling curve is dependent on cardiac output, which is calculated by measuring the area under the curve using a bedside computer.

Cardiac output can also be measured by a variety of non-invasive methods, including oesophageal Doppler probe, pulse waveform analysis, bioimpedance and echocardiography. Stroke volume can be calculated using 2D echocardiography from

**Box 13.23 Constriction**

**Typical patient**

- Middle-aged or elderly patient with a long history of progressive debilitation

**Major symptoms**

- Dyspnoea and weight loss with abdominal discomfort

**Major signs**

- Fluid retention with raised JVP (prominent 'x' and 'y' descents), hepatomegaly and peripheral oedema; diastolic filling sound ('pericardial knock'); paradoxical rise in JVP with inspiration (Kussmaul's sign)

**Diagnosis**

- High level of clinical suspicion
- CT or MRI scan: increased pericardial thickness (±calcification)
- Cardiac catheterization: equalization of diastolic pressures in the four cardiac chambers

**Additional investigations**

- Chest X-ray: often normal, but occasionally shows pericardial calcification on lateral film

**Comments**

- Tuberculosis no longer most common cause in developed countries, where most cases are idiopathic

**Box 13.24 Tamponade**

**Typical patient**

- Either middle-aged or elderly patient with malignant disease (usually breast or lung), or patients of any age with tuberculosis

**Major symptoms**

- Dyspnoea and variable circulatory collapse

**Major signs**

- Low-output state (hypotension, oliguria, cold periphery); tachycardia; paradoxical pulse; raised JVP with rapid 'x' descent; paradoxical rise in JVP on inspiration (Kussmaul's sign)

**Diagnosis**

- This is a clinical diagnosis, confirmed by echocardiographic demonstration of pericardial effusion.

**Additional investigations**

- ECG: low-voltage QRS complexes with alternating electrical axis
- Tests for aetiological diagnosis: these might include serological tests for rheumatoid and systemic lupus erythematosus, and tests for malignant or tuberculous disease.

**Comments**

- Any cause of pericardial effusion may produce tamponade, malignant disease being the most common cause in developed countries. Pericardiocentesis relieves the tamponade. Pericardial fluid should always be sent for cytological and bacteriological analysis.

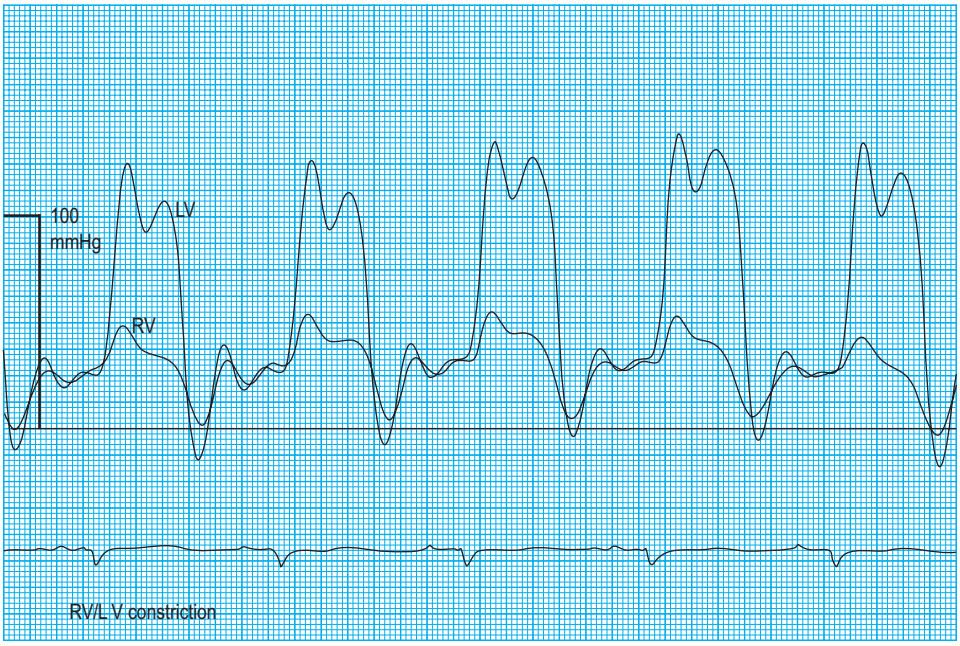



**Figure 13.60** Pericardial constriction: left (LV) and right (RV) ventricular pressure signals. Like tamponade and restrictive cardiomyopathy, constriction usually affects both ventricles equally and the diastolic pressures must rise and equilibrate to maintain ventricular filling. Thus, in diastole, the pressure signals are superimposed with a typical 'dip and plateau' configuration ('square root' sign).

the estimated difference in left ventricular volumes at the end of systole and diastole. Alternatively, Doppler echocardiography can be used to measure stroke volume.

Cardiac output measurement is performed most commonly in intensive care units to characterize cardiovascular status and monitor responses to treatment. It is particularly valuable in the shocked patient whose intravascular volume status cannot be determined by clinical assessment.

# Pathology laboratory support

## Haematology laboratory

Anaemia exacerbates angina by adversely affecting the myocardial oxygen supply-demand relationship, and also exacerbates heart failure by increasing the cardiac work necessary to meet the oxygen demands of metabolizing tissues. Rarely, anaemia is a consequence of, rather than a contributor to, heart disease. In infective endocarditis, a normochromic normocytic anaemia is almost invariable, reflecting the adverse effects of chronic illness on erythropoiesis. Chronic haemolysis occasionally occurs in patients with mechanical prosthetic heart valves and is caused by traumatic erythrocyte damage. Anaemia is usually low grade; if severe, it may require removal of the prosthesis and substitution with a xenograft.

## Biochemistry laboratory

### Cardiac enzymes and other markers of myocardial injury

Following myocardial infarction, enzymes and structural proteins released into the circulation by the necrosing myocytes provide biochemical markers of injury. In the UK, for many years, creatine kinase (CK) was the most widely used marker, the CK-MB isoenzyme being the most specific for myocardial injury. However, newer serum markers of myocardial injury, particularly the highly specific troponins T and I, have replaced enzymatic markers. Indeed, these form the basis of the updated definition of myocardial infarction, which is diagnosed in any patient with raised circulating troponins who presents with cardiac chest pain, or who develops regional ST elevation on the 12-lead ECG. Implicit in this definition is the fact that myocardial infarction may be diagnosed in patients without ST elevation. In non-ST elevation myocardial infarction (non-STEMI), the increase in circulating troponin concentration is a useful measure of risk, troponin concentration correlating directly with rates of recurrent myocardial infarction and death; it is widely used to guide decisions about cardiac catheterization and coronary revascularization.

Increased ventricular wall stress (which occurs in heart failure, left ventricular hypertrophy, etc.) causes release of natriuretic peptides from the

heart (brain natriuretic peptide—BNP and its precursor—N-terminal pro BNP), causing dilatation and natriuresis. High levels are associated with worsening symptoms and prognosis. They can be used to determine whether breathless patients need further cardiological investigations or need to be assessed for lung disease. Untreated patients with low levels of BNP are unlikely to have heart failure.

### Renal function

Renal function should always be measured in the cardiac patient. Renal dysfunction is a major cause (and consequence) of hypertension whereas, in heart failure, progressive deterioration of renal function is almost inevitable as perfusion of the kidneys becomes threatened. Renal function may also deteriorate in response to treatment with diuretics and angiotensin converting enzyme (ACE) inhibitors, and careful monitoring is essential as these drugs are introduced. Cardiac drugs that are excreted through the kidneys (e.g. digoxin) should be used cautiously if renal function is impaired. In the patient with established renal failure, accelerated coronary artery disease and heart failure commonly occur, reflecting the combined effects of dyslipidaemia, hypertension, arterial endothelial dysfunction, anaemia and volume loading on the cardiovascular system. Cardiovascular disease is the major cause of death in patients with renal failure.

### Electrolytes

Patients with hypokalaemia are at risk of lethal cardiac arrhythmias, whereas hyperkalaemia may cause bradyarrhythmias and heart block. Patients presenting with acute coronary syndromes commonly have hypokalaemia, owing to the effects of sympathoadrenal activation on membrane-bound sodium–potassium adenosine triphosphatase (ATPase), and this may require correction. In patients on thiazide or loop diuretics, potassium supplements (or potassium-sparing diuretics) are usually necessary to protect against hypokalaemia unless ACE inhibitors are also given. The combination of an ACE inhibitor and an aldosterone antagonist is used in patients with heart failure and can be complicated by hyperkalaemia. Serum potassium should always be measured in patients with hypertension, not only to provide a baseline before the introduction of diuretic therapy, but also as a simple screening test for primary aldosteronism. Hyponatraemia is common with chronic diuretic use.

### Glucose and lipids

Blood glucose and lipid profiles (triglycerides, total cholesterol and high-density lipoprotein (HDL) and low-density lipoprotein (LDL) cholesterol) should be measured in all patients with suspected vascular disease. In high-risk individuals, particularly those with diabetes, and in all patients with established coronary artery disease, statin therapy is essential regardless of the baseline lipid profile; it reduces the cardiovascular event rate by about 25%. Statins lower LDL ('bad') cholesterol, but if HDL ('good') cholesterol is low, treatment with nicotinic acid or fibrates should be added. Treatment of diabetes protects against microvascular complications, and in hyperglycaemic patients with acute myocardial infarction, infusion of insulin and glucose is usually recommended.

## Bacteriology laboratory

### Blood culture

In suspected infective endocarditis (Box 13.25), treatment must not be delayed beyond the time necessary to obtain three to four blood samples for culture (Table 13.3). Aerobic, anaerobic and fungal cultures should be performed. Occasionally, bone marrow cultures are helpful for detection of *Candida* and *Brucella* endocarditis. *Coxiella* and *Chlamydia* can never be cultured from the blood and must be diagnosed by serological tests. Failure to detect bacteraemia may be owing to pretreatment with antibiotics, inadequate sampling (up to six

**Box 13.25 Infective endocarditis**

**Typical patient**

- Elderly man or woman with mitral or aortic valve disease (often not previously recognized)
- Younger patients with congenital heart defects (usually ventricular septal defect or patent ductus arteriosus)
- Patient of any age with prosthetic heart valve or history of intravenous drug abuse

**Major symptoms**

- Non-specific feverish ('flu-like') illness

**Major signs**

- Fever and heart murmur (usually aortic or mitral regurgitation)
- Splinter haemorrhages and vasculitic rash may occur
- Clubbing, Osler's nodes and Roth's spots are rare.

**Diagnosis**

- Blood culture: usually provides bacteriological diagnosis
- Echocardiogram: usually reveals valvular regurgitation ± vegetation

**Additional investigations**

- Haematology: leukocytosis; normochromic, normocytic anaemia
- Inflammatory markers: raised erythrocyte sedimentation rate and C-reactive protein
- Urinalysis: haematuria

**Comments**

- Formerly a disease of young adults, now seen more commonly in the elderly
- Diagnosis should always be considered in patients with fever and a heart murmur.

**Table 13.3** Organisms implicated in endocarditis

| Organism | Typical source of infection | First choice of antibiotics (pending sensitivity studies) |
|---|---|---|
| *Streptococcus viridans* | Upper respiratory tract | Benzylpenicillin: gentamicin |
| *Streptococcus faecalis* | Bowel and urogenital tract | Ampicillin: gentamicin |
| *Anaerobic streptococcus* | Bowel | Ampicillin: gentamicin |
| *Staphylococcus epidermidis* | Skin | Flucloxacillin: gentamicin |
| Fungi: *Candida*, histoplasmosis | Skin and mucous membranes | Amphotericin B*: 5-fluorocytosine* |
| *Coxiella burnetii* | Complication of Q fever | Chloramphenicol*: tetracycline* |
| *Chlamydia psittaci* | Contact with infected birds | Tetracycline* and erythromycin |
| **Acute disease** | | |
| *Staphylococcus aureus* | Skin | Flucloxacillin: gentamicin |
| *Streptococcus pneumoniae* | Complication of pneumonia | Benzylpenicillin: gentamicin |
| *Neisseria gonorrhoeae* | Venereal | Benzylpenicillin: gentamicin |

*These drugs are not bactericidal, and valve replacement is nearly always necessary to eradicate infection.

blood samples should be taken over 24 hours) or infection with unusual microorganisms.

## Serology

If a recent streptococcal throat infection can be confirmed by demonstrating an elevated serum antistreptolysin O titre, Jones's criteria may be used for the diagnosis of rheumatic fever (Box 13.26). The presence of two major criteria, or one major and two minor criteria, indicates a high probability of rheumatic fever. In suspected viral pericarditis or myocarditis, the aetiological diagnosis depends on the demonstration of elevated viral antibody titres in acute serum samples, which decline during convalescence. Virus may sometimes be cultured from throat swabs and stools.

**Box 13.26** Jones's criteria for the diagnosis of rheumatic fever

**Major criteria**

- Carditis
- Polyarthritis
- Erythema marginatum
- Chorea
- Subcutaneous nodules

**Minor criteria**

- Fever
- Arthralgia
- Previous rheumatic fever
- Elevated erythrocyte sedimentation rate
- Prolonged PR interval